1. Excellence

1.1 Objectives

The increasing use of bioplastics in the packaging sector is driven by the continuous demand for sustainable products by consumers and brands alike, due to a growing awareness of the environmental impact and the need to reduce the fossil resources dependency. The bioplastics industry´s continuous innovations make the number of manufacturers, converters and end-users increasing steadily. New and innovative biopolymers such as polyhydroxyalkanoates (PHAs) show a very high relative growth rate. With a global production capacity of 25.300 Ton in 2019, it is expected an increase up to 160.000 Ton in the next 5 years[1] (from 1.2% to 6.6% of bioplastics´ global production capacity). In this context, new ways of producing PHAs are going to take a great importance in a short future, being one of the challenges open to source it from 2nd-3rd generation raw materials instead of raw sugars[2].

PHAs are an important polymer family, they are 100 percent bio-based, with versatile biodegradability properties in most environments exceeding those of any other bioplastic[3], recyclable and featuring a wide array of physical and mechanical properties depending on their chemical composition, from very flexible poly(3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBV) to rigid polyhydroxybutyrate (PHB), showing similar properties to some fossil-based materials like polypropylene (PP) and polyethylene (PE)[4] and being a good biodegradable and compostable alternative in packaging applications[5]. However, further improvements are needed to provide cost effective solutions with suitable performances for demanding packaging applications, and alternative environmentally sustainable end of life (EoL) that maximises the retention of the added value of these biopolymers in a new circular plastic economy.

On the other hand, spent organic materials have yet to reach an industrial level of market potential related to a value hierarchy (Figure 1). It is the case of brew spent grains (BSG), the most abundant by-product from the brewery industry[6], available at low or no cost and produced in large quantities[7]. Being high valued, its use as a feedstock is limited at present[8], primarily as animal feed (70%), bioethanol (10%) and landfilled (20%). Main challenges are a high moisture and low fermentable sugar content (mainly C6 sugars), making microbial growth and spoilage possible in only 7–10 days, thus being an environmental problem. Despite of them, BSG by-product is one of the most promising candidates as a PHA feedstock, because: i) high amounts are produced in Europe every year and in almost all European countries, ii) it is not seasonal, iii) it is highly stable in comparison with other food by-products, iv) 2G sugars isolation from BSG is more efficient since the complex structure of lignocellulose is damaged during malting and mashing and, therefore, the material is much more prone to enzymatic hydrolysis (less recalcitrant)[9], v) BSG's proteins act as a nitrogen source avoiding the use of expensive nitrogen supplements in the fermentation process, vi) BSG's lipids can be also carbon source for PHA production. Although the brewing industry tends to be more environmentally friendly than other industries[10], the increasing demand for sustainability in all operations could be satisfied by reducing the amount of BSG still landfilled or used in low added value applications. BioSupPack **strategy and innovations**, shown in Figure 2, have been well defined to overcome the described limitations and deliver novel packaging that will respond to end users' needs and European sustainability policies.

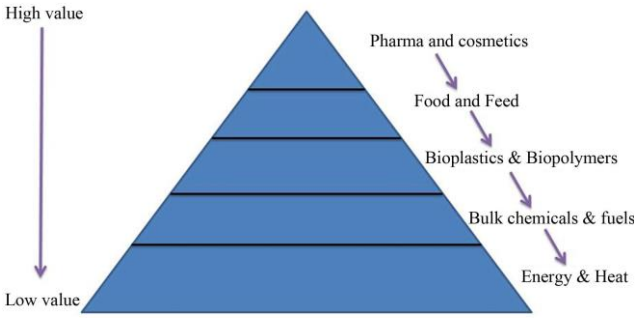

Figure 1: Biomass cascade *(Source: Peter Westermann, Copenhagen Institute Technology)*

In terms of packaging performance
PHA has been selected for the production of novel BioSupPack packaging due to its good mechanical performance in rigid packaging, its ability for being applied as wet coating on paper/cardboard substrates, and its better gas, fat and liquid barrier properties compared to other bioplastics like polylactic acid (PLA) and even some fossil-based materials[11]. Other requirements like squeezable properties and suitability for hot filling and microwavable applications are also addressed. The following approaches will led to high-performing bio-based packaging: For **primary packaging**: i) production of **tailored PHA compounds** for rigid packaging for food, cosmetic and homecare products; ii) use of already developed **plasma technology** to improve **barrier performance** of food and

[1] https://docs.european-bioplastics.org/publications/market_data/Report_Bioplastics_Market_Data_2019.pdf
[2] Bugnicourt e al (2014) PHA: Review of synthesis, characteristics, processing and potential applications in packaging. Express Polym Lett 8, 791-808.
[3] Debbarma et al (2017) Comparative in situ biodegradation studies of polyhydroxybutyrate film composites. 3 Biotech 7(3), 178.
[4] Jedra F. (2014) Polyhydroxyalkanoates: various properties, various applications. Arnhem: KCPK.
[5] Daraniet al (2015) Application of PHA in food packaging: Improvements by nanotechnology. Chem Biochem Eng Q. 2, 275-285.
[6] Buffington, J. (2014) The Economic Potential of Brewer's Spent Grain (BSG) as a Biomass Feedstock. Adv Chem Eng Sci 4, 308-318.
[7] Fărcae al (2014) Brewers ' spent grain -A new potential ingredient for functional foods. J Agroaliment Process Technol 20, 137–141.
[8] Gupta et al (2010) Barley for Brewing: Characteristic Changes during Malting, Brewing & Applications of Its By-Product. Compr. Rev. Food Sci Food Saf 9, 318-28.
[9] Robertson et al (2010) Profiling Brewers' Spent Grain for Composition & Microbial Ecology at Site of Production. LWT Food Sci & Tech 43, 890-96.
[10] Ishiwaki et al (2000). Development of high value uses of spent grain by fractionation technology. MBAA Tech Q.37, 261–5.
[11] G. Keskin et al (2017) Potential of polyhydroxyalkanoate (PHA) polymers family as substitutes of petroleum based polymers for packaging applications and solutions brought by their composites to form barrier materials. Pure Appl Chem 89(12), 1841–1848.

cosmetic packaging, mainly in terms of oxygen barrier, avoiding the use of multi-layered packaging structures and facilitating their posterior EoL; iii) optimization of recently developed patent-pending **PHA based wet coatings** to protect rigid service packaging based on paper pulp/cardboard from moisture, liquids and fats; iv) **grafting of fibre-based** PHA coated substrates to protect food packaging from gases, fat and liquid; iv) production of PHA based **compounds suitable for squeezable packaging** for food and cosmetics products. For beverage **secondary packaging**, specifically beer glass bottles multipack, a PHA based compound will be produced for rigid clips that will replace current packages (rustic wood packaging or plastic rings to carry six-bottle packs).

In terms of biobased packaging materials from biomass feedstock

BSG will be valorised to produce PHAs, specifically PHB. The whole process contemplates **innovations in upstream BSG pre-treatment, fermentation and downstream PHB separation and purification** stages focusing on the development of an economically feasible, **highly efficient and sustainable bioprocess**: i) enhancement of BSG to sugar conversion yield using the already developed **plasma technology** towards lignocellulose **pre-treatment** at low temperature combined with sequential enzymatic hydrolysis of cellulose and hemicellulose. Lipid and protein hydrolysis will be also assessed to supply carbon and nitrogen sources to the fermentation stage; ii) PHB production via fermentation with conventional mesophiles and novel **extremophilic bacteria** for reduced risk of contamination in non-aseptic cultures; iii) **combined use of sugars and 3-hydroxybutyric acid (3HB) monomer** derived via enzymatic recycling of used PHA packages as carbon sources in fermentation; iv) extraction of PHB from bacterial biomass by **'green' solvent and recyclable switchable anionic surfactants** to facilitate PHB extraction of high purity and yield, avoiding any PHB degradation leading to retention of the molecular weight (Mw) produced during fermentation. The PHB obtained will be used for the development of tailored compounds (physical modifications by blending with necessary additives, and commercial PHAs if required) for rigid packaging for food, cosmetics, homecare and beverage. Physical modifications of selected commercial PHA grades aimed at wet coatings for food pulp based rigid packaging (service packaging), will address the need for alternatives to PE coated paper that cannot be neither recycled nor degraded in nature.

In terms of packaging EoL sustainability

The inherent properties of PHA will be exploited not only in the design and production of packaging prototypes, but also in the selected EoL by **enzymatic recycling.** Enzymatic degradation of PHB is easier and faster compared to other biopolyesters like PLA or polybutylene succinate (PBS)[12]. This EoL will provide **profitable precursors** to be reintroduced in the value chain, maximizing the retention of the biopolymer added value in a new circular plastic economy. Specifically, derived 3HB will be an **additional carbon in fermentation** for PHB production, besides BSG´s sugars and lipids, allowing producing the **original biopolymer with the same properties**. Whereas compostability can be suited for applications whose mechanical recycling is extremely difficult or impossible, e.g. packaging waste containing remains of the product packed or PE coated paper as PE cannot be separated from the paper fraction being the packaging not recyclable, enzymatic recycling using **selective enzymes** prone to hydrolyse PHAs (vs. product remains, paper) is a very suitable EoL. While considering recycling as a favourable option for PHA upcycling, it may require adaptation of the sorting lines. As such, photonic based solutions will be delivered to assess the **correct sorting** towards industrial recycling streams. The proposed EoL will contemplate: i) use of **plasma** as **pre-treatment** of the packaging waste, reducing the polymer Mw and modifying its surface properties to facilitate subsequent enzymatic hydrolysis; ii) exploit known microorganisms producing PHB depolymerase releasing monomer 3HB during PHB hydrolysis; iii) select organisms from currently available metagenomic databases created from extreme environments that produce PHB depolymerase with higher 3HB production rate than existing ones; iv) implement enzyme engineering to improve PHB depolymerisation efficiency to 3HB; v) employ cloning and heterologous enzyme expression to overproduce the targeted PHB depolymerase; iv) optimise enzyme cocktails to maximise PHB depolymerisation efficiency. BioSupPack also considers the **mechanical recycling of industrial PHA scraps** to be reintroduced in the conversion processes loop, increasing circularity and making the whole process more cost-efficient and environmentally sustainable. For PHA based coated paper, BioSupPack will address the need for enzymes and bacteria during the repulping of such packaging materials, so that they can be recycled with the simple addition of specific microorganisms or enzymes to the recycling process.

[12] Tokiwa, Y.; Calabia, B.; Ugu, C.; Aiba, S. (2009) Biodegradability of Plastics. Int J Mol Sci 10(9): 3722–3742.

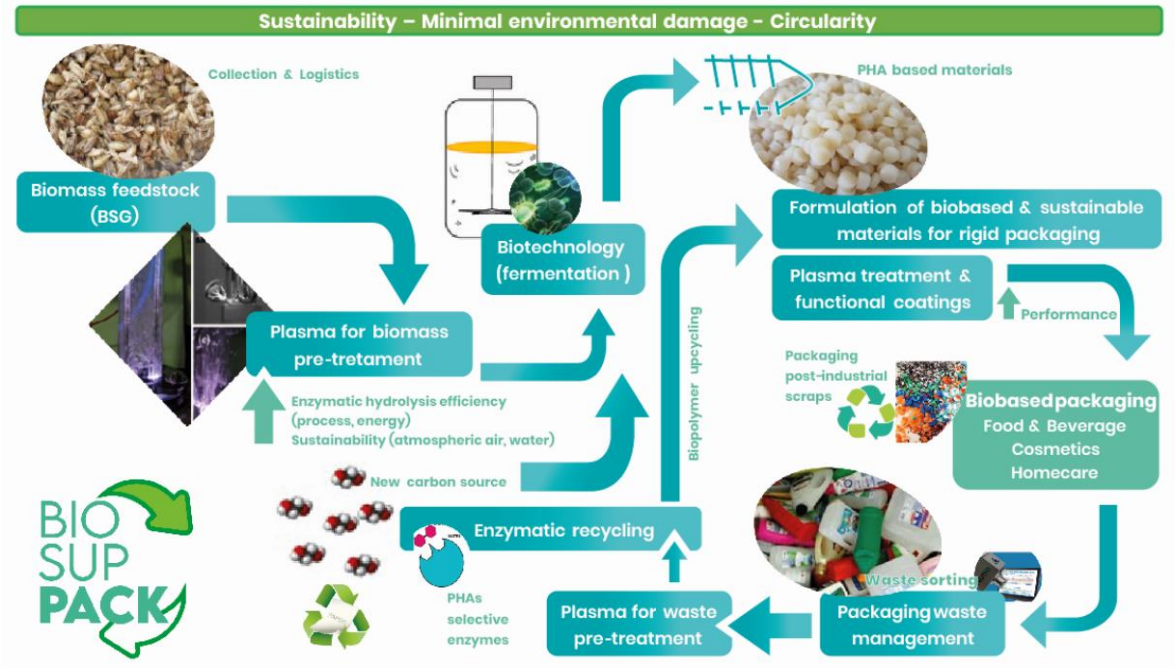

Figure 2. BioSupPack strategy and main innovations

The project contemplates the production of different **rigid packaging prototypes** representing large volumes on the packaging market, and where current performance of biobased biodegradable materials, waste management and EoL contexts must be improved (Figure 3). It is addressed to applications where the novel packaging can influence the consumers´ perception about the use of plastic packaging. The innovative materials, which will replace conventional polyolefins like PE and PP, will cover the food, beverage, cosmetic and homecare packaging of early adopters present in the consortium, but their relevance goes far beyond this, with potential impact in the brewery and the biotechnology industries, covered by our advisory board (AB).

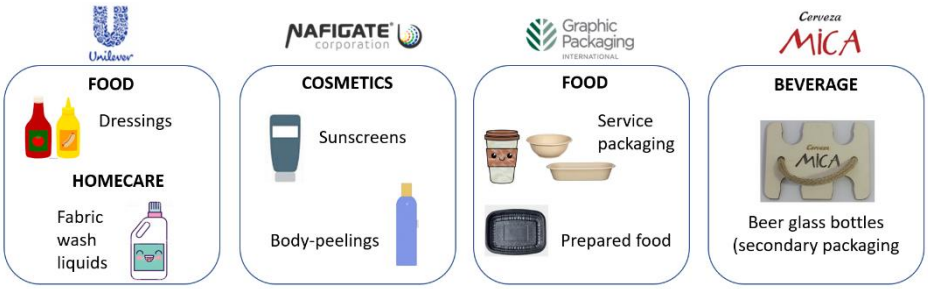

Figure 3: Packaging prototypes to be produced in BioSupPack

The main goal of BioSupPack is to **deliver novel, cost-competitive and versatile bio-based packaging solutions based on PHA, that demonstrate high-performance for the packaging of food, cosmetics, homecare and beverage products as well as no environmental damage** during and after their use, by means of:

1. Optimization of PHA based formulations based on significantly **use of >85% w/w of renewable resources**.
2. **Upscale the different conversion processes as well as post-consumers waste sorting and recycling** following advanced industry 4.0 approaches.
3. **Broad the range of rigid packaging applications** by tailoring biobased materials and packaging properties through the optimization of the formulations in combination with plasma technology or grafted coatings.
4. **Integrate plasma technology** in three different points of the value chain (biomass pre-treatment, packaging production and packaging waste pre-treatment) increasing: i) PHB production yield, ii) PHB purity, iii) packaging performance (high oxygen and water barrier) and iv) effectiveness and yield of enzymatic recycling.
5. **Demonstrate and increase recyclability** through: i) setting up a real-time monitoring system for **the selective separation** of developed packaging after its use; ii) **mechanical recycling** of industrial scraps and **selective enzymatic recycling of packaging waste as the best EoL**, with the final recovery of carbon sources for feedstock fermentation. Thus, the new packaging will be environmentally safe (sustainable and improved value from enzymatic recycling) and will **contribute to Circular Economy (CE)**.
6. **Establish a new value chain** including the development of logistics and management of both the brewery waste and the packaging waste.

Complying with the **industry needs** (cost/performance competitive vs. fossil-based non-biodegradable counterparts and legislative compliance), as well as with **consumers´ awareness**, BioSupPack will have a **great impact** on EU bioplastics & end users' sectors, the biorefineries and biotechnology industries and on the society.

To realise this over a **42-month** work plan, the following **specific objectives**, linked to their respective WPs and key performance indicators (KPIs), later used as milestones and check points for project implementation, will be fulfilled:

Objective 1: To obtain PHB from BSG feedstock and monomers from enzymatic recycling of PHA packaging waste (WP1, 5)

Production of nutrient-rich fermentation medium from BSG for PHB production via lignocellulose pretreatment with non-thermal plasma will be followed by enzymatic hydrolysis. Cellulose, hemicellulose, lipid and protein hydrolysis will be targeted. Crude hydrolysates will be used as fermentation feedstock for PHB production using either conventional (i.e. Burkholderia sacchari) or extremophilic (e.g. halophilic Halomonas halophila CCM 3662, thermophilic Schlegelella thermodepolymerans DSM 15344) bacterial strains. Crude 3HB derived via enzymatic recycling of used PHA-based packages will be used as feeding solution in fed-batch cultures to avoid concentrating the sugar-rich hydrolysate, leading to efficient up-cycling of the new materials after EoL. Downstream PHB separation and purification will be based on disruption of bacterial cell walls and solving PHB. Novel green and recyclable solvents or biosurfactants will be employed to achieve the most efficient and sustainable process of PHB production in terms of productivity, yield, purity and PHB properties to be scaled-up. **KPI**: By M24, PHB production process enabling reaching desired product yields in pilot plant scale with the following objectives: (i) BSG cellulose and hemicellulose to sugar conversion yields higher than 70%; (ii) higher than 50 g/L PHB concentration, 0.25 g/g yield and 1 g/L/h productivity in fed-batch fermentation; (iii) PHB purity higher than 99% with Mw of 500-600 kDa.

Objective 2: To obtain tailored PHA based compounds and wet coatings for packaging prototypes (WP2)

PHA based compounds with tailored properties will be obtained to produce rigid packaging (bottles) by extrusion blow moulding (EBM). Previous know-how in the development of PHA based compounds and considering the barrier and mechanical properties of the PHB produced from BSG, final PHA based compounds will show similar water vapour barrier as PE, being able to replace it in most of the target packaging applications. Patent pending PHA based coatings will be fined tuned for delivering hydrophobic properties and improved barrier to pulp-based packaging (bowls, trays and coffee cups). In the case of bottles for food and cosmetics, squeezable properties will be tailored according to the compound's formulations are EBM parameters will be optimized to obtain the desired effect. **KPI**: By M24, optimized PHA compounds and wet coatings will be available.

Objective 3: To produce rigid packaging prototypes at pilot scale and improve their performance (WP2)

Production of rigid packaging prototypes integrating the system package/closure will be undertaken at a demonstrative level (pilot scale) by using the produced materials in WP1 by EBM in the case of bottles, and wet coating application by rotogravure on pulp-based substrates in the case of pulp-based bowls/trays and coffee cups. Barrier properties will be improved by surface modification of the obtained packaging using plasma technology, applied both in bottles for food and cosmetics and on the PHA coated pulp. It is expected to improve barrier performance about 20% respected untreated samples. In coated structures, water vapour barrier will be in the same ranges as PE, PE coated cardboard and PP current packages. In addition, a bio-based nano-scale coating will be applied to fibre-based substrates to enable deposition of monomolecular layers of fatty acids that will drastically improve the substrate barrier properties by a factor of up to 20. The objective is to reach barrier values suitable for selected modified atmosphere packaging applications. For squeezable bottles, the combination between materials composition and parameters in EBM processing technology will be crucial. Closure systems will be produced by injection moulding technology. The consumer & market requirements for the new packaging solutions for each target application will be considered, as well as the strategy to be implemented to maximise the overall economic/environmental viability of the final packaging. **KPI**: By M30, prototype series with suitable properties of: i) PHB based bottles and secondary packaging will be available for their validation with food, fabric wash liquids and beverages; ii) PHA-coated fibre-based bowls, trays and cups will be available for their validation with ready-to-take food and beverages.

Objective 4: To demonstrate performance and assess consumer perception/acceptance of new packaging (WP3)

Performance of packaging prototypes will be validated by the end-users according to each application following the corresponding standards. Squeezable properties in dressings and cosmetics products will be assessed, as well as packaging suitability for hot-filling processes in the case of dressing bottles and microwave resistance for prepared food. Migration tests will also be carried out on food and cosmetics packaging to check fulfilment of current legislation in these applications and guarantee packaging safety (linked to objective 11). For beer secondary packaging validation, the involvement of a big distributor company (stakeholder Calidad Pascual) is also contemplated. Furthermore, the consumer perception of the new packaging will be assessed in collaboration with the project end users. **KPI**: By M40, comparable shelf life and suitable consumer acceptance vs. benchmark non bio-based packaging.

Objective 5: To set up a real-time monitoring system for selective separation of new packaging after its use (WP4)

As prior sorting is needed for addressing BioSupPack packaging waste to recycling, the ability to sort the different packaging developed in the project will be assessed priorizing the use of conventional photonic approaches available in packaging waste facilities such as near infrared (NIR). **KPI**: By M34, establishment of the ability to identify at least 90% of BioSupPack materials vs. conventional plastic and fibre-based packaging waste streams.

Objective 6: To develop logistics and management of the biomass for polymer production and management of the packaging waste for enzymatic recycling feasibility as EoL (WP4)

The logistical steps of locally and regionally source of the brewery waste (malt bagasse), including time distances costs, needs to be improved to ensure sustainability, security and reliability. The same happens with the new packaging waste management. This will contribute to the minimization of the environmental impacts, to the creation of the new value chain and to the transformation of a traditional linear economy into a circular economy. **KPI**: By M12, a report providing technical and functional architecture of best logistical system, and by M40, a report providing the design of the packaging waste management.

Objective 7: To demonstrate feasibility of post-industrial waste reintroduction in the productive processes (WP5)

Reintroduction of new materials´ industrial scraps will be assessed to guarantee economic feasibility of the processes and increase circularity. To guarantee the maximum utilization of the developed materials until their properties downgrade too significantly, the number of cycles for reprocessing the new materials will be assessed. In the case of the PHA coated fibre-based substrates, a re-pulping process will be assessed to recycle and reuse the cellulose fibres in new products after an optional enzymatic removal of the coating (in link with next objective). **KPI**: By M40, the ability to mechanically recycle PHA developed materials, blending less than 20% of virgin materials in any new production cycle targeting application of similar value as previous life cycle will be shown.

Objective 8: To demonstrate feasibility of enzymatic recycling as BioSupPack packaging waste EoL (WP5)

Used PHA-based packages will be pre-treated with plasma focussing on PHB surface modification and potential reduction of Mw. Then 3HB monomers will be produced via optimisation of enzymatic hydrolysis using enzyme cocktails containing existing and/or efficient PHB depolymerases mixed with lipase. Existing PHB depolymerase producing 3HB monomers will be exploited (e.g. from Bacillus megaterium ATCC 11561). BioSupPack will select new PHB depolymerases genes with 3HB producing ability. Improved variants of native enzymes will be obtained via combined directed evolution and rational engineering involving site-directed mutagenesis. After selection and cloning of PHB-degrading enzymes, recombinant plasmids will be developed for large-scale production in Escherichia coli and Pichia pastoris production hosts. 3HB-rich hydrolysates produced via enzymatic hydrolysis will be used as carbon source in fermentation processes for PHB production. **KPI**: (i) achieve more than 50% of used PHB-based package hydrolysis into 3HB monomer recovery; (ii) complete utilisation of 3HB-rich hydrolysate as carbon source in fermentative PHB production leading to at least 30% carbon source supply via recirculation in the fermentation process. By M40, feasibility of selective enzyme depolymerization for monomer recovery will be shown.

Objective 9: To demonstrate environmental & socio-economic sustainability of new compounds/products (WP6)

LCA (life cycle analysis), s-LCA (social LCA) and LCC (life cycle cost) analysis will be performed. It will include the study of all involved process steps and developed products, including the mechanical recycling assessment of the new materials scraps. The results of the suitability for the targeted end of life scenario will be also taken into account. **KPI**: By M42 reduced environmental and economic impact vs. selected benchmarks feeding into post project industry, stakeholders and policy waste management recommendations to be delivered in WP8. Technology acceptance will be demonstrated through S-LCA results.

Objective 10: To define sustainable circular value chains & cross-sectorial connections (WP4, 7, 8)

The project will build a new value chain by developing sustainable, highly productive biomass collection and supply systems and making better use of biomass feedstock, including co-products, side streams and residual streams from various sources. A value chain sustainability analysis will be performed based on existing best practices in the field of bio-economy, markets for the products, as well as the readiness of key actors to shift to such bioplastic solutions. The value chain study will also address how the results could tap into the much broader potential of the technologies developed. Valorization of BSG´s additional components like phenolics or proteins derived as additives to different sectors (e.g. antioxidants for the bioplastics industry, animal feed or feeding supplements) is also considered. **KPI**: By M40, a report will be delivered on the interconnection across the involved sectors.

Objective 11: To guarantee safety of the new biobased packaging (WP3,7)

The packaging solutions delivered by BioSupPack project must comply with all applicable regulations (e.g. food, cosmetic packaging) and safety regulations in particular. REACH compliance will be considered at all development stages. Packaging prototypes for food, beverage and cosmetics applications will be validated not only in terms of performance, but also in terms of fulfilment of legislative aspects involved. Migration tests will be carried out to ensure safety according to the corresponding European standards. A regulatory assessment will be done during the project duration, in order to be updated about possible legislative changes or novelties related to BioSupPack and the final products obtained. **KPI**: By M38, food, beverage and cosmetics packaging will be fully validated.

Objective 12: To maximise the innovation impacts by contributing to the post-project development and uptake of the project results for growth and jobs (WP7, 8)

The project knowledge and results will be managed and protected. Business and exploitation plans will be delivered for all Key Exploitable Results (KERs) to map the post-project phase and business models will be set up. A professional communication campaign to outreach, engage with and mobilise our key stakeholders will be designed and executed. In addition, significant efforts will be devoted to encouraging policy makers to support beneficial (or at least non-discriminatory) legislative frameworks for BioSupPack solutions and contribute to the development of new or improved industry standards. Regular knowledge transfer activities will be organised among the partners to secure the post-project stage. Besides, specific trainings will be targeted to scientists, end user industry and policy makers to guarantee upskilling and knowledge exchange. Furthermore, BioSupPack will participate in the Innovation Radar Assessment, the EC initiative to identify high potential innovations and innovators in EU-funded research and innovation projects. ***KPI:*** By M6, a robust communication plan to support the recognition of BioSupPack as a reference in the bioplastics marketplace will be available, as well as to secure stakeholder engagement with the project. From M36 to M42, at least 3 training sessions will be held to ensure knowledge exchange. By M42, final plans will be in place for the transition to the post-project phase to advance the TRLs and reach the market, sources of further funding will be analysed focusing on applications where a return on investment is feasible based on the assessment performed in WP6, as well as policy and regulatory recommendations.

1.2. Relation to the BBI-JU Annual Work Plan

The BioSupPack proposal, addressing the topic BBI.2020.SO3.D4 – ***Demonstrate superior bio-based packaging solutions with minimal environmental damage,*** will contribute significantly to the KPIs set out in the Strategic Innovation and Research Agenda (SIRA)[13] and is fully aligned with the BBI roadmap as will be further discussed in section 2. Although our overriding goal **aligns with** its **4 strategic orientations (SO)** (Figure 4), **covering the full value chain**, it is **mainly focused in SO3** *Developed innovative biobased products for identified market applications*. The alignment with the different SO is justified as follows: SO1) we will use biomass derived feedstocks (>85%), although minor non bio-based compounds, e.g. processing aids or additives may be used if no alternative exists, specifically waste from agro-industrial value chains: BSG from the brewery industry. According to the expected increase in PHAs global production in a short time and the current difficulties in getting suitable commercial PHA grades in Europe in terms of cost and performance, new ways of producing PHAs become of a great of importance, being necessary promotion of 2nd and 3rd generation raw materials to source it. SO2) we will include different approaches in both the production of new biopolymers from biomass and the enzymatic recycling of the new packaging waste focused on the increase of cost-efficiency and environmental sustainability; SO3) we will developed bio-based rigid packaging with suitable performance for food (dressings, prepared food, service food), cosmetics (body-peeling, sunscreen), homecare (fabric wash liquids) and beverage packaging (beer secondary packaging), meeting the identified market applications' needs in terms of performance and a more sustainable EoL compared to existing packaging solutions through: i) improved barrier performance biobased materials (oxygen, moisture, liquids/fats); ii) squeezable, hot-filling and microwavable properties; iii) recyclability of packaging waste containing rests of the packed products by selective enzymatic degradation; iv) recyclability of new fibre-based service packaging (SUP) vs. PE coated paper (SO4) the high volumes that the selected applications represent in the packaging sector together with the direct involvement of end-users in the project and the planned awareness raising campaigns, will allow us to meet the needs of our EU end user industries and the consumers' demand for greener products.

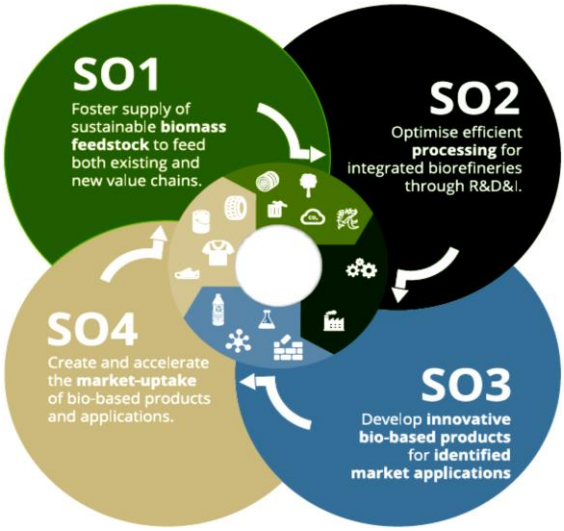

Figure 4: The 4 SO in the BBI SIRA

Besides creating socio-economic value, the project will also help the EU in implementing better waste management strategies as well as reducing the major societal challenge reflected in the European Plastics strategy. The detailed relation of the proposal to the specific challenge and scope addressed by the topic are outlined in Table 1.

In keeping with the **expected scope of proposal under BBI 2020.SO3.D4 topic,** BioSupPack has a budget of ca. €8.79 million and an EC requested contribution of ca. €6.4 million (within the range guided by the call text, including 38.2% in kind contributions by partners as described in Section 2).

[13] Strategic Innovation and Research Agenda (SIRA) BIO-BASED INDUSTRIES for Development & Growth in Europe – May 2017

Table 1: How BioSupPack relates to the scope and challenges of the BBI 2020.SO3.D4 topic

Topic text	How BioSupPack addresses these challenges/scope
Upscale the production of sustainable and high-performing bio-based packaging solutions that do not create environmental damage during and after use.	BioSupPack will deliver at least 2 biobased PHA materials to produce high-performing rigid packaging: PHA based compounds and grafted PHA coated fibre-based materials. They will provide up to 8 sustainable packaging solutions (Figure 3) in the food, cosmetics, homecare and beverage sectors. Packaging prototypes will be produced at a demonstration level in terms of polymer production, plastics converting and specific technologies to improve barrier performance. Sustainability is addressed in the whole value chain, from the routes and technologies to produce the new materials and packaging, to the recyclability of post-industrial scraps and packaging waste, which respectively, will follow mechanical and enzymatic recycling as the most suitable EoL, maximizing the retention of the new biopolymer´s added value in a new circular plastic economy by the production of precursors for their reintroduction in the value chain and the PHA upcycling. Moreover, the extraction of additional BSG components, specifically phenolics to be used as antioxidants (e.g stabilizers in PHB processing) and proteins to be addressed to animal feed and food supplements, complement packaging sustainability. See Section 2. Environmental damage after use is also addressed in case of accidental littering by exposing our developed solutions for service packaging based on PHA coated paper/cardboard, to biodegradation trials in marine and soil environments as contingency plan.
Superior packaging performance to fossil-based alternatives and to existing bio-based material such as paper at a demonstration level.	BioSupPack PHA based rigid packaging will show superior performance in terms of barrier properties (oxygen, moisture, liquids and fats) when compared to standard fossil-based materials and biobased biodegradable materials on the market, as described in Section 2. The use of PHA as based material, in combination with the technologies addressed to increase performance (plasma and wet coating grafting) will led to superior packaging performance in simplified structures (PHAs monomaterial or coated structures), avoiding multilayer structures which combine non compatible materials and tie layers with limited recyclability, or the use of non-recyclable PE coated paper where paper cannot be separated from PE. Other properties like squeezability, compatibility with hot filling and microwaving will also be provided.
Address only one of the following two objectives: *A.Produce superior, reusable or recyclable, bio-based plastic solutions for specific applications that demand reusability or recyclability as the best end-of-life.* *B.Produce superior bio-based biodegradable/compostable plastic solutions for specific applications that demand biodegradability/compostability as the best end-of-life.*	BioSupPack addresses the objective of producing superior recyclable biobased plastic packaging solutions for specific applications (food, cosmetics, homecare and beverage) that demand recyclability as the best EoL. Although being packaging solutions based on PHAs, and therefore biodegradable/compostable, recycling increases the fossil carbon reduction potential by keeping the renewable carbon in the material cycle even longer, thus, being a more sustainable solution in general terms. Provided both recyclability of PHA materials and the specific applications addressed, BioSupPack will demonstrate increased sustainability of the mechanical recycling of post-industrial scraps and enzymatic recycling of packaging waste that may contain rests of the products. Thus, selective enzymatic recycling will contribute to packaging recyclability without detriment to the environment after their use and will contribute to increase retention of the biobased materials added value by their upcycling. PHA based materials´ upcycling will be done by two ways: the reintroduction of the obtained precursors (3HB) from enzymatic recycling in the fermentation process as additional carbon source to BSG´s sugars and lipids and PHB polymerization to obtain original PHB for the same packaging applications. In addition, for PHA coated paper/board packaging, this EoL will allow separation of the fibre-based materials from the coating, allowing their recycling. In the same line, although paper/board is compostable, it is currently one of the most recycled materials, BioSupPack will align the best EoL for coated PHA fibre-based materials, providing a recyclable solution to current PE coated paper/board packaging.
The demanded proof should meet accepted standards for performance, recyclability, degradability/compostability. The scale-up solutions must comply with all applicable regulations (e.g. food, cosmetic packaging) and with safety regulations in particular.	Standards that the processes and the new products need to comply with will be considered along the whole project. REACH compliance will be considered at all development stages. In the case of food, beverage and cosmetics packaging, safety will be established by the packaging prototypes validation through the corresponding migration tests according to European legislation. The rest of the validation tests (e.g. mechanical and barrier properties) will be carried out considering the corresponding standards. The recyclability of the products obtained in the project will be analysed through standards about recyclability of packaging as UNE-EN 13430:2005 Packaging – Requirements for packaging recoverable by material recycling. The use of recycle materials in the new packaging produced will also be considered, as PHB produced from enzymatic recycling could be used in food and cosmetic applications.
Pave the way to improved consumer perception and greater acceptance of bio-based solutions.	The project contemplates, as a part of the packaging prototypes validation, a study on consumer perception, confidence and acceptance of the new and sustainable packaging solutions for the different applications. It will ensure the sustainable orientation of BioSupPack to the actual market needs. Thus, the time of introduction to the market of the proposed solutions in form of the developed prototypes can be minimized and market success can be optimized. An additional objective is to check whether the packaging specifications defined in the beginning of the project and the prototypes developed meet the consumers' requirements. Expert interviews, literature research and brand owner surveys will be used as multipliers to extrapolate consumer perception and consumer acceptance tests will be carried out with the developed prototypes. The end-users also will also take part as experts in their corresponding sectors to which the new packaging are addressed. NFG partner, also as a producer of cosmetic formulations, will test the customers' response to the new bio-based packaging on their products according to their own methodology based on three market research levels: business-to-customer, business-to-business and small marketing. MICA partner will validate the beer secondary packaging trough one of the project stakeholders, Calidad Pascual, a big distributor of MICA product. ULK partner will coordinate consumer testing (functionality, appearance and home use) by using current professional methodology of consumer research experts. Additionally, an approach of consumer perception for the environmental footprint of the novel packaging and value perception will be included. The evaluation of consumer perception is complemented by the social life cycle assessment (S-

	LCA). To profile the differences in perception and acceptance of bio-based packaging among women and men, the gender of the respondents will also be considered in the analysis.
Proposals should include waste-management actors as beneficiaries for the design of the value chain, including in the end-of-life phase. This is a necessity if the developed packaging solution does not integrate into existing waste-management infrastructure.	BioSupPack counts with the presence of a waste manager actor as a partner (SAV) with more than 120 years in waste and treatment collection, for both the logistics and management of the biorefinery waste (from MICA) to be valorised in the project and the management of the packaging waste addressed to recycling, so that the logistics for collecting BioSupPack specific packaging and bring them to the recycling actors is assured. SAV is a large Spanish enterprise focused on providing environmental services including bioprocesses and counts with their own bio-process plant in Valencia. Its large expertise will be high valuable in the design of the whole value chain created in the project.
Proposals including feedstock supply by primary sector should involve actors from the related primary sector as strategic partners in the value chain and as beneficiaries. They should participate in the design of the targeted value chain and benefit from it. *Offer solutions to problems in the logistics, transport modes and associated infrastructure.*	Beer produces annually around 3.4 MT of BSG in EU and 38.6 MT globally. Due to its high-water content (60-80%) it experiments high moisture levels, provoking a deterioration and spoilage of its proteins and polysaccharides in 7-10 days, being able to cause smell and even health hazards. Also, the actual pathways under exploit the BSG due to different issues, related to its specific indicators and scarce tradition in drying the BSG. Then, the main problem is its characterization as a residue and its deficient management, while it could be an excellent raw material (or feedstock) supply. BioSupPack will valorise biomass from BSG that will be supplied by MICA project partner as feedstock for PHA based production as the new biobased materials. MICA will work in close collaboration with SAV in the design of the targeted value chain, considering logistics, transport models and associated infrastructure. MICA will also take advantage of the new biobased materials as being one of the packaging end users in the project. This collaboration is considered in WP4 together with stakeholder Vitoria-Gasteiz City Council, very active in CE and social awareness in these new types of packaging solutions.
Proposals must ensure that the biomass supply chain is sustainable. In the case of biomass and supply from the agricultural and marine sectors, proposals must show they can integrate with the food chain and are not in competition with it but can help to enhance it.	BioSupPack will recover the leftovers by the brewer industries destined for feeding livestock, used for generating bioethanol, but still a significant amount is wasted, being disposed of in landfills (about 20% of the by-product generated by the industry[14]). The project aims to encourage the use of this residue as a raw material in the bio-industry sector, eliminating the quantity that ends up in the landfill and producing a biopolymer that is a higher market/environmental value commodity product than ethanol. Besides, the logistical solution contemplates IoT & blockchain technology to reduce the carbon footprint during the transport process. As a feedstock, BSG has the potential of the development of a biorefinery concept of Zero Waste, where MICA valorises all BSG components by: i) hydrolysis of sugars, lipids and proteins, where lipids are an additional carbon source to sugars in fermentations and the protein hydrolysate will be used as a nitrogen source to avoid commercial nitrogen sources; ii) extraction of phenolics to be used as additives by the bioplastic and the cosmetic industries; and iii) extraction of protein isolate from remaining BSG solids to be used as food supplement or animal feed. See Section 2.
Demonstrate the benefits versus the state-of-the-art technologies. Include evidence of new processing solutions and/or new products obtained.	Validation of demonstrators (bottles for food, cosmetics and homecare, fibre-based trays, bowls and cups for food and secondary packaging for beverage) will be performed by final users according to their established quality controls. It will encompass the testing of the processability of the new materials by the converters, demonstrating their compatibility with existing converting technologies and packaging lines, and the comparison with existing biopolymers and fossil-materials and counterparts on the market, showing their outperforming functionality. An increased efficiency and sustainability will be also demonstrated by: i) PHB production via enhanced BSG conversion into PHB, no chemical usage for BSG pre-treatment, reduced energy demand; ii) enzymatic recycling with carbon source recovery and no chemical usage; iii) recyclability, by the implementation of selective sorting devices in PHAs based packaging waste management.
Assess the environmental and economic impacts of the developed processes or products on the different stakeholders and actors involved in the value chain, via an LCA and an LCCA. Analyse also social impacts.	BioSupPack will valorise BSG and will address the new packaging to recyclability (mechanical recycling of post-industrial scraps and enzymatic recycling of packaging waste for PHA upcycling). New materials, technologies and the creation of the new value chain, in which circular economy is promoted, will have environmental and socio-economic impacts to be assessed (LCA, LCCA and S-LCA methodologies) in WP6. Project partners and stakeholders along the value chain will have an important role in these analyses, also to detect points or aspects to be optimized for maximum environmental and cost efficiency of logistics procedures, processes and management, being linked to WP4 and 5 activities and the business plan in WP8. The new packaging will be applied in large markets for target applications, having a potential great impact in socioeconomic terms. High-performing packaging should be cost competitive compared to current packaging solutions but also accepted by consumers. The study of consumers´ perception and acceptance in WP3 will be linked with the social impact assessment in WP6.
Proposals for projects on recyclability should include the phase of their lifecycle where the envisaged materials or products are captured/separated for recycling.	BioSupPack addresses recyclability as the best EoL of the packaging waste. Specifically, it considers industrial enzymatic recycling as a favourable option for PHA upcycling, as it will be demonstrated in WPs 4 and 6 by connecting the results from its feasibility with their environmental and socio-economic impacts. In this context, as it may require adaptation of the sorting lines, the project will deliver in WP5 photonic based solutions. A sorting prototype will be designed and built to be adapted to the actual needs of waste management facilities, being fitted to a conveyor. It will be validated at pilot scale under simulated industrial conditions and it should guarantee the 90% identification target. For the validation, packaging waste provided by stakeholder Romei, together with samples of the packaging prototypes produced in the project will be used.

[14] https://lifebrewery.azti.es/

1.3 Concept and methodology

Overall concept

Today resource-efficient plastics are present in an infinite range of products and applications helping us to save energy, carbon dioxide emissions, water and even food. They contribute to circularity, to health and safety and to mitigate climate change. However, to make the most of these materials, challenges related to the EoL of certain products, and particularly plastic packaging, still need to be addressed. The increasing use of biopolymers in packaging applications is aligned with this necessity, when alternative EoL to landfilling, incineration or mechanical recycling are considered, like biodegradability or compostability. But bioplastics are not only used for this purpose, but others like the use of non-fossil based raw materials. Linking both purposes and considering the fact that in Europe we still landfill nearly 40-50% of plastic waste[15] and only the 25-31% is currently recycled, bioplastics are a good alternative to conventional polymers, covering the current EU´s policies in terms of the use of non-fossil based resources, the recyclability rate increase of materials and packaging products (when using non-biodegradable/compostable bioplastics) or the look for the most favourable EoL according to the final application. BioSupPack project addresses these EU´s policies through the use of brewery waste, available in large amounts and with a suitable composition for the production of PHA based bioplastics in high yield fermentation processes[16], but also goes a step further when considering EoL of the new biobased packaging, as it will boost recyclability of the biobased materials and products, by enzymatic recycling of the packaging waste. This EoL allows getting more value to the packaging materials compared to biodegradability or compostability, as precursors suitable for feeding the fermentation processes or even polymerize PHA, increasing circularity. In addition, BioSupPack addresses also the specific need for recyclability of current single use packaging (service packaging) made of PE coated paper/cardboard, difficult to recycle due to the complex separation of both materials, by the use of PHA based coatings and the use of enzymes that degrade then after the packaging life, leaving the paper/cardboard 100% available for recyclability. Being paper and carboard the packaging material with the highest recycling rate in Europe (85,8% across the 28 EU countries)[17], the project will allow integrating this waste stream in the recycling loop.

The BioSupPack project will:

1. Develop and **produce at a demonstrative level** a range of **bio-based material formulations with tailored properties** for identified packaging applications, derived from available agri-food by-products (PHA/PHB based compounds) and from commercially available PHAs (wet coatings).
2. **Validate** the use of the resulting biobased materials in several **foods, cosmetic, homecare & beverage packaging** and **their environmental impacts and economic feasibility** compared with existing benchmarks.
3. **Provide recommendations for suitable business approaches** for the new packaging materials and derivates from BSG, from an integrated bio-based **value chain perspective** including, among others, consumers & end users requesting more sustainable solutions, feedstock logistics and management and waste management bodies through a series of outreach activities exchanging knowledge between the relevant actors.

The considered technologies are summarised in Table 2, together with their purpose.

Table 2: BioSupPack bio-based materials, packaging and processes innovations to achieve project goals

Life Cycle Step	Technology to be applied	Effect	Addressed to
Sourcing	Non-thermal air cold plasma technology	Increased BSG to sugar conversion yields in subsequent enzymatic hydrolysis. Efficient pre-treatment at ambient temperature and pressure.	Increase efficiency and sustainability of PHB production via enhanced BSG conversion into PHB, no chemical usage for BSG pre-treatment, and reduced energy requirements.
	Fermentation routes with extremophilic bacteria	Reduced risk of contamination and energy demands.	
	Extraction of PHB from bacterial biomass by 'green' and recyclable solvents and switchable anionic surfactants	PHB extraction of high purity and yield and no Mw reduction via PHB degradation.	
Packaging production	Air cold plasma technology	Oxygen and water vapour barrier for food and cosmetic packaging, homecare.	Improve packaging performance.
	PHA wet coating and wet coating grafting on pulp-based substrates	Water vapour and fat/liquids barrier for service and prepared food packaging.	
	Tailored PHAs compounds to be applied in packaging production.	Squeezable packaging for sauces and cosmetics. Microwable packaging / Hot-filling properties.	
EoL	Combination of several spectroscopic technologies (e.g. NIR and visible HSI)	New sorting technology for PHA based BioSupPack packaging waste.	Selective separation, increase recycling efficiency.
	Cold plasma technology	Surface modification, reduced Mw of PHA used packaging for enzymatic hydrolysis. Increased used packages to 3HB conversion yields.	Increase efficiency and sustainability in enzymatic recycling as EoL, with carbon source recycling and no chemical
	Exploit existing strains and currently available metagenomics libraries from	Production of highly efficient PHB depolymerase for used PHB-based package	

[15] http://ec.europa.eu/environment/waste/plastic_waste.htm
[16] Mussato, S.I. (2014) Brewer's spent grain: A valuable feedstock for industrial applications. J Sci Food Agric 94(7), 1264-1275.
[17] EU figures ahead Global Recycling Day

	extreme environments for selection of PHB depolymerase	hydrolysis into 3HB monomers profitable by the industry.	usage for PHA recycling, causing no damage to the environment.
	Directed evolution and rational design of enzymes. Use of enzyme cocktails	Increased PHA depolymerization rate of selected enzymes and enhanced 3HB production efficiency from used packages.	
Logistics along the value chain	Blockchain	Traceability from waste feedstock to end product. Increase transparency to stakeholders.	Certify feedstock streams along the supply chain.
	QR code	Tracing the information from the blockchain and the characteristics of the feedstock.	Facilitate traceability information access to stakeholders.
	Eco-driving	Reduction in a 20% of GHG generation: 17% due to the efficient conduction system, focusing on a less aggressive driving system and a route optimization; 3% due to an intelligent system by the incorporation of IoT.	Reduction on climate impact and optimization of fuel and time consumption in transport operations.
	Shared logistics	Shared routes and transport resources depending on the feedstock needed, the places where recollecting, and the destiny of the same.	Smart management of transport resources.
	Cloud Platform	Centralization of real-time information in a unique platform.	Generate the blockchains and provide real-time info. about the available.

In BioSupPack, our trans-disciplinary team, formed by 17 partners from 8 countries across Europe, combines all the relevant skills and organisations needed to address the full extent of the challenge of upscaling the production of innovative sustainable bio-based compounds and coatings in plastic packaging solutions. The consortium is very strong with mature suppliers pioneering bioplastic & packaging converting companies, innovative technology companies and RTDs, as well as SME and large end users. The participation of leading industry players proves the actual soundness of the concept to generate large impact and will maximize the consortium capability in creating and accelerating the market uptake of the new biobased materials and packaging. The participation of key Stakeholders with specific expertise involved in an Advisory Board (AB), will provide high value to the project results and impacts, collaborating in different points of the value chain (details in section 4.5 and support letters annexed to section 5).

Positioning of the project

Most BioSupPack activities will be implemented **starting at TRL5,** and even some of them, like plasma technology for improving packaging performance or PHA based compounds production, start with a more advanced **TRL6,** with a **target of a consolidated TRL7 and TRL8** respectively at the project end, when our technologies will be combined in the final project demonstrators to be produced first at RTDs pilot scale and finally at the industrial facilities of the plastic converters, with the subsequent validation by the end-users in operational environment. In the specific cases of the use of extremophilic bacteria in fermentation processes, the development of squeezable packaging, and the use of plasma technology in both BSG and packaging waste pre-treatment as well as the fermentation routes for PHB production from PHB monomers from recycling, start at **TRL4** and they will achieve a **TRL6**. Proposed innovations are based on prior research from involved partners for the same or other base biopolymers for packaging applications, and on technologies defined at the pilot level or in other industrial sectors, as Table 3 shows.

Table 3: Summary of the TRL advancements through BioSupPack project

Technology	Initial TRL	Final TRL	Justification of current TRL
Plasma technology for BSG pre-treatment.	5	7	PLT have successfully scaled their plasma power source from their commercialised lab unit (400W) to a commercial scale 5,000W design. This system will facilitate the TRL jump from 5 to 7 for the proposed plasma bubble reactor for the pre-treatment of BSG. PLT has demonstrated its plasma-bubble reactor at small pilot scale (TRL5). Bubble technology is an established technology in the chemical industries which will further support the development.
Fermentation routes for PHB production using conventional meshophiles.	5	7	AUA has participated in 4 national research projects focusing on PHB production (up to 5 kg) using crude renewable resources (wheat milling, biodiesel industry and fruit processing by-products, and sugar beet pulp) with *Cupriavidus necator* and *Burkholderia sacchari* in 2 to 30 L fed-batch bioreactor cultures involving also technology transfer to industrially relevant environments (TRL5). NFG has experience with the scale-up of PHB production from used cooking oil in its pilot plant (TRL8) where bioreactor up to 1500 L is used. NFG is involved in other H2020 project where fermentation of PHB from coffee oil will be scaled-up.
Fermentation routes for PHB production using extremophilic bacteria.	4	6	BUT has experience with utilizing various thermophilic bacteria for PHA production, showing the process feasibility in the lab-scale (5L fed-batch reactor) and the process has been described and published (TRL4). NFG will use its experience as described above to scale-up to TRL 6.
Compounding for development and production of PHA based compounds for rigid packaging.	6	7	SABIO has already designed and produced formulations suitable for packaging in pilot scale and industrial scale plant. Additionally, the packaging prototype obtained after scale-up, meet requirements validated in industrial environment**.**
Plasma technology for barrier properties improvement.	6	8	PLT have successfully scaled their plasma power source from the commercialised lab unit 400W to a commercial scale 5,000W design. PLT is highly experienced in engineering of atmospheric pressure plasma reactors. The system will drive the required treatment to reach TRL8.
PHA coating in pulp-based packaging for moisture, liquids,	5	7	The patented plastisol technology is established for different applications by CTB at TRL5. It will be tailored to application on fibre-based substrates. Developed coating will be applied in an

fats protection.			industrial pilot environment at GPI.
Grafting of PHA coating for oxygen protection of pulp-based packaging.	5	7	Paper grafting is industrially available. Bio-based grafting solutions currently developed in BionTop project for application on PLA will be tuned to application to paper and PHA coated paper in BioSupPack. Developed grafting will be applied in industrial pilot environment at GPI.
Technology for squeezable packaging.	4	6	ILAB has dealt with several lightweight projects in both ISBM and EBM technologies, producing packages with very thin layers, to the point of having self-collapsing bottles to empty the bottle by gravity, without compromising its structure. The materials developed from SABIOMATERIALS are already technology validated to produce several packaging items. Thus, developing tailored materials for squeezable packaging will satisfy the TRL jump from 4 to 6.
EBM technology for PHA based bottles production.	5	6	ILAB has been developing hundreds of bottles through EBM processes for over 20 years, experimenting with different materials, from traditional HDPE and PP to several PHA and PHB.
Photonics technology for selective optical sorting of biobased materials.	5	7	IRIS has already developed a prototype for the discrimination of most abundant polymer in multilayer materials for sorting applications. The systems applied advanced data analytics merging ML and chemometrics for the proper data treatment. IRIS will extend the materials range of applications beyond the fossil-based ones and will train and optimize the models to increase the ability of the final device to be used in real industrial environment.
Plasma technology for packaging waste pre-treatment.	4	6	PLT have successfully scaled their plasma power source from their commercialised lab unit (400W) to a commercial scale 5,000W design. This system will facilitate the TRL jump from 4 to 6 for the proposed plasma waste reactor. We have demonstrated our gas plasma and plasma-bubble reactors at small pilot scale (TRL4).
PHB depolymerase enzyme selection producing 3HB monomers, engineering, cloning and development of recombinant expression strains.	4	6	AUA and BUT have produced PHB-depolymerizing enzymes using existing microorganisms in lab scale (TRL4). AUA has developed metagenomics libraries from extreme environments containing various organisms, that will be used for the selection of organisms producing PHB depolymerase with 3HB producing ability (TRL4). AUA has expertise in enzyme cloning, expression, engineering and redesign, using rational and combinatorial methods (directed or random evolution approaches), development of recombinant expression systems (*E. coli*) and metagenomic technologies in various R&D projects commercially oriented (TRL6).
Enzyme production using recombinant expression strains.	5	7	ASA has expertise in heterologous enzyme production using *E. coli* and *P. pastoris* strains at large scale using a 1.000 L reactor and will carry out enzymatic production in the project (TRL7).
Bioprocess development for enzymatic hydrolysis of used PHB-based packages	5	6	AUA has established expertise in the development of lab-scale enzymatic hydrolysis (e.g. amylase, protease, lipase, cellulase, xylanase etc.) and enzymatic biocatalysis processes using various crude feedstocks and crude or engineered enzymes in various R&D projects in many cases commercially oriented (TRL5). SAV will conduct the enzymatic hydrolysis process to produce at large scale 3HB monomer using a 1.500 L reactor, in an own pre-industrial bioprocess plant. This will allow SAV to set the requirements in an industrial environment (TRL6).
Fermentation routes for PHB production from PHB monomers from recycling	4	6	AUA has produced PHB from 3-hydroxybutyric acid at lab-scale cultures (TRL4). NFG will validate the technology in the 300 L fermentor, reaching TRL6 within the project.

National or international research and innovation activities linked to the project

BioSupPack will build on prior research activities and IP belonging to several project partners. Table 4 lists the most relevant projects participated by the partners, derived expertise, methods or processes that will feed in this project.

Table 4: Relevant research and innovation activities (additional ones can be found in the respective partner profiles)

Project	Partners involved	Opportunity for carryover to BioSupPack
LIFE Bread4PLA Demonstration Plan Project to produce Poly-lactic acid (PLA) Biopolymer from waste products of bakery (LIFE+) (2011-2014)	AIMPLAS	Proofs of feasibility to obtain LA and derived PLA from 3rd generation LA (bread waste).
PHBOTTLE New sustainable, functionalized, competitive PHB material based in fruit by-products getting advanced solutions for packaging and non-packaging applications (FP7-NMP-280831) (2012-2016)	ILAB, AIMPLAS	New PHB production from the wastewater from juice processing industries to produce PHB bottles, caps and sleeves prototypes.
BIOBOTTLE Biodegradable solutions for packaging of liquid dairy products (FP7-SME-606350) (2014-2016)	AIMPLAS	New biopolymer blends development and processing for dairy packaging applications (pouches and bags).
AGRIMAX Agri & food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications (H2020-BBI-PPP-720719) (2016-2020)	IRIS, AIMPLAS	Bio-based blends and composites with PLA, PHAs, bio-based additives and natural fibres.
RES URBIS: RESources from URban BIo-waSte (Grant Agreement 730349; H2020-CIRC-05-2016)	SABIO	Integration into a single facility and to use one main technology chain for the conversion of several types of urban bio-wastes into valuable bio-based products (including PHAs polymers), while also minimizing any residual or consequent waste to be disposed of.
USABLE PACKAGING (Grant Agreement No. 836884; BBI.2018.SO3.R10)	SABIO	Development of high-performance plastic packaging through a sustainable and fully circular value chain, where the biomass raw material sourcing derives from food processing.
Valor plus: Development of processes for the production of polyhydroxyalcanoates (EC-FB7; 2013 – 2017)	ASA	Improving of bacterial strains and processes for the synthesis of PHA.

BIOnTop Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings (BBI 837761)	ASU, IRIS, AIMPLAS, CTB,ENCO	Development of novel packaging in different formats using new copolymers based on PLA and other bio-based diacids as well as innovative barrier coatings.
NENU2PHAR Initiative to upscale production of PHA from renewable carbon sources at a competitive cost. (H2020-BBI-JTI-2019) (2019-2023)	CTB	Availability of PHA of different types.
Utilization of bio-refinery concept for valorization of waste biomass of wood processing industry (COST CZ project LD15031)	BUT	Experience with conversion of lignocellulose- based substrates for PHA production.
Evolutionary Engineering of Polyhydroxyalkanoates Producing Bacteria (Czech Science Foundation Standard Project GA 19-20697S).	BUT	Experience with development of strain for PHA biosynthesis, extremophiles for PHA production.
Transfer of Hydal biotechnology to China (Technology Agency of the Czech Republic TF02000062)	BUT, NFG	Scale-up of PHA production to pilot scale.
WaysTUP! - Value chains for disruptive transformation of urban biowaste into biobased products in the city context (H2020-SFS-2018-2020) (2019–2023).	NFG, SAV	Experience in PHA production from coffee oil extracted from spent coffee grounds (NFG) and in lipids and proteins production from food and fertilizers from SCG (SAV).
HOOP Hub of circular cities boosting platform to foster investments for the valorisation of urban biowaste and wastewater (H2020-FNR-2020) (2020–2024).	NFG, SAV	Promoting circular economy through the production of innovative bio-based products.
Valorisation of agricultural sugar beet residues and sugar beet pulp for the production of PHA-based biodegradable packaging (2020-2023, MIS5069983, GSRT. Greece)	AUA	Bioprocess development for PHB and PHBV production from sugar beet leaves and pulp at 2 to 30 L bioreactor scale
Valorisation of wheat milling by-products for the production of PHB-based biodegradable and active packaging (2018-2021, MIS5032796, GSRT. Greece)	AUA	Bioprocess development for PHB production from wheat milling by-products at 2 to 30 L bioreactor scale
TetraSOD® - food supplement enriched in the enzyme superoxide dismutase from microalgae Tetraselmis chui (www.tetrasod.com)	AUA	AUA has developed various enzyme-based products currently in the market via R&D commercially oriented projects funded by enterprises.
Erwinaze(TM) and Pegcrisantaspase used as antileukemic drugs - the recombinant L-asparaginase from Erwinia chrysanthemi	AUA	
SAFEBAG Novel continuous in-pack decontamination system for fresh produce & MEATPACK A novel packaging system for meat safety and shelf-life extension (FP7-SME)	IRIS IRIS	Use of plasma technology for packaging decontamination and food shelf life extension. These projects allowed forming of PLT as a company.
MULTICYCLE (H2020 SPIRE) Advanced & sustainable recycling processes and value chains for plastic-based multi-materials.	IRIS, CTB, AIMPLAS	Use of packaging SRM in textile, optical sorting, etc. Dissemination strategy around a Circular Plastic Helix that can help creating synergies.
Development of enzymatic and microbial processes for the degradation of plastic compounds (BMWA; 2013 – 2016)	ASA	Searching for enzymes for the degradation of PET, PLA, PCL, PBS.
BiogasEnzymes: Assessment of enzymes for the use in biogas plants from laboratory to industrial scale (BMBF, 2010-2013)	ASA	Optimization of the depolymerization of biopolymers in BSG.
BIOICEP Bio innovation of a Circular Economy for Plastics (H2020-NMBP-BIO-CN-2019)	AIMPLAS	Pre-treatment methods for further biodegradation. Enzymatic plastic depolymerization. PHB upcycling.
4EverGreen industry alliance to boost the contribution of fibre-based packaging in a circular economy.	GPI	Sustainability strategy especially in terms of substituting plastic coatings by biodegradable solutions. Recycling of fibre-based packaging.
VALUEWASTE Unlocking new value from urban biowaste (H2020 CE-SFS-25-2018 Grant Agreement 818312) (2018-2022)	SAV	System development for urban biowaste valorisation through the development of three value chains that can be integrated following a cascading approach.
WaysTUP! Value chains for disruptive transformation of urban biowaste into biobased products in city context (H2020 SEP-210494567)	SAV, NFG, AIMPLAS	Establishment of new value chains for urban bio-waste with pilot demonstrations.
HOOP Hub of circular cities bOOsting Platform to foster investments for the valorisation of urban biowaste and wastewater (2020)	SAV	Implementation of profitable bio-processes from private investments.
PlastiCircle Improvement of the plastic packaging waste chain from a circular economy approachGrant agreement ID: 730292 (2018–2021).	SAV	Develop and implement a holistic process to increase recycling rates of packaging waste in Europe.
RECOVER Development of innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries BBI JU (2020-2024)	SAV	Demonstrate and upscale novel bio-based approaches to dealing with the problem of agri-food waste plastics (AWPs).

Overall approach and methodology

BioSupPack is a 42-month project. The consortium has designed a work plan structured into 9 work packages (WPs). Figure 5 provides an overview of the content and interrelationships of the respective WPs and overall impact. **WP1** will upscale the production of new PHA based compounds and coatings. In **WP2**, packaging prototypes will be produced at demo level. **WP3** will focus on prototypes´ validation by the end-users, with participation of some of the stakeholders, also addressing consumer´s perception and acceptance issues. **WP4** is a strategic WP focused on all the logistics aspects regarding the management of the brewery by-product for PHAs production and the new packaging waste for recycling and the creation of the new value chain. **WP5** will optimize and demonstrate recycling viability of the new packaging (mechanical and enzymatic recycling) with participation of the stakeholders, and also upcycling of PHAs by reintroduction of derived monomers in the polymers production. The environmental and socio-economic

impacts will be assessed in **WP6**. **WP7**, will address the Communication and Dissemination activities foreseen in the project, together with the main regulatory concerns. **WP8** will manage the exploitation and business model activities, with a perspective of moving towards the commercialization process, based on the contributions of all the industrial partners in the consortium. **WP9** will address the overall technical, administrative, legal and financial management of the project, as well as its oversight.

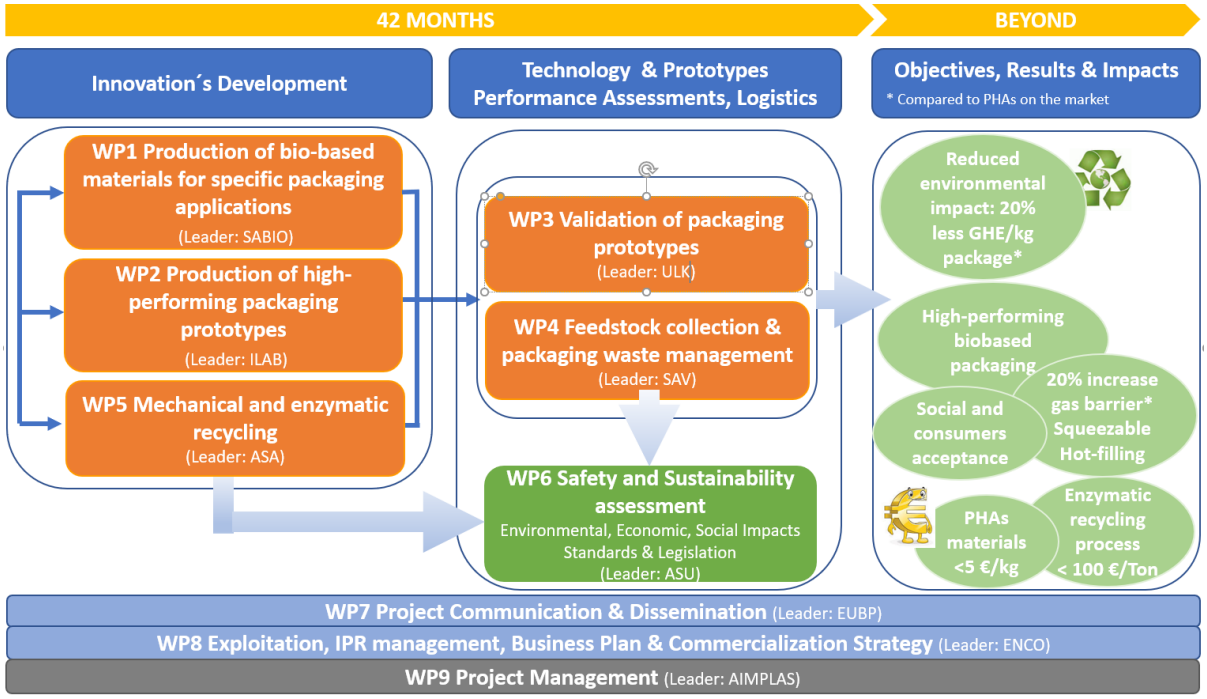

Figure 5. Overview of the interconnected WPs and overall impact of BioSupPack

How sex and/or gender analysis is taken into account in the project's content

Gender analysis concept is relevant to the content of the project in several aspects. To this end, innovation will consider the gender dimension and cater for the needs, motivations and differences of all genders where possible. Taking gender into account in the innovation activities will help yielding new results and spark creativity, opening new avenues for future research as recognized in the past for those who give it serious attention[18]. Gender analyses will be highly relevant in consumer perceptions for biobased solutions study (task 3.4) that would differ depending on gender and to the preponderant role of women in household waste pre-sorting. The outcome of the study will be used in the communication and dissemination context to ensure our message is appropriate to reach all genders. For example, the consideration of environmental factors, related to the source of new biobased materials and the new packaging EoL, is likely to vary depending on the gender. Women are predominantly involved both in the purchase and waste sorting of packaging at household level. In addition, female inputs to new business models will also be considered. Indeed, while the presence of females in the plastic converting sectors is traditionally lower than of males, there is a strong presence of females along the value chain of the bioeconomy industries where they play an increasingly prevalent role in consumer decision-making.

The gender dimension of our consortium and our measures to align with article 16 of H2020 programme are presented in section 3.3. It will ensure education and training of the next generation of scientists, technologists and others needed in the EU to foster the future of a circular bioeconomy and maximise its creativity and innovation potential.

1.4 Ambition

Advance beyond the state-of-the-art and the extent to which the proposed work is ambitious

The use of bio-based materials is a very interesting way to decrease the environmental impact of end users and packaging industries. BioSupPack project has the **ambition to upscale the production of novel rigid packaging products in different formats (bottles, trays, bowls, coffee cups and secondary packaging for glass bottles) using new PHA based polymers with improved properties, able to preserve the environment during and after their use,** leading to significant knowledge advancements of the state-of-the-art in packaging materials.

State of the art and progresses in cold atmospheric plasma technology applications

Plasmas can be categorized according to their thermodynamic properties to thermal (hot) and non-thermal (cold) plasmas[19]. **Cold plasma** equipment includes DC glow discharge, radio frequency (RF) discharge, dielectric barrier

[18] Londa Schiebinger (2008) Gendered Innovations in Science and Engineering. ISBN: 978080475814
[19] Fridman et al. (2004) Plasma Physics and Engineering. Boca Raton, FL: Taylor & Francis.

discharge (DBD), atmospheric pressure plasma jet (APPJ), microwave discharge and pulsed power discharge[20], and they have been widely studied on microbial decontamination, pest control and food and package functionalization.[21]

Lignocellulosic biomass pre-treatment via cold plasma

Cold plasma can disrupt linkages within lignocellulosic matrix for maximizing the efficiency of enzymatic hydrolysis[22], offering environmentally friendly and sustainable alternatives to conventional thermochemical processes, as it employs ambient temperatures and pressures and dry gases instead of chemicals and solvents. **DBD plasma reactors** have been employed due to its simplicity, robustness, and proven scalability[23]. Pretreatment has been carried out producing reactive oxygen species (ROS), such as O_3, H_2O_2, and hydroxyl radicals (•OH) and reactive nitrogen species (RNS), such as NO_2, NO_3, HNO_3, and N_2O_5[24]. Emission of UV by plasma and ozonolysis have also been widely studied for lignin degradation and removal to facilitate enzymatic hydrolysis of polysaccharides[25,26,27]. Combination of O_3, UV, •OH, ROS and RNS constitutes advanced oxidation and an efficient and sustainable lignocellulose pretreatment[28].

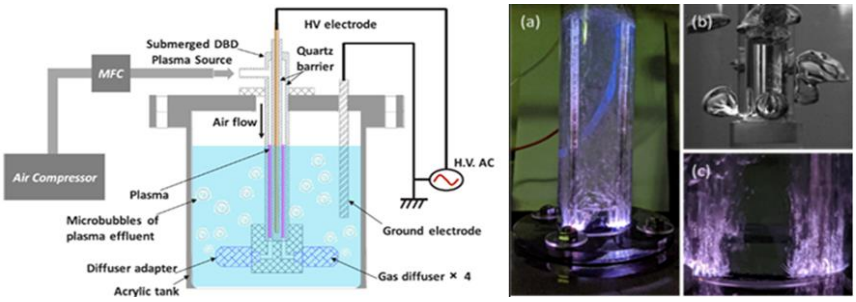

Figure 6. PLT patented DBD air plasma reactor for BSG pre-treatment

Figure 6 presents a scheme of a DBD air plasma reactor used by PLT in BSG pretreatment[29]. Crucial parameters influencing pretreatment efficiency are bubbles size, maintenance of a homogeneous biomass suspension, duty cycle of power supply, plasma temperature, reactive species produced and treatment duration. The patented quartz bubble reactors improve both process and energy efficiency.

A BSG to sugar conversion yield of 0.163 g/g (less than 30% of theoretical conversion yield) has been reported while the control enzymatic hydrolysis of untreated BSG was 0.076 g/g. PLT has demonstrated the potential of using plasma at low doses to enhance fermentation. Given the ubiquitous employment of bubble technology in the chemical industries including fermentation, the design inherently lends itself to effective scale-up.

Beyond the state of the art, BioSupPack will enhance BSG to sugar conversion yield to higher than 70% of the theoretical BSG to sugar conversion yield by optimising the operation of the DBD air plasma reactor and using highly efficient enzyme cocktails provided by ASA. Furthermore, BioSupPack will demonstrate that at least 20% cost reduction will be achieved as compared to conventional thermochemical pre-treatment processes.

Polymer surface property modification and degradation via cold plasma treatment

Surface functionalization of packaging polymers can improve polymer wettability, sealability, mechanical resistance, printability, dye up-take, antimist properties, glazing resistance, adhesion, gas and solvents barrier and impart antimicrobial properties, without compromising the bulk polymer properties[30]. The low surface energy of PE and PP has driven most of the research in cold plasma towards surface modifications to overcome problems in printing, coating and lamination[31,]. PLT has evaluated atmospheric plasma jet to efficiently modify the surface of polyethylene terephthalate (PET) to change their hydrophobic properties to more hydrophilic[32]. Efficiency of air cold plasma treatment of biopolymers has also been confirmed like in PLA[33]. PLT has characterised the changes induced in commercial PLA films to achieve in-package food decontamination.[34] In PLA/PHA blends plasma surface treatment showed an improvement due to the cross-linking effect achieved[35]. Significant improvements in barrier properties of starch/PHA composite films due to cross-linking effects have been reported[36] (23% in water vapor permeability and

[20] Lu et al Cold Plasma Sources. Cold Plasma in Food and Agriculture: Fundamentals and Applications (Eds: N. N. Misra, O. Schlüter, P. J. Cullen), Elsevier, Amsterdam, the Netherlands 2016, Ch. 4.
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[29] Ravindran et al (2019) Improving enzymatic hydrolysis of brewer spent grain with nonthermal plasma. Bioresour Technol 282, 520-524.
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[32] Scally et al (2017) Impact of atmospheric pressure nonequilibrium plasma discharge on polymer surface metrology. J Vac Sci Technol A: Vacuum, Surfaces, and Films 35, 03E105
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[34] Pankaj et al (2014) Characterization of polylactic acid films for food packaging as affected by dielectric barrier discharge atmospheric plasma. Innov Food Sci Emerg Technol 21, 107-113
[35] Omaníková et al (2020) Influence of composition and plasma power on properties of film from biodegradable polymer blends. Polymers 12, 1592;
[36] Sun et al (2018) Effects of various cross-linking agents on the physicochemical properties of starch/PHA composite films produced by extrusion blowing. Food Hydrocolloids 77, 964-975.

45.6 % to 73% in oxygen permeability). The advantages of plasma treatment over other polymer modification methods has been described, not only considering the modification of the polymer surface layer without upsetting the bulk polymer's properties[37], but also speed of treatment, ability to involve a limited surface layer, flexibility concerning the plasma reactors that can fit the specific needs from small to large scale, low capital investment, easiness and efficiency in implementation in already existing industrial lines and option for use in batch or continuous mode of operation.

Current-state-of-art plasma systems can be used to aid in the degradation of polymeric materials. Although the colonization of polymer surfaces by microorganisms depends on the functional groups present on the surface, samples with higher hydrophilicity may give rise to an easier colonization process, as PLT reported for LDPE[38].

Beyond the state of the art, BioSupPack will scale-up air plasma treatment technologies for (1) BSG pretreatment, (2) PHB-based packaging functionalization and (3) used package pretreatment to increase the efficiency of enzymatic recycling. The operation of the innovative DBD atmospheric plasma bubble technology developed and patented by PLT for BSG pre-treatment (current TRL5) will be optimised and scaled-up to TRL7.

State of the art and progresses in the field of the use of biomass for PHA biopolymers production

Various crude cellulose and hemicellulose and bacterial strains have been used for PHB production[39,]. Among the bacterial strains used, *Burkholderia sacchari* emerges as the most interesting one. Wheat straw hydrolysates have been used in 2 L fed-batch bioreactor cultures of *B. sacchari* to produce 105 g/L PHB (72% intracellular content) with 0.22 g/g yield and 1.6 g/L/h productivity[40]. Hardwood hydrolysate has been employed in fed-batch bioreactor cultures by *Paraburkholderia sacchari* for the production of 34.5 g/L PHB (58% intracellular content) at 0.15 g/g yield and 0.72 g/L/h productivity.[41] Apart from this mesophilic bacterium, utilization of extremophiles agrees with concept of Next Generation Industrial Biotechnology which was recently formulated by prof. Chen[42]. Its use brings many benefits associated with robustness of cultivation process against contamination by common mesophilic microflora. So, the process can be operated at reduced or minimized sterility precautions with substantially reduced energy demands which improves its economic feasibility and LCA impact. Halophilic *Halomonas halophila* CCM 3662[43] and thermophilic *Schlegelella thermodepolymerans* DSM 15344[44] bacterium have been identified for PHA production of lignocellulose-based resources.

Sustainable PHB recovery technologies focus on the utilization of environmentally benign solvents for PHB dissolution directly from wet bacterial mass followed by PHB precipitation. Ideal solvents should meet requirements as favorable toxicity and safety profile[45], low boiling point for easy recycling, water miscibility to ensure efficient PHB precipitation, efficient PHB dissolution at relatively low temperature, high PHB recovery yield and purity from wet bacterial mass, and no PHB Mw and property degradation. Water should be used for PHB precipitation from the dissolved PHB. Besides, sustainable PHB recovery could be also achieved by selective dissolution of non-PHB bacterial mass using switchable anionic surfactants (SAS) that switch from water-insoluble neutral forms into anionic water-soluble compounds through pH shift that is achieved by simple addition or removal of CO_2. Ammonium dodecanoate (laurate) is a relatively low-cost SAS that can provide high PHB recovery (99%) and purity (90%), while 98% recycling of the surfactant is achieved by simple pH reduction by CO_2 addition[46].

Beyond the state of the art, BioSupPack will develop efficient fermentations for PHB production using *B. sacchari* and thermophilic bacterial strains using BSG hydrolysates pretreated in a DBD air plasma reactor. DSP will be achieved using green solvents or SAS to ensure efficient PHB recovery with high purity and yield and low Mw and property degradation.

State of the art and progresses in the field of PHA processing by conventional conversion technologies and derived compounds and coatings for packaging applications

The limited thermal stability of PHAs narrows their processability window[47], specifically in the case of PHB, the most well studied polymer of the family. The thermal degradation proceeds very quickly at temperatures very close to its melting point, causing a rapid decrease in its Mw at the processing temperatures. Moreover, due to its high

[37] Visakh P. M. Chapter 1 Polyhydroxyalkanoates (PHAs), their Blends, Composites and Nanocomposites: State of the Art, New Challenges and Opportunities in *Polyhydroxyalkanoate (PHA) based Blends, Composites and Nanocomposites*, 2014
[38] Scally et al (2018) Significance of non-thermal plasma treatment on LDPE biodegradation with *Pseudomonas aeruginosa*. Materials 11, 1925.
[39] González-García et al (2019) Tequila agave bagasse hydrolysate for the production of polyhydroxybutyrate by burkholderia sacchari. Bioengineering 6 (4), art. no. 115.
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[46] Samorì et al (2015) Dimethyl carbonate and switchable anionic surfactants: two effective tools for the extraction of PHAs from microbial biomass. Green Chem 17(2), 1047-1056.
[47] Li et al (2016) Polyhydroxyalkanoates: opening doors for a sustainable future. NPG Asia Mater 8, e265.

crystallinity, PHB is stiff and brittle resulting in very poor mechanical properties with low extension at break, which hinders its wide applicability.[2,48] A further factor of high relevance for the processability is the degree of purity achieved after the biosynthesis and extraction of the PHAs, which affects both the thermal stability during processing and the resulting mechanical properties.[49,50] Methods to improve the processability of these biopolyesters in the melt state are needed for efficient processing, e.g. by melt extrusion. Different approaches range from the in vivo synthesis of PHA copolymers[51], such as poly(3-co-hydroxyvalerate) (PHBV)[52,53,54] and poly(3-hydroxybutyrate-co-4-hydroxybutyrate) [P(3,4HB)][55,56], to the enhancement of post-synthesis PHB-based material performances[51], thus blending with other polymers or plasticizers[57] or by reactive polymer blending with other bio-based polymers. All of them may offer opportunities to improve processability by lowering the processing temperature and reducing brittleness, broading the range of applications in the medical, agricultural, and packaging fields[58]. Furthermore, in vivo biosynthetic methodologies aimed to modulate polymer Mw and cost competitiveness of the overall process, and these blends are very convenient as industrial materials due to their lower cost. PLA-PHB blends have been extensively studied during the last years.[59] The ability of PHB to act as a nucleating agent for PLA improves its mechanical resistance and barrier performance.[60] Blends with other biodegradables polymers including starch, cellulose derivatives, lignin,[61,62] PBS[63], PBAT[64], and polycaprolactone[65] have also been widely studied. Also, blending of different types of PHAs offer the potential for alternating the different properties, improving mechanical performance and facilitating the expansion of applications. In melt compounding these blends have also been studied.[66] SABIO partner, with a wide expertise in the development of tailored biobased compounds owns a patent application concerning durable PHA compositions (see Table 7) that can be melt processed similar to PP into various forms such as moulded articles, films, fibers and nonwovens.

There are many reasons for coating paper with polymer, like reducing hydrophilicity, changing mechanical properties or better surface. Techniques such as extrusion, dispersion coating and solvent coating can be exploited[67,68,69,70,71].

A major drawback in circular economy is the lack of coated paper properly recyclable. There is an evolution away from PE coated paper/cardboard towards PLA extrusion coatings for this application. Although LCA studies have proven to be more eco-friendly, PLA coated paper is nor home compostable neither mechanically recyclable (furthermore industrial composting facilities are insufficiently available). For drinking cups, these LCA studies conclude that reusable cups are preferable[72,73], but for the use of PE or PLA coated paper in packaging there are little

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[72] Comparative LCA of 4 types of drinking cups used at events, D/2006/5024/06, OVAM, public waste agency of Flanders.
[73] Single-Use Paper Cups Circularity Improvement and Environmental Impact Mitigation Measures for Lappeenranta University of Technology Campus, V. Suskevice and K. Grönman, Proceedings, 3 September 2019.

alternatives available. There are currently little to no commercial examples of PHA grades for coating applications. Metabolix and CJ CheilJedang companies launched a hotmelt grade of PHA, which was retracted from the market, as it turned out to be difficult to process and sometimes resulted in an uncontrollable foaming. There are nonetheless several descriptions of PHA coatings in patent literature (see Table 6 at the end of this section). The majority describe the use of PHA in hotmelt or extrusion coating processes typically using low PHA Mw grades. PHA water-based dispersions are not as common but exist. Kaneka describes the production of aqueous PHA dispersions or as blends with other polymers. Powder based dispersions are described but focus mainly on the dispersibility.

A novel approach for PHA coatings was developed by CTB based on plastisol coatings, in which a polymer powder is mixed with plasticisers that upon heating of the formulation dissolve/gelate the polymer that will form a solid film. The method was optimized using biobased and biodegradable plasticisers compatible with PHA. It has not been described previously and is patent pending[74]. The technology is at TRL 5 and specific optimization towards barrier properties will be included in BioSupPack. Current formulations already proved moderate water barrier and good oil barrier (Figure 7).

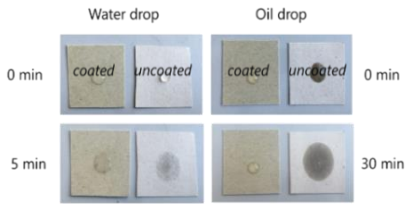

Figure 7. Water and oil drops applied PHA coated vs. uncoated

Beyond the state of the art of PHA compounds, BioSupPack will bring an advancement to the methodology for the design of biobased formulations for rigid packaging. The innovative approach will take in consideration the technical requirements of all the stages of the Circular value chain with the aim of delivering improvements in terms of overall productivity and overall reduction of environmental impacts. In this context production the process of biobased formulations for packaging will be brought to TRL 7.

BioSupPack will bring specific optimization of the PHA coatings based on plastisols towards barrier properties by the use of novel crosslinking chemistries based on environmentally friendly crosslinkers (e.g. starch aldehydes, organic acid based or carbodiimides) and should mainly improve water barrier at higher temperatures. It will bring the technology from TRL5 to TRL7 on paper and cardboard substrates of GPI with improved water vapour, oxygen, oil and water barrier properties. The process will also be tailored to allow its industrial use.

State of the art and progresses in the field of nanoscale surface modification by chemical grafting for improved hydrophobicity and to optimize barrier properties

PHA have excellent oxygen barrier properties but for sensitive to moisture absorption or release applications the water vapour barrier needs to be further improved. With this purpose, polar substrates are generally combined with hydrophobic layers, such as PE layers, by wet lamination (with adhesives), extrusion lamination (without adhesives), wet and extrusion coating. For the "wet" processes, water-based options are preferred for environmental reasons. Such solutions and dispersions of waxes, starches or poly (vinyl alcohol) (PVOH) are available. In addition, physical and chemical vapor deposition (plasma coating) on a planarization layer is possible. It can consist of bio-ORMOCER, an organic-inorganic hybrid polymer, which is only partially based on renewable resources and offers no intrinsic barrier without inorganic coatings. However, the inorganic coatings can be scratch-sensitive and thus be protected by an additional top layer.

For an innovative approach it is also necessary to improve the water vapour barrier properties of bio-based films and coatings. This is possible by green chemical grafting with fatty acids (FA) from renewable resources[75,]. The chemical grafting will be applied on polymer-based films and coatings by the so-called transfer method which was developed for the surface grafting of PVOH-based films[76]. The main attractions of this solvent free grafting process are its low cost, its use of existing industrial bio-based reagents and its potential to be applied on a printing machine at high speed and high efficiency. It is also a clean chemistry process that does not give rise to by-products or to polluting effluents. In addition, the FA green grafting is expected to provide easy emptying properties due to significantly reduced surface energy of the grafted surfaces. These rebalance properties are comparable to emerging non bio-based market available solutions such as LIQUIGLIDE™ developed by the Massachusetts Institute of Technology, Toyal Lotus® developed by Toyo Aluminium K.K. or AsahiGuard E-Series developed AGC Chemicals.

Beyond the state of the art and current development for PLA based packaging in the BionTop project, BioSupPack will provide additional improvements of the water vapour of precoated board by the grafting of nanolayer of bio-based fatty acid on top of the precoated internal side and on the external side of the cardboard substrates. This technology, already applied on different coatings at ASU, allows a high repellence of liquids (e.g. contact angles above 100º against water) and water vapour barrier improvement above a factor 20[77]. It will be adjusted to the target packaging structure in BioSupPack.

[74] Belgian patent application 2020/5152 "Dispersion aqueuse de biopolymères thermoplastiques", March 2020.
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State of the art and progresses in the field of packaging recycling

Plastic waste sorting

The recovery of innovative biopolymers through recycling[78] requires a pre-calibration of the used identification techniques which has to be developed for them, for which the right identification approach has to allow both its subsequent recycling and industry acceptance for new biopolymers in the mainstream of currently recycled materials. BioSupPack packaging will be biobased monolayer structures with improved performance due to PHAs properties and the use of plasma (physical treatment) and FA grafted coatings technologies (very thin layer with no need of identification when applied on paper/carboard substrates), being not necessary to deliver multilayer structures to imitate performance of conventional fossil-based materials, and consequently, thus without the well-known inconveniences of multilayer structures in the recycling streams, being materials´ detection simpler. IRIS has experience in developing tailored fast online VIS/NIR spectroscopic tools for plastic processing monitoring to lead to precisely identify their composition and traces of unwanted substances[79,80].

Beyond the state of the art, BioSupPack will demonstrate the capacity for collecting the PHA based developed packaging to be diverted to the recycling streams. To tackle the different challenges related with sorting of new materials, we will optimize the hardware, combining several spectroscopic technologies (possibly NIR and visible HSI), software thanks to advanced chemometrics analysis and possibly data fusion.

Enzymatic recycling of biopolymers

Enzymatic depolymerization could evolve into a sustainable route for bioplastics recycling where conventional recycling schemes cause significant loss in quality and value[81]. PHB depolymerases and lipases are the most extensively studied enzymes[82]. They can decompose PHB to produce 3HB monomers, dimers, oligomers or longer Mw chains depending on the depolymerase producing strain[83]. However, the native intracellular PHB depolymerase produced by *Bacillus megaterium* ATCC 11561 produces only 3HA monomers[84]. *Schlegelella thermodepolymerans*, apart from PHA production, is capable of extracellular hydrolysis of a wide range of PHAs[85]. Commercial lipases, such as Lipopan® 50 BG, have also been used for enzymatic depolymerization of PHA into oligomers.[86] Most studies focus on the isolation and characterization of enzymes depolymerising different biopolymers, but limited studies have focused on enzymatic bioprocess development for biopolymer recycling.[87] Bioprocess development for enzymatic depolymerization of PHB-based materials has not been reported yet. However, the population of aliphatic polyester-degrading microorganisms has been reported to decrease in the order of PHB>PBS>PLA. Few specific PLA degrading enzymes have been described compared to PHB depolymerases, indicating that hydrolysis is highly favored in case of PHB enzymatic degradation, so a higher conversion efficiency can be expected for PHB. The monomers or oligomers produced can be used to repolymerize, obtaining a low Mw PHB (Mw ≈ 2000)[88].

Geothermal springs and biofilms on marine plastic debris provide access to microbial communities with wide diversity in extremophilic microorganisms[89,]that may serve as potential sources of PHB depolymerases[90], which are expected to exhibit higher efficiency in plastic/bioplastic degradation.

AUA has expertise in high-throughput functional microbial and viral metagenomic technologies for complex microbial community analysis and mining new enzymes as well as the development of recombinant expression systems (e.g. *E. coli*) in various R&D projects many of which are commercially oriented (e.g. funded by Prince Sattam bin Abdulaziz University in Saudi Arabia, H2020-MSCA-RISE-2017, Bayer CropScience AG in Germany, Sea Spread Ltd in Hong Kong etc.) involving the use of modern techniques including protein chemistry, kinetics analysis, biophysics and omics technologies (genomics, transcriptomics, proteomics, bioinformatics).

Limitations associated with the industrial use of native enzymes (low stability, activity and less-favoured kinetic properties and specificity) are overcome by protein engineering[91]. AUA has wide expertise in improving variants of

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[78] www.plasticseurope.org/documents/document/20120821135405new_trends_within_polymer_sorting_to_tackle_for_tomra_sorting_solutions.pdf
[79] http://www.optiject.eu
[80] B2B contracts for the detection of good batches of recyclates as well as for HSI-based monitoring solution for plastic identification in recycling line.
[81] Soroudi & Jakubowicz (2013) Recycling of bioplastics, their blends and biocomposites: A review. Eur Polym J 49(10), 2839-2858
[82] Dieter Jendrossek and René Handrick (2002) Microbial Degradation of Polyhydroxyalkanoates. Annu Rev Microbiol 56, 403-432
[83] Polyák et al (2018) Enzymatic degradation of poly-[(R)-3-hydroxybutyrate]: Mechanism,kinetics, consequences. Int J Biol Macromol 112, 156-162.
[84] Chen et al (2009) Identification and Characterization of a Novel Intracellular Poly(3-Hydroxybutyrate) Depolymerase from *Bacillus megaterium*. Appl Environ Microbiol 75(16), 5290–5299
[85] Elbanna et al (2004). Studies on the biodegradability of polythioester copolymers and homopolymers by PHA-degrading bacteria and PHA depolymerases. Arch Microbiol 182, 212-225.
[86] Rodríguez-Contreras et al (2012) Enzymatic degradation of PHB by a commercial lipase. Polym Degrad Stab 97, 2473-2476
[87] Panyachanakul et al (2019) Development of biodegradation process for Poly(DL-lactic acid) degradation by crude enzyme produced by *Actinomadura keratinilytica* strain T16-1.Electron J Biotechnol 40, 52-57.
[88] Kaihara et al (2005) Enzymatic transformation of bacterial polyhydroxyalkanoates into repolymerizable oligomers directed towards chemical recycling. Macromol Biosci 5(7), 644-652.
[89] Sheludko & Fessner (2020) Winning the numbers game in enzyme evolution-fast screening methods for improved biotechnology proteins. Curr Opin Struct Biol 63:123-133.
[90] Pinnell & Turner (2019) Shotgun metagenomics reveals the benthic microbial community response to plastic and bioplastic in a coastal marine environment. Front Microbiol 10, 1252.
[91] FH Arnold (2018) Directed Evolution: Bringing New Chemistry to Life. Angew. Chem Int Ed Engl 57(16), 4143-4148.

enzymes by directed evolution and rational engineering[92]. Although either rational redesign or directed evolution can be very effective, a combination of both strategies represents the most successful route to improve the properties and function of the depolymerases that will be the target of the present project.

Beyond the state of the art, BioSupPack will scale-up an enzymatic bioprocess for recycling 3HB monomers from PHB-based used packages through the following steps: (1) selection of efficient PHB depolymerase from existing strains (e.g. *B. megaterium*) and existing metagenomic libraries from geothermal springs and plastic decontamination sites producing 3HB monomers; (2) Enzyme engineering via directed evolution and rational engineering; (4) cloning of PHB depolymerase for enzyme overproduction in *E. coli* and *P. pastoris*; (5) plasma treatment for improving enzymatic hydrolysis of PHB; (6) optimisation of enzymatic hydrolysis of used PHB-based packages using PHB depolymerase also or in combination with lipase; (7) utilisation of 3HB monomers in PHB production. A PHB to 3HB conversion efficiency higher than 50% will be achieved by the end of the project.

Innovation potential

Feedstock recovery for PHAs production and other potential applications

BSG accounts for about 31% of the original malt weight, representing around 15-20 kg of BSG per 100 L of beer produced[93]. Its composition[94,95] (Table 5) varies with barley variety, harvest time, and malting and mashing conditions. Being commonly used as animal feed, it has been used as crude renewable feedstock to produce various fermentation products. However, despite its already described potential (low cost, large availability throughout the year and rich composition in polysaccharides and protein)[96], about the 20% is still landfilled. BSG valorization into biopolymers like PHA could represent a potential new business for both the biorefinery and biotechnology industries derived from BioSupPack project.

Table 5. BSG composition

Component	*Cerveza Mica*	*Literature*
Moisture	72.5 ± 0.49	75
Free sugars	0.5 ± 0.03	-
Cellulose	23.2 ± 0.54	12-25
Hemicellulose	35.4 ± 0.82	22-40
Xylan	22.3 ± 0.75	-
Arabinan	8.4 ± 0.06	-
Mannan	2.8 ± 0.11	-
Galactan	1.9 ± 0.03	-
Lignin	13.04 ± 0.47	12-28
Protein	16.8 ± 0.29	14-27
Lipids	6.5 ± 0.56	4-13
Ash	3.2 ± 0.10	1-5

BSG represents approximately 85% of the total by-products of the brewery industry. Considering the annual beer production per EU country and the production of ca 15-20 kg BSG per 100 L of beer, then ca 10.8 BSG MTon (wet basis) was produced in 2016 (Eurostat). Considering 50% cellulose and hemicellulose BSG content and 0.25 g/g conservative practical sugar to PHB conversion yield, as well as the fact that 20% of the by-product is landfilled, then ca. 270.000 Ton PHB could be theoretically produced in the EU.

Other interesting BSG components have other potential markets that can be exploited, like phenolics and protein isolates. The predominant phenolic acids in BSG are ferulic acid and p-coumaric acid, with sinapic, caffeic and syringic acids being present in lower quantities. Phenolics can be used as antioxidant additives in bioplastics and proteins in the food sector by the use of derived proteins as animal feed or food supplements.

Sustainable packaging

When referring to bioplastics, packaging leads the market with more than 53% (1.14 MTon) of the total in 2019[97].

The **European food packaging market** size is expected to grow at a CAGR of 3.1% over the forecast period 2020 to 2025[98]. Market analysis by material segmentation indicates that paper and paperboard segment is envisioned to record the most substantial revenue in the future. Their raising applications in the packaging of edible products, fresh vegetables and beverages are the significant factor boosting this market demand. The plastic with its features like lightweight, flexibility, affordability and protection is a prominent material. European paper packaging market is expected to grow at a CAGR of 4.5% over the forecast period 2020 to 2025. The way consumers view and interact with packages is altering. Owing to the rising focus on sustainability, consumers are preferring easily recyclable carton packaging over plastic or tin packaging[99]. This has been supported by many leading EU food brands and retailers – including Nestlé, Unilever and Carrefour – publishing ambitious corporate sustainability goals that aim to minimise plastics in their packaging. This is stimulating demand for innovative paper-based alternatives that can substitute effectively for plastic formats, while demonstrably illustrating to the consumer the brands commitment to a greener future. Key areas of focus include coated paper and paperboard formats for flexible packaging formats such as pouches; and paperboard or moulded paper trays for retail and food service applications[100]. In this last case of application, GPI partner has already on the market some service packaging coated with bio-PE and PLA, but there is

[92] NE Labrou (2010) Random mutagenesis methods for in vitro directed enzyme evolution. Curr Protein Pept Sci 11(1), 91-100.
[93] Moreirinha et al (2020) Antioxidant and antimicrobial films based on brewers spent grain arabinoxylans, nanocellulose and feruloylated compounds for active packaging. *Food Hydrocolloids*: 105836.
[94] Lorente et al (2019) Analysis and optimisation of a novel "bio-brewery" approach: Production of bio-fuels and bio-chemicals by microwave-assisted, hydrothermal liquefaction of brewers' spent grains. Energy Convers Manag 185, 410-430.
[95] Ioannidou et al (2020) Sustainable production of bio-based chemicals and polymers via integrated biomass refining and bioprocessing in a circular bioeconomy context. Bioresour Technol 307, 123093.
[96] Lynch et al (2016) Brewers' spent grain: a review with an emphasis on food and health. J Inst Brew 122 (4) 553-568.
[97] Bioplastics Market Data 2019 (European Bioplastics).
[98] https://www.marketdataforecast.com/market-reports/europe-food-packaging-market
[99] Mordor Intellingence Report
[100] https://packagingeurope.com/four-growth-opportunities-packaging-europe-smithers-pira/

still a strong gap to reach solutions with versatile biodegradability.

The global **cosmetic packaging market** registered itself at a CAGR of 4.76% over the forecast period 2016-2027.[101] Cosmetic packaging acts as a crucial factor in product marketing and has become more natural and simpler. Commonly, plastic is most preferred the material utilized for packaging of cosmetic materials.

The global **home care packaging market** is expected to grow at ~ 8.5% CAGR during the forecast period 2020-2024[102]. Innovations have been numerous, particularly in laundry care. While sales are stagnating in developed countries, developing markets are driving volume sales through small pack sizes that make laundry detergent affordable to all. Furthermore, offering more functional packaging remains key, globally, while consumers are demanding more eco-friendly packaging and brand owners are working toward a more circular economy[103]. Plastics are the major packaging material being used by companies, driven by advantages like durability, cost benefits and versatility. New polymers like PLA, biodegradable PHA (polyhydroxyalkanoate), bioPTT (polytrimethyl terephthalate), etc are slowly getting acceptance, however, cost remains a deterrent.

Overall benchmarking and IP position

ANNEX II gathers the results from the patent search carried out focused on the main innovation area of the novel IP to be generated in BioSupPack project. The table shows the conclusions in terms of the patents´ impact on the project.

On the other hand, some of the BioSupPack partners own patents that are relevant to the project, positioning the project in terms of intellectual property and in starting TRL of some of the technologies to be applied for delivering the new biobased packaging solutions. These patents are shown in Table 6.

Table 6: Patents owned by BioSupPack partners relevant to the project

Patent (reference, title, *status*)	Relevance to BioSupPack	Conclusion in terms of impact on the project
Dispersion aqueuse de biopolymères thermoplastiques, BE 2020/5152 pending, partner CTB.	Describes the formulation of a PHA coating for different substrates.	The novel and cost-effective technique described is scalable towards the innovation goals of the project (from TRL level 5 to TRL level 7).
WO2013124361 A1 "Durable polyhydroxyalkanoate compositions". Partner SABIO.	Relates to polymeric compositions based on PHAs polymers, which are biodegradable and renewable bioplastics, and that can be melt processed like polypropylene based compositions into various forms of articles.	SABIO owns a unique expertise in the field of Melt Mixing Process for producing biobased materials, including PHAs. We have an own specific IP about durable materials containing biobased polymers. BIODURA® is the brand of our line of sustainable materials conceived for both durable high-end articles and compostable packaging.
Provisional Application: AU2019904694 Plasma Water Treatment. Partner PLT.	Applies to the formation of plasma discharges in forming bubbles underwater or other liquids. It has direct application for both BSG sugar production and waste treatment.	The technology offers an electron driven means for both enhancing sugar extraction and waste product treatment. Such an approach can be driven by renewable energy and employ air as the feed gas, making the process completely sustainable.

How the proposal covers the whole value chain.

BioSupPack will enable know-how transfer through its ***value chain***: actors from the primary sector providing biomass (MICA), bio-based raw materials producers from the biotechnology industry (NFG in close collaboration with BUT, AUA), biobased compounds and coatings producers (SABIO, GPI in close collaboration with AIMPLAS, CTB and ASU), plastic/packaging convertors (ILAB, GPI), end users of the new biobased packaging (UNL, NFG, MICA), as well as consumer representatives (ASU), enzyme producers for fermentation processes and enzymatic recycling (ASA, in close collaboration with AIMPLAS, AUA, BUT), feedstock and waste manager (SAV), enzymatic recycling (SAV) and providers of specific technology to be applied in different points of the value

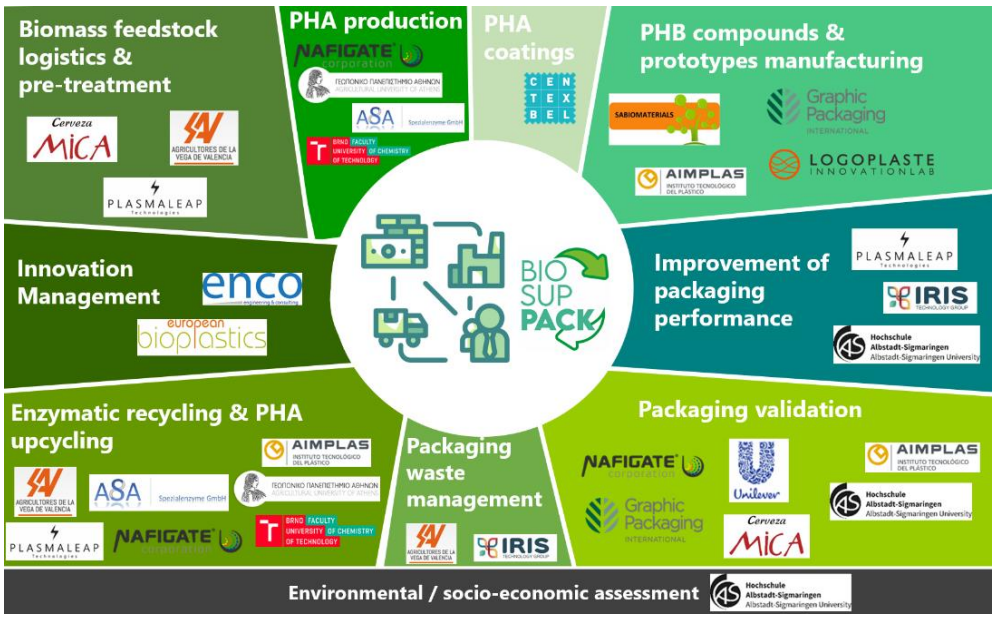

Figure 8. BioSupPack value chain and actors

[101] https://www.crystalmarketresearch.com/report/cosmetic-packaging-market
[102] https://www.marketresearchfuture.com/reports/home-care-packaging-market-5203
[103] https://www.mordorintelligence.com/industry-reports/homecare-packaging-market

chain: i) to enhance processes effectiveness, yields as well as packaging performance (PLT providing and upscaling plasma technology), ii) to assure collection and management of the new packaging waste streams so that they follow their corresponding pathways for recyclability (IRIS providing NIR based technology for the implementation of selective sorting devices) and finally sustainability assessment providers (ENCO, ASU, AUA) and specialists in project results transfer and innovation (EUBP, ENCO).

Stakeholders represented in our AB, actively supporting the project and covering also the value chain will be a strong support to the project partners: food and beverage packaging end users and big distributors like *Calidad Pascual* will collaborate with the validation of the beer secondary packaging, plastic waste recyclers like *ROMEI* will provide packaging waste from conventional plastic materials to validate the developed sorting device for PHAs waste, *Vitoria-Gasteiz City Council*, the association *GOPHA!* which will help support in promoting the development of the new materials and packaging into the bioplastics industry, and stakeholders from other sectors like *La Unió*, interested in the valorization of proteins for animal feed, also form part of it.

2. Impact

2.1 Expected impacts

In a world of finite resources, the transition to a knowledge-based post-petroleum economy is imperative. To support this evolution towards a low-carbon Europe, reducing its heavy dependency on fossil resources, the EC presented in 2012 a roadmap targeting the establishment of sustainable and competitive bio-based industries and value chains[104].

The **Bioeconomy Strategy and its Action Plan** pave the way to a more innovative, resource-efficient and competitive EU society, being reviewed in 2017[105] & subsequently updated in 2018[106], to accelerate the deployment of a sustainable EU bioeconomy so as to maximise its contribution towards the **2030 Agenda** and its **Sustainable Development Goals (SDGs),** as well as the **Paris Agreement**, and the new EU policy priorities, in particular the renewed **Industrial Policy Strategy, the Circular Economy Action Plan and the Communication on Accelerating Clean Energy Innovation**.

That reconciles food security with the sustainable use of renewable resources for industrial purposes, while generating new incomes and ensuring job creation, within a Circular Bioeconomy Society. In this context, the BBI's SIRA[107] aims to accelerate, among other things, the **development of bio-based packaging and plastic sector**, decoupling economic growth from resource depletion and environmental impact, and contributing to Vision 'The circular bio-society in 2050'. Perfectly in line with this overall goal, **BioSupPack** addresses the need to bridge technology gaps, generating the necessary enabling knowledge and technologies to build and boost the new value circle from bio-monomers to end markets. In this framework, the bio-based industries play a strategic role in complementing and replacing product streams from fossil oil and providing innovative bioplastic solutions for different applications. In the strategic area related to **bio-based materials,** according to *SIRA 2030-draft2 (pag 18): 'The BIC SIRA 2017 set 25% in 2030 as a target for the share of bio-based or renewable feedstock of the total volume of organic chemicals raw materials/feedstock used by the chemical industry'*.

BioSupPack contributes to 3 of 9 identified product groups with high potential for bio-based applications, i.e. plastics/polymers, cosmetics, and coatings to be used in the packaging of several food and non-food products. Moreover, the bio-based raw used in BioSupPack are **locally, renewable and sustainably sourced chemicals, cellulosic materials and bioplastics leading to close to 100% renewably sourced packaging.** In addition, increased consumer awareness and acceptance of bio-based products, concerted policy and labelling will make an important contribution to improve market penetration. As such, BioSupPack packaging constitutes an **enormous business opportunity** helping to nurture the realisation of the new EU CE[108] that also requires the optimal management of bioplastic waste.

Nowadays, plastics are used in various sectors with a demand of 51.2 MT per year. Packaging makes up for 39.9% of this use, of which food packaging accounts for around 90%. In this sector, **the EoL treatment of plastics is underperforming**, especially when compared to more homogenous materials such as metals or glass, traditionally used in food packaging. According to Plastics Europe[109], in 2018, 32.5% of plastic waste was recycled in Europe; nevertheless, there are many opportunities to increase recycling ratios. The recently adopted EC's plastics strategy aims at decoupling plastics production from virgin materials, reducing their leakage into the environment, improving

[104] COM(2012) 60 final. The Commission Communication of 13 February 2012 entitled "Innovating for Sustainable Growth: A Bioeconomy for Europe".
[105] https://op.europa.eu/en/publication-detail/-/publication/c2f36c72-2e59-11e8-b5fe-01aa75ed71a1/language-en/
[106] https://ec.europa.eu/knowledge4policy/publication/updated-bioeconomy-strategy-2018_en. COM/2018/673 final.The Commission Communication of 11 October 2018 entitled "A sustainable Bioeconomy for Europe: Strengthening the connection between economy, society and the environment".
[107] SIRA-Strategic Innovation & Research Agenda – 2030, for a Circular bio-based EU. Draft 2, March 2020
[108] Adoption of 11 March 2020, of the entitled 'New Circular Economy Action Plan', one of the main buidling blocks of the European Green Deal-COM(2019) 640 final. https://ec.europa.eu/environment/circular-economy/pdf/new_circular_economy_action_plan.pdf
[109] https://www.plasticseurope.org/application/files/1115/7236/4388/FINAL_web_version_Plastics_the_facts2019_14102019.pdf

the economics and uptake of recycling to support the transition from a linear to a circular economy model.

BioSupPack is a **high impact** project for the improvement of the **Circular Bioeconomy** in EU. Thus, it is hinged on delivering **cost competitive packaging solutions that can be enzymatically recycled**. Moreover, industrial scraps can be mechanically recycled. In this way, **the sustainability from cradle to cradle of several new bio-based high-volume applications will be maximised.** This impact will be achieved by performing **innovation and demo work** with the objective of making this concept **commercially viable** in practice, analysing the potentially **successful exploitation pathways,** and **assessing and managing any related risks**.

- Creation of at least 1 new cross-sectorial interconnection in circular bio-based economy cluster (KPI 1)

With **BioSupPack** we **will establish one new bio-cluster circular connection** as represent the follow scheme:

BSG → PHA/PHB →Packages → Collection/sorting → Enzymatic recycling

and further **intensify 4 connections,** among the sectors of **agri-food, bioplastics´ raw materials, pulp & paper and primary/secondary packaging from different industries** (i.e. food & beverage, cosmetics, homecare):

- BioSupPack will reinforce the **strong bond that the bioplastics sector has with the packaging sector** (which is by far the largest consumer of biodegradable and bio-based plastics) by providing new demonstrators in food, beverage, homecare and personal care applications. This new link is for example illustrated at the level of some of our compounding and large packaging converters, SABIO, GPI, ILAB, which so far have had good commercial biopolymers for other applications, but who wish to overcome the gap with the market to widen the new improved bio-based packaging in their product portfolio. The participation in BioSupPack will ensure them to drive the development towards actual market needs.
- Relying on the transition from linear to circular value chains, i.e. value circles, we **close the loop connecting biomass generators, with bioplastics producers and the new enzymatic recycling alternative,** which will provide the new secondary carbon source for bioplastics production. This way a new up-stream links with producers of non-valorised food by-products (BSG) will be created, opening up new business opportunities, *especially for primary sector*, becoming a crucial actor in this bio-based value circle and giving them a win-win solution. The BioSupPack aims to deliver >85% of bio-based formulations using highly sustainable biomass feedstock. As such, the current waste challenge turns into an enabler for innovation allowing a significant decrease in the environmental impact of the agri-food sectors and of the value chains of the BioSupPack solutions. We will have the agri-food processing industry playing a strategic polyvalent role: packer (end user) and supply raw material (BSG). Incontrovertibly, this magnifies the strength of the new links and the resonance of our solutions in the new bio-based value circle, which as an overall picture, it covers 7 stages: i) degradation of BSG to generate free fermentable sugars (with PLT plasma treatment and ASA as an enzyme supplier), ii) production of PHB from BSG sugars and 3HB produced in the enzymatic recycling of PHA based packaging waste, iii) Modification of PHB polymer by melt compounding by SABIO to produce bottles (ILAB) and commercial PHA by CTB to produce carboard bowls, trays and cups (GPI) in standard processing equipment, iv) to improve surface and barrier properties, produced packages will be treated with plasma by PLT partner and grafting of PHA coatings by ASU partner, v) food, beverage, homecare and cosmetic packers (ULK, NFG and MICA with the support of stakeholder Calidad Pascual) will validate the packages in their current products, GPI will validate their coated cardboard packages (with the potential involvement of customers like Starbucks o KFC) vi) IRIS and SAV are in charge of the logistic related collection and sorting to have a homogeneous source of recycled plastics for further treatments, vii) the post-consumer PHA based packages will be recycled by SAV using enzymes provided by ASA, the substances from the enzymatic recycling will be used to produce new PHA based polymers for similar applications, in order, to close the circle.
- Besides with PHA based coatings in the pulp-based packaging, the link **between the bioplastic packaging and the pulp/paper sector** will be reinforced for some primary packaging. Indeed, while conventional plastic-based coatings are now used, with BioSupPack, new bio-based nano-scale grafting and PHA coating for the 2 sides of fibre-based materials (solvent free, only water born) will allow improved barrier properties. The use of enzymatic recycling will allow the removal of the plastic coating from the cardboard-based packages and reuse the clean pulp to other non-food packages.
- **Cross-sectorial interdependencies will be created** through the synergies among the needs of the sectors of the 8 packaging solutions-demonstrators (e.g. similar materials serve the needs for food & beverage including service packaging, homecare and personal care, and the chosen 2^{ndary} packaging). We will build on such synergies to reach out to sectors beyond the consortium, like pharma packaging or other non-food packaging. Further links with other applications in packaging sector can also be established though the discussion of the post project potential of the new materials with enhanced barrier, hydrophobic, microwavable or squeezable properties. All these links, considered when building our AB, can contribute to the outmost exploitation of our results.

BioSupPack will create an unprecedented systematic, cross-border and cross-sectorial, connection among all industrial actors from 'cradle' to 'cradle' for the considered bioplastics-based end use applications. Indeed, **our consortium covers in the cradle-side, i.e. biosourcing**, represented by the bio-based feedstock (MICA), bioplastic producers (NFG and SABIO) **and packaging producers** (ILAB, GPI)**.** But BioSupPack consortium, once **going through the end-user** (MICA, ULK, NFG and GPI customers), also **covers the new route 'from former grave to new cradle-side, closing the loop, within a circular bioeconomy', i.e. from waste management** (SAV) **and waste sorting** (IRIS), **going through the enzymatic recycling** (SAV) **to obtain the raw material to new polymers production.** Complementary technologies will be proposed such as surface grafting of PHA coatings (ASU) and cold plasma (PLT) to treat the BSG, packages and waste materials prior enzymatic recycling and specific enzymes for the hydrolysis of the BSG and waste packages at the EoL (ASA). Their presence favours the generation of positive spill over effects among their affiliates leading to push-pull effects for the new BioSupPack technologies.

While the current recycling of biodegradable bioplastics is almost inexistent industrially, this new connection will allow launching concerted advocating for an improved framework for circular bio-based plastics. Potentially including incentive schemes both at EU and national levels and providing valuable input for on-going standardisation efforts at EU level, the project will aim to reduce the use of conventional petrol-derived plastics by promoting the use of bio-based plastics, which are required to be biodegradable and compostable[110].

- Creation of at least 1 new bio-based value chain based on the innovative products developed (KPI 2)

BioSupPack aims to produce of a suite of novel bioplastic packaging solutions that respond to unmet demands in terms of functionality and novel EoL and that requires connecting a critical mass of actors able to generate value in a synergistic collaboration. We aim to create **3 new stable and effective value chains not only by new players interconnections at the upstream of the chain** (to obtain the new bio-based raw materials and process polymers) but also **remodeling existing value chains**, becoming them value circles (from compounders, plastics converters onwards) that will substitute their petrol-derived products with our bioplastics solutions. As illustrated in Figure 9, upstream players crossover the 3 targeted packaging sub-sectors.

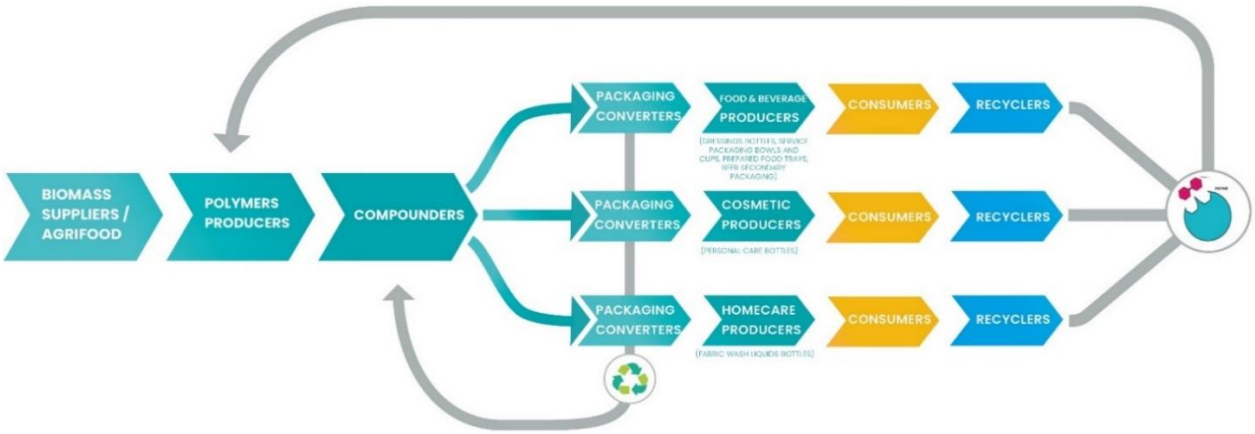

Figure 9. BioSupPack new effective value chains

The BioSupPack actors will work towards the same goals and a proper proportion of large end users, who will test and validate the new bio-based solutions in the selected target demonstrator applications, are included as early adopters at the end of the value chain to stimulate the demand. This will particularly benefit SMEs (PLT, SABIO, NFG, MICA, IRIS, ASA) as it will provide them with new up- and down-stream partners. Cooperation among BioSupPack members will be particularly important in the post-project phase, as results of the project are pre-competitive and further additional investment will be required before BioSupPack is ready for commercialisation. As addressed through specific tasks in WP8, setting the basis for a robust value circle will be key to generate real value on the market. We are confident that our developments **fulfil market demands and they can convert into a sustainable supply & value chain of bio-based material with functionalities that outperform existing plastic materials at a competitive market price**.

-Demonstrate at least two new bio-based materials for packaging (KPI 5)

BioSupPack will deliver at least **2 biobased PHA materials** to produce high-performing rigid packaging:

1. PHA based compounds to produce bottles by EBM and the secondary packaging by IM.
2. PHA based coatings to be used in the protection of grafted fibre-based packages such as bowls, trays, and cups.

[110] For example, a new law in Germany encourages the use of materials from renewable resources or recycled materials for the production of packaging, supports the use of certified bio-based and compostable plastic bags for the mandatory separate collection of biowaste. Likewise, in 2016, France adopted its 'Green Growth and Energy Transition law' and stipulated that bio-based, home-compostable packaging was mandatory for many applications. A Decree specified percentages of bio-based content to be reached by defined dates: 30% bio-based content by 2017, 40% by 2018, 50% in 2020, and 60% in 2025 and such initiative is now extended to Europe under the SUP legislation under definition.

PHB will be produced from fermentable sugars of BSG and from the carbon substances produced during the enzymatic degradation of the PHA based waste (mainly 3-hydroxybutyric acid-3HB) in a synergic combination. PHB fermentation is divided into two stages, i.e. bacterial growth and PHB accumulation. Using BSG hydrolysate containing carbon sources, nitrogen sources and minerals will facilitate bacterial growth with minimal commercial nutrient supplements. Using 3HB during PHB accumulation, a higher carbon source to PHB conversion yield will be achieved as the microorganism does not grow any longer and only accumulates PHB. 3HB can also be used for growth, so it can also be used together with BSG hydrolysate if higher carbon source concentration is needed at the beginning of fermentation. Both raw materials have a great advantage in comparison with other sources of 2G sugars for fermentation process as we detail in the following text. Based on the different EU food industries statistics, the beer industry is among the most important ones in terms of production capacity as Table 8 shows. Regarding 2G sugar availability derived from the food industry by-products produced by the different food industries[95], as figure 10 details BSG constitutes a considerable source of 2G sugars with only sugar beet pulp and wheat milling by-products being able to provide 2G sugars (wheat milling by-products contains also a considerable amount of starch) at higher quantities than BSG. Oilseed meals could also provide comparable quantities of 2G sugars with BSG, but these are used as predominant animal feed in EU. BioSupPack is focusing on one of the major sources of 2G sugars in food industry by-products available in almost all EU countries and around 20% are landfilled.

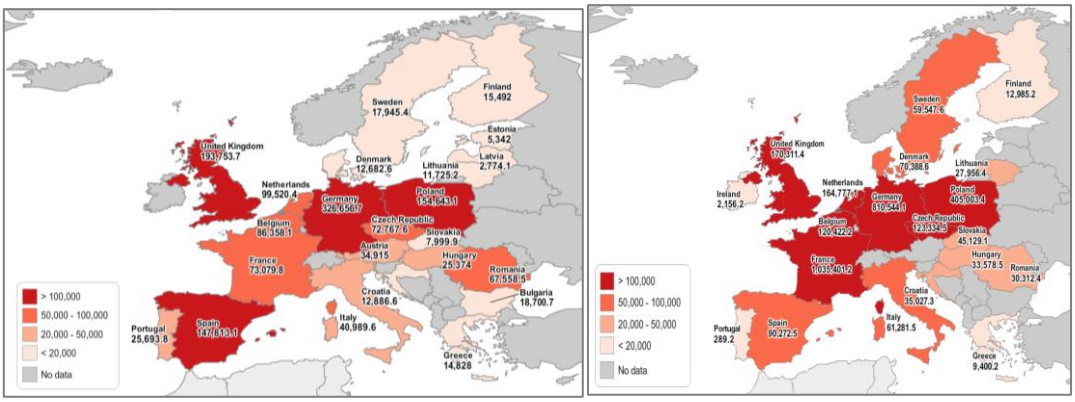

Figure 10. Geographical distribution in EU-28 in 2016 of potential fermentable sugar availability derived from BSG (left) and Sugar Beet Pulp (right)

BSG has lower availability compared to other lignocellulosic feedstocks (agricultural and forest residues), which however, are more recalcitrant to pre-treatment/hydrolysis and the associated logistics/collection/transportation is complex. However, 2G sugars isolation from BSG is more efficient since the complex structure of lignocellulose is damaged during malting and mashing and, therefore, the material is much more prone to enzymatic hydrolysis. BSG indeed contains a moisture content of 70-75%[6], if the transportation from the breweries to the bioprocessing plant is carried out in a relatively short time after production (few days) and it will befree of contaminants, and thus, transport be done without any need for temperature or moisture control. This is a significant advantage contrary to other food industry by-products (e.g. cheese whey is easily contaminated, sugar beet pulp must be dried for transportation).

BSG contains **other value-added products** such as a polyphenols, lipids, lignin and proteins to be used in the plastic industry or other industrial sectors[96]. This will permit to **reduce the final treatment costs and to achieve a zero-waste refinery**. Depending on the final quality of the obtained lignin in BSG plasma treatment, they can be sale to companies that produce vanillin or lignosulfonate-based additives, being a huge market for these complex molecules. BSG proteins can be used not only in animal feeding but also in food supplements- there is high demand for proteins in modern diet and for non-synthetic additives, whereas BSG phenolics can be evaluated as antioxidants in PHA/PHB formulations for the plastics industry. Lipids and proteins can be used also in the fermentation of PHB, the lipids can be used as additional carbon source and it is possible to hydrolyse the proteins and use them as additional nitrogen source in PHB production. In our experience, introduction of protein hydrolysates boosts up significantly PHB production. In conclusion, **to use a synergic blend of carbon sources (BGS hydrolysates with proteins and 3HB) can easier achieve economies of scale of PHB production from BSG**.

As reported in Section 1, according to EUB the global production capacity of PHA is expected to increase in more than the 500% from 2019 to 2924. Indeed, PHA will contribute the most to the growth of biopolymers in that period. BioSupPack paves the way to widen bioplastic market, providing innovative PHA bio-based products.

Current production of PHA[111] is mainly in the US (CJ Bio CheilJedang (5kton), Danimer Scientific (5kton)) followed by Asia (Kaneka, Japan; 5kton), TianAn Biological Materials Co. Ltd. (China; 2kton), Tianjin GreenBio Materials 9kton). In despite the excellent properties of this materials and the huge market growth, the production capacity in Europe is very limited. The most important criterion for large-scale production of PHA is sustainability in terms of

[111] https://www.globenewswire.com/news-release/2019/10/09/1927137/0/en/The-global-polyhydroxyalkanoate-PHA-market-size-is-estimated-to-grow-from-USD-57-million-in-2019-to-USD-98-by-2024-at-a-CAGR-of-11-2.html

supply and cost. The sustainable production of PHAs could be achieved by utilization of renewable, inexpensive carbon substrates and adopting efficient extraction processes. The operational cost of PHAs production process can be significantly minimized by using high yielding strains and optimization strategies of production and downstream[112]. ***The use of two complementary carbon sources with limited contaminants, as BioSupPack suggests, can increase the yield and decrease the downstream needs, in consequence, it reduces significantly the investment required to produce large amounts to have a competitive cost by economy of scale.*** *According to the studies of Choi et al.[113], the industrial production (>1,000 t/year) of R3HB though P3HB depolymerization via chemical methods is twice expensive than the use of enzymatic routes. Additionally, if purification process of R3HB is not required due to the new P3HB will be also produced via biological methods and the selected fermentation bacteria are specific to work with 3HB, the cost of production of P3HB will be expected lower than current technologies that uses only sugars and lipids as a feedstock.*

- Creation of at least two new demonstrated 'consumer' products based on bio-based chemicals and materials that meet market requirements (KPI 6)

BioSupPack will deliver 2 main new demonstrated 'consumer' packaging products based:

1) **High-performing rigid mono-material packaging**, which will show **excellent performance in terms of barrier properties** (oxygen, moisture, liquids and fats), squeezability and/or compatibility with hot filling, equivalent to fossil and better than another bio.
2) **Coated and grafted fibre-based packages: wet coated and grafted paper structures,** with unique functionalities (i.e. microwaving), still compatible with optimal EoL of derived coated materials.

Those improvements can enhance the consumer's experience, facilitating the product release from the packaging, reducing waste and maximising products organoleptic characteristics. BioSupPack materials will replace the current plastic packaging products made up of conventional PP & PE (for sauces, cosmetic and homecare applications), and the service packaging composed of board base, with PE coating. The consumer is given an important position in our bio-packaging value circles, as direct user of all BioSupPack products.

Based on the current price of PHA, the **cost of the proposed monolayer packages versus fossil based or other non-biodegradable bioplastics will be about 4-5 folds more expensive compared to fossil alternatives**. Current, monolayer packages made with PP and PE are fully mechanical recyclable but with downgrading, but the recycled post-consumer packages cannot be used to produce new food packages, they only can be used for non-food packages or for low-cost applications like pipes, waste bags or pots, whereby the price of the material lost 70% of their value after their reprocessing. In BioSupPack, the enzymatic recycling and the obtained product (mainly 3HB) will be used to produce a new polymer for the same application as in 1st life cycle. Loss of value is therefore significantly lower as 3HB is mainly used to PHB accumulate in the bacteria.

Current prices of PHA vary between 7-15 €/kg. Because there are recent announcements by several companies to increase PHA production, and there are several start-up companies and R&D efforts to make production more efficient, it is estimated that a price of 5-8€/kg will be achieved in near future. Specifically, for the coated PHA materials, the patent-pending coating pastes are water-based dispersions in which also relatively high amounts of inexpensive food-contact-approved plasticisers are used. This way, raw chemicals used for the coating pastes are estimated to be 2 – 4 €/kg, including pricing, marketing, formulation and sales this will lead up to 4.5 – 8 €/kg for a coating paste with about 40% solid content. According to different market studies, PHA barrier coatings will be about 2-3 folds more expensive compared to fossil alternatives, depending on the desired properties. It will therefore be important to highlight the additional benefits PHA coatings will offer, such as recyclability and biodegradability.[114]

[112] Geeta Gahlawat et al. Polyhydroxyalkanoates Biopolymers: Production Strategies. 2019. Springer Nature, 70 páginas, ISBN 3030338975.
[113] Choi, et al (2004). Process analysis and economic evaluation for the production of enantiomerically pure (R)-3-hydroxybutyrate by degradation of bacterial poly (3-hydroxybutyrate). Korean Chemical Engineering Research, 42(3), 345-354.
[114] http://www.tianan.com/en/

PHA-based packages have attracted interest for food packaging application due to its renewability, biodegradability, and water vapor barrier properties. PHB acts as a better light barrier in the visible and ultraviolet light regions than PLA. Despite the comparable properties of PHB with synthetic thermoplastics[115], its widespread usage has been limited by drawbacks of cost, limited commercial availability and narrower melt processing window due to brittleness and low thermal stability in the molten state[116]. PHAs have properties close to traditional plastics such as low-density polyethylene (LDPE), PP which makes them suitable for industrial uses especially in packaging[11] (see Figure 11).

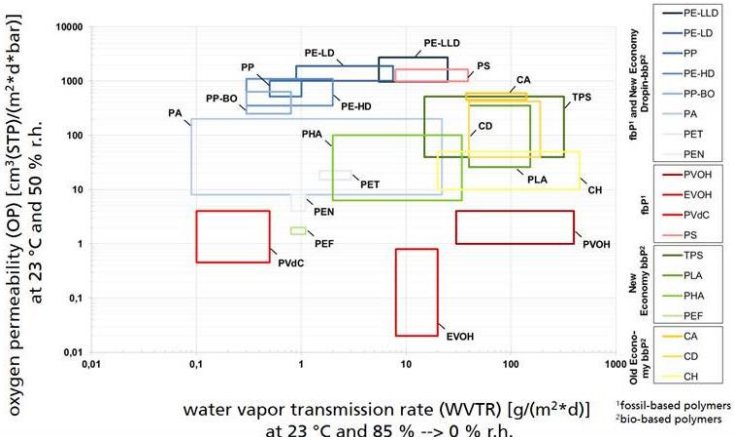

Figure 11. Polymer barrier properties (oxygen & water vapour)

PHBV offers the suitable gas barrier properties for use in coated paper and films. Non-biodegradable PE coating is currently used for this purpose preventing either biodegradation of the paper or recycling of the cellulose fibres. PHBV coated paper has been shown to be completely biodegradable in all main environments and easier to recycle compared to conventionally coated paper.

PHB has also better barrier properties than other bioplastics such PLA, OTR and WVTR of PLA is 30-300 cm^3/ (m^2 day bar) and 35-200 g/ (m^2 day bar) respectively whereas OTR and WVTR of PHB is 8-100 cm^3/ (m^2 day bar) and 3-30 g/ (m^2 day bar) , for the selected applications polymers with good barrier properties is encouraged, and for this reason we select PHA for coatings an PHB for rigid packages. This would also help to optimize thickness of the final packaging (thickness reduction) or avoid the use of multilayer structures to fulfil requirements.

ENVIROMENTAL IMPACT: Minimise landfill or incineration of the packaging material after use.

Our objective is to reduce significantly the amount of packages that are not properly managed at the EoL, by the implementation of technologies of **enzymatic recycling** that allow to transform PHA/PHB in 3HB that will be useful as a carbon source in the production by fermentative/enzymatic routes of new PHA/PHB, that can to be used in same application.

In the case of **monolayer bottles made with polyolefins** are collected and mechanical recycled, but due to the final quality and food packaging regulations, the do not reach the security level requested in the manufacture of new food packages and their final use is in the production of low added value products with very low industrial profit. The increase of use of post-consumer recycled polyolefins in profitable applications is limited, and alternatives, as BioSupPack proposes, are needed, aligned with EU Directive 2019/904.

Pulp-based packaging that combines carboard and plastic, the percentage of their mechanical recycling is very low due to the insufficient capacities of specialised recycling mills which mainly focus on recovering beverage cartons and technical difficulties to separate plastic and pulp making it impossible to process by standard paper mills and the low value of post-consumer recycled materials coming from the treatment of pulp-based packaging. **BioSupPack aims to improve the recyclability of pulp-based packaging by the use PHA coatings instead of the use of polyolefins and to implement the enzymatic recycling, where a better-quality paper can be obtained (cellulose fibers and PHA will be transformed in new PHA by fermentative routes at the enzymatic recycling facilities).**

ENVIROMENTAL IMPACT: Reduce greenhouse gas (including CO2) emissions (expressed in CO2 equivalents) in the overall value chain compared with the state of the art;

The main benefits of using bioplastics vs. conventional plastics are generally in terms of saving non-renewable resources and low toxicity. According to the BBI JU's SIRA, bioplastic has been identified as a major sector that determine the economy decarbonisation. In terms of **resources**, bio-based plastics have the unique potential over conventional plastics for reducing GHG emissions or even being carbon neutral (1 metric ton of **bioplastic** generates between 0.8 and 3.2 fewer metric tonnes of CO_2 than 1 metric ton of **petroleum-based plastics**)[117,118].

Figure 11 left shows two ellipses, separated from each other, that represent the clusters of GHG results. The ellipse on the upper right, which contains data based on using fossil resources of more than 70 MJ/kg of plastics and greenhouse gas emissions of partly clearly more than 3 kg CO_2 equivalent/kg of plastics, correlates with petrochemical plastics. The other ellipse illustrate the **results of the bio-based plastics PLA and PHA/PHB, the data of which for the use of fossil resources are lower than 70 MJ/kg of plastics**. At the same time the **GHG of**

[115] Thellenet al. J A J Polym Environ 16 (2008) doi: http://dx.doi.org/10.1007/s10924-008-0079-6
[116] Erceg et al., Polym Compos 31 (2010) 272
[117] Fondazione Istituto Italiano di Tecnologia, Bioplastics and Bioelastomers
[118] www.mckinsey.com/business-functions/sustainability-and-resource-productivity/our-insights/the-new-plastics-economy-rethinking-the-future-of-plastics

bio-based plastics amount to clearly less than 3 kg CO_2 equivalents/kg.

Figure 12 right shows that the production of bio-based polymers in comparison with the production of petrochemical plastics in most cases also leads to greenhouse gas emission savings. For PHA, the average savings potential in this case amounts to 5.8 ± 2.7 kg CO_2 eq/kg of plastic.

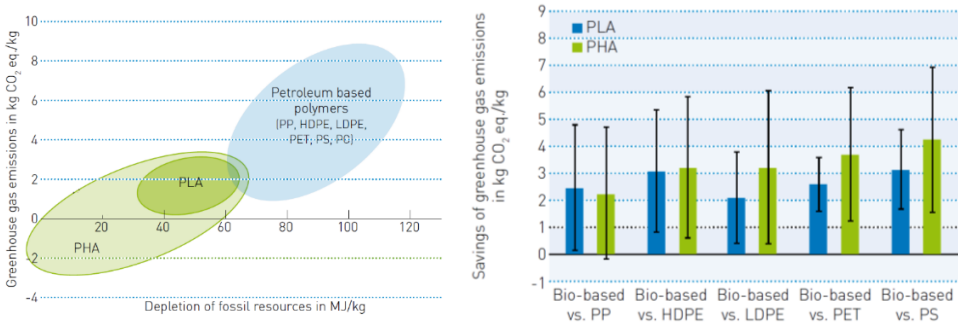

Figure 12. Comparison of environmental impacts of bio-based and petrochemical polymers[119]

In comparison with PP and HDPE, considerable savings potentials ranging between 2.2 and 3.2 kg CO_2 equivalents are found. Table 7 shows the differences in the greenhouses emissions between a rigid package produced using a polyolefin and PHB. The expected greenhouses gases reduction is estimated in the range of 3.2 to 5.5 *CO_{2equi}/Kg of package depends on the compared EoL system.*

Table 7: GHG Emissions of packages made with HDPE and PHB

Greenhouse gas emissions (CO_2 equivalents)[120]	***HDPE***	***PHB***
Production of the polymer and the package (CO_{2equi}/Kg of package)	*5.7*	*2.5*
Collection, transport, and sorting (CO_{2equi}/Kg)	*0.2*	*0.2*
Mechanical recycling of the package (CO_{2equi}/Kg)	*0.35*	*0.35*
Enzymatic recycling of the package (CO_{2equi}/Kg)[121]	--------	*0.45*
Energy recovery of the package (CO_{2equi}/Kg) as Municipal Solid Waste (MSW)	*2.8*	-------
Total CO_{2equi}/Kg per package emissions based on mechanical recycling (other EoL)	***6.25 (8.7)***	***3.05 (3.15)***

But not only the use of bioplastics but also the recovery and valorisation of the 20% amount of landfilled BSG will promote environmental benefits in terms of GHG emissions. As every tonne of BSG in landfill releases 513 kg CO_2 equivalent of GHG[14] and considered the amount of BSG available for recovery in the EU, nearly 350 Ton CO2 equivalent of GHG release could be avoided.

ENVIROMENTAL IMPACT: Contribute to the EU's 2050 long-term strategy for a climate-neutral Europe by replacing fossil-based material with bio-based, renewable material.

Contributions to the Green Deal Goals. BioSupPack will contribute the follow key policies and measures:

-Mobilising industry for a clean and circular economy (section 2.1.3). Energy-intensive industries, such as steel, chemicals and cement, are indispensable to Europe's economy, as they supply several key value chains. The decarbonisation and modernisation of this sector is essential.

While the circular economy action plan will guide the transition of all sectors, action will focus on resource-intensive sectors such as textiles, construction, electronics, and **plastics**. The Commission will develop requirements to ensure that all packaging in the EU market is reusable or recyclable in an economically viable manner by 2030, will develop a regulatory framework for biodegradable and bio-based plastics, and will implement measures on single use plastics. *BioSupPack aims to produce PHA bioplastics for different package applications including service packaging that are single use. Furthermore, PHA will be enzymatically recycled to be used in the same application and reduce the loss of value at EoL.*

-A sustainable product policy also has the potential to reduce waste significantly. Where waste cannot be avoided, its economic value must be recovered and its impact on the environment and on climate change avoided or minimised. This requires new legislation, including targets and measures for tackling over-packaging and waste generation. In parallel, EU companies should benefit from a robust and integrated single market for secondary raw materials and by-products. This requires deeper cooperation across value chains, as in the case of the **Circular Plastics Alliance**. The Commission will consider legal requirements to boost the market of secondary raw materials with mandatory recycled content (for instance for **packaging**, vehicles, construction materials and batteries). To simplify waste management for citizens and ensure cleaner secondary materials for businesses, the Commission will

[119] Essel (2012): "Meta-analysis of 30 LCAs"; Bioplastics Magazine (Vol. 7/2012-02),

[120] *In despite that the density of PHB is higher than HDPE, due to the better strength properties of bio-polyester, we can reduce the package thickness maintain the same mechanical and thermal properties as the Layman report of ECOBIONET project details* www.ows.be/wp-content/uploads/2013/10/LAYMAN_ecobionet-report_vfinal.pdf

[121] *Enzymatic recycling is done by selected enzymes that work at moderate temperatures, require low investment and OPEX than chemical process, in consequence the CO2 emissions will be significant lower than chemical process or simple biodegradation in a industrial compost plant.*

also propose an EU model for separate waste collection. *BioSupPack aims to optimize new routes of recycling to maximize their economic value and to contribute to improve the knowledge in waste separation technologies.*

-A zero-pollution ambition for a toxic-free environment (section 2.1.8). 'Creating a toxic-free environment requires more action to prevent pollution from being generated as well as measures to clean and remedy it'. *All processes in BioSupPack are toxic free and thanks to the optimization of enzymes, reactors and their processing conditions, we expect a significant reduction of direct and indirect emissions of GHG in the different process involved*, as we detailed before (under the previous environmental impact).

Presently, there are bio-based alternatives for many industrial applications, including packaging. In the case of PHAs, currently there are applications in bio-refinery, medical, agriculture, construction and packaging (though still in niches). The comparative cost analysis of PHA from by-products and enzymatic recycling with commercially available ones and potential to attain circular economy is one of the main advantages to be demonstrated in BioSupPack, considering that our PHA will come from 2nd & 3rd generation raw materials, instead of raw sugars. The reasons for this shift are connected to the non-renewability of fossil-based resources, the deteriorating environmental impacts, and the lack of biodegradability of the petroleum-produced materials but also to the avoidance of food competition as in case of current PHAs.

Using our novel BioSupPack compounds and coatings will ensure that **the impact of plastics on the environment** is minimised by offering alternative & more sustainable recycling solutions to the fossil fuel materials. The selected **EoL**, by industrial **enzymatic recycling**, **will be the environmentally sustainable EoL, for PHA upcycling,** as an alternative the current recycling of the counterparts materials, coming from the fossil fuels, thus contributing to the EU's 2050 long-term strategy for a climate-neutral Europe.

Although mechanical recycling normally emerges as the superior waste processing option by life cycle analysis, all plastic materials have a limited mechanical recycling lifetime (while some materials are simply non-recyclable) and therefore materials need to also be **chemically or biologically recyclable if a "recycling loop" is truly to be implemented[122] especially when targeting food contact applications**. Moreover, there will always be a certain level of leakage of materials to the environment, regardless of the improvement in our collection systems. Biodegradable plastics in mild conditions as PHAs would help to alleviate the impact of this leakage. In a **holistic approach to mitigate current EoL challenges for PHA**, besides delivering new materials, **BioSupPack will also adapt specific sorting lines that will ensure the right material sorting towards enzymatic recycling streams to maximize the final value of the polymer**.

Enzymatic recycling is the most sustainable EoL, when the post-consumer waste still contains rests of food or other products. Keeping with the waste management hierarchy, we will also show that the recycling of BioSupPack packaging, post-industrial waste and rigid fractions is viable contributing to increase the recycling rates of plastic in the future CE (target <10% landfill in municipal solid wastes by 2030) and reducing landfill or incineration. The need to explore the establishment of industrial recycling facilities which are dedicated to non "drop-in" bio-based plastics, currently very limited, due to their current use is very limited in comparison with petrol-based polymers. On the other hand, enzymatic recycling is a technology that require a limited CAPEX[123] due to selected natural enzymes works at mild conditions, has high efficiency, and require low enzymes concentration (less downstream needs).

Finally, in terms of use, **the barrier packaging (functionalized PHA coated paper and plasma treated EBM packaging) ensuring a longer shelf life will have a hugely positive impact on the food chain.** Likewise, whereas packed food typically binds much more resources than the packaging itself, the reduction of products waste due to squeezability provided will have a large effect on the LCA results expected in the project. All in all, in BioSupPack, the **carbon footprint** of the considered bio-based products will be decreased along the new value chains as will be shown by LCA during the project in WP4.

ECONOMIC IMPACT: Lay the basis for market-acceptable production costs of the targeted bio-based products.

In different OECD countries, packaging waste represents a significant, yet increasing, share of total household waste. This represents a significant economic burden too. The applicable EoL options and related fees vary per type of packaging material and mainly depends on how easy it is to separate it in the waste management stage. Gate fees for EoL management for similar treatment options can vary substantially at national level but also at regional and local level. Due to the specificities of plastic packaging, the costs linked to plastics EoL treatment tend to be significantly higher than the costs related to the treatment of other materials. Despite the recycling rate is increasing following a shift towards a CE, currently only 32.5% of plastic is recycled and more than 42,6% incinerated (energy recovery) and 24.9% landfilled[124]. These last two EoL options lead to significant resource depletion. Indeed, even the energy recovery achieved by incineration plants only has an average efficiency of 35%. Therefore, gate fees at material recovery facilities are substantially lower than those for incineration and landfilling. Usually, factors influencing gate

[122] Biodegradable Plastic in Solving Plastic Solid Waste Accumulation www.sciencedirect.com/science/article/pii/B9780128131404000194
[123] Tournier et al. An engineered PET depolymerase to break down and recycle plastic bottles. Nature 580, 216–19 (2020).
[124] https://www.plasticseurope.org/es/resources/publications/1804-plastics-facts-2019

fees are mainly related to the landfill tax (which enables competing treatments to raise their gate fees as the tax increases); the quality and quantity of material being presented; energy costs, etc.

According to the EUBP, more than 90% of materials currently disposed-off in incinerators and landfills could be reused, recycled and composted. According to EPA, a well-run curb side recycling program can cost anywhere from €40 to more than €122 per ton whereas trash collection and disposal programs, cost from €57 to more than €160 per ton. Recyclers pay variable amount for post consumers and post-industrial plastic wastes depending on their cleanliness and purity (e.g. from 0 to 700 € per ton) less than three times lower than virgin ones. Then they have to separate, clean, grind and reprocess the compounds and subsequent materials vs. operative costs involved, a positive balance can be obtained only if such materials can be sold above 600€/t. To find customers paying this plastic cost to produce low added value products is in most case complex.

Table 8: Summary of differences of cost of monolayer rigid packages produces using HDPE and PHB.

Cost	*HDPE Bottle*	*PHB bottle*
Production of the polymer (€/Kg).	***1.50***	***5.00***
Production of the package (€/Kg).	***0.40***	***0.45***
Plasma treatment (€/kg)	***0.00***	***0.02***
Collection, transport, sorting (€/Kg).	***0.10***	***0.10***
Mechanical recycling (€/Kg).	***0.20***	***------***
Enzymatic recycling (€/Kg).	***--------***	***0.1***
Value of the plastic at the EoL		
Sales of recycled polymer or 3HB (€/Kg).	***0.40***	***1.00***
Difference of cost	***-1.80***	***-4.75***

BioSupPack aims to produce two main types of packages: monolayer bottles for food and non-food products and pulp-based packaging both produced using PHA/PHB as the main plastic material. Current PHA/PHB cost are several times higher than petrol based. However as demonstrated in a preliminary economic study (detailed in Annex-B.P.), the packages produced using the materials and the technology optimized in BioSupPack will be cost competitive compared with the current petrol-based material, if the full process is considered**. One of the key points to be cost competitive is to reduce the cost of PHB to values below 5.0 €/kg. To fulfil this, we plan to scale up the production of PHB using BSG hydrolysates BSG proteins as a source of nitrogen and 3HB as a synergic source of carbon that it is accumulated directly by the bacteria in form of PHB.**

Furthermore, BSG is a low-cost lignocellulosic biomass, but it is less recalcitrant than other such as straw (i.e., its lignocellulosic structure is damaged during malting and mashing and, therefore, BSG is much more prone to enzymatic hydrolysis).

This expected cost will represent a significant reduction in comparison with the current state of the art, **where we reduce from actual 4-5 folds in comparison with petrol-based materials to near 2.5 folds, as showed the calculations in the following point.**

ECONOMIC IMPACT: Increase business opportunities for new, innovative, and sustainable packaging solutions that have no negative impact on the environment.

BioSupPack valorises residual streams from primary sector, BSG. **Estimations on the 2 main bio-based materials & products (according to KPI5&6) are detailed in Annex-Business Plan (B.P.), showing the viable commercial costs for the 2 project lines.** Herein economic breakdown for the new business line based on PHA compounds + High-performing rigid mono-material packaging is included, as it represents the highest impact (ca. 95%) in terms of PHAs amount introduced in the market. All calculations are based on the foreseen scheme for the Business plan of the Industrial plants (see descriptions in annex B.P.) and data and assumptions provided by the industrial partners. Table 9 shows the details of the **Industrial Plant 1- bottles whole process production**, and the source of the data to calculate the comparisons between an average bottle weight as data provided by ULK, as a reference HDPE, 30g weight (supposedly now made up of BioSupPack PHB). Table 10 shows the details of **Industrial Plant 3-Collection, Sorting & Enzymatic recycling** -to close the value chain. The cost that SAV will pay for having the PHAs waste sorted results in about 0.1 €/kg, and the same amount (i.e. 0.104 €/kg) is the cost that SAV will assume to perform the enzymatic recycling. 3HB will be the new raw material to be supplied NGF.

Table 9: Industrial Plant 1-bottles whole process

	NFG	SABIO	ILAB	TOTAL
INDUSTRIAL PLANT 1,000 t/year of PHB production, PHB compouning and bottle production.				
CAPEX (10years)	23.000.000,00	1.950.000,00	4.500.000,00	29.450.000,00
OPEX/year	1.300.700,00	650.000,00	450.000,00	2.400.700,00
				31.850.700,00
TOTAL OPEX/CAPEX/year	3.600.700,00	845.000,00	900.000,00	
Cost €/t	3600,7	845	900	
Cost €/kg	3,6007	0,845	0,9	5,3457
Contingency plan (15%)	4,140805	0,97175	1,035	6,147555
Bottle weight 30 g PHB				0,184
Bottle weight 30 g HDPE				0.075

Table 10: Industrial Plant 3-Collection, sorting & enzymatic recycling

	SAV	SAV/IRIS
INDUSTRIAL PLANT 460 t/year of PHB enzymatic recycling		
CAPEX (10 years)	185.000,0	150.000,0
OPEX/year	29.637,0	31.728,0
Production (t/year)	460,0	460,0
TOTAL OPEX/CAPEX	48.137,0	46.728,0
3HB €/ton	104,6	101,6

Other social and environmental impact (including contribution against climate change)

BioSupPack will also contribute to other impacts especially in terms of pollution and consumer health.

Microplastics Sea pollution: A study based on measures taken during 24 expeditions, estimates the presence of a minimum of 5.25 trillion plastic particles in the oceans, corresponding to a mass of 270 ktons.[125] Recent concern about plastic pollution has focused on microplastic particles. Microplastics are likely the most numerically abundant items of plastic debris in the ocean today, and quantities will inevitably increase, in part because large, single plastic items ultimately degrade into millions of microplastic pieces. Microplastics are of environmental concern because their size (5 millimetres) renders them accessible to a wide range of organisms at least as small as zooplankton, with potential for physical and toxicological harm. Moreover, plastic waste can also damage coral reefs and marine invertebrates and contribute to the spreading of invasive species as they drift in the oceans, possibly causing ecosystems' destruction.

PHB polymers are fully biodegradable in soil and sea water conditions, and the bioplastics products incorporating the PHB coming from BSG valorisation developed in BioSupPack project can consistently contribute to the environmental risk related to microplastic formation in case they are not well managed at the end of their life. Additionally, a lot of studies detail the use of PHB or PHBV as ingredient in fish feed formulations due to their "prebiotic effect"[126]. Replacing a small amount of the diet with PHB improved weight gain, specific growth rate (SGR) and survival in the fish fingerlings[127-128].

Unquestionably, our **quality of life** is linked to the quality of our environment and measures are needed for lowering the environmental burden of our activities and consumption patterns. According to a report of the Ellen MacArthur foundation[129], only a very low amount of plastic materials is recycled (14%) while high amounts end up in landfills (40%). It is important to highlight the most critical point is **the amount of plastic wastes leaking from our waste management streams shows as much as 32% of plastic ending up in our environment**.

Long term plastic persistence in the environment is aggravated by a **cumulative effect,** due to both accidental and intentional littering. The slow decomposition of plastic into small pieces leads to damages to the fauna via ingestion and entanglement[130],[131]. Recognized as one of the major international environmental challenges by the European Environment Agency (EEA)[132], **plastics account for 89-95% of marine litter** (roughly 8 million tons every year), in particular packaging (bottles, lids, etc.) leading to irreversible consequences for marine ecosystems. Besides, **25 Mt of plastic waste is accumulated annually in soil worldwide**, being detrimental to the environment (irreversible soil contamination, degradation of soil quality, pollution of rivers and groundwater) and human health (heavy metals, dioxins and furans released during plastic burning are associated with endocrine disruption, heart disease, cognitive and motor disabilities, and human carcinogen)[133]. At the EU level, the recent BOHEMIA study[134] recognises ocean and marine pollution as key priorities to be tackled by the EU post-H2020 R&I framework programme (HORIZON EUROPE). Concerted actions are required to sustainably solve the huge plastic waste accumulation issue.

Promoting the use of recycled plastics in value added application, BioSupPack solutions will influence the increase of the quantity of plastic waste enzymatically recycled to produce polymer for food and non-food applications Furthermore, BioSupPack PHA/PHB biodegradability in mild conditions is a step towards the existence of plastic packaging that can biodegrade also when **accidentally littered in the environment, mainly on sea waters, where the PHA is has biodegradable, has prebiotic effect in fish fauna and do not produces microplastics.**

BioSupPack will use harmless/innocuous substances which derive from food (by-products from BSG that contains proteins, holocellulose fermentable and antioxidants) and paper. Also, barrier packaging will provide consumers with healthy food along longer shelf life. Furthermore, in terms of coated pulp-based packages, our new PHA coatings will impart durable hydrophobicity enabling industry **to replace polyolefins which are currently used in hydrophobic coatings, but they are difficult to manage at the end of life as previously detailed.**

BioSupPack packaging demand is expected to be driven by green- and health- conscious consumers concerned about the environment and especially unsustainable persistent littering and resulting microplastics[135]. Furthermore, societal benefits will be assessed by the active involvement of partners such as ASU and ULK, that are working on consumers' needs and perceptions. **BioSupPack will ensure a strong coverage at EU level, on consumers attitudes on the issue and their requirements which will contribute to encourage the knowledge of the positive benefits of bio-**

[125] Sailing seas of plastic, cf. //app.dumpark.com/seas-of-plastic-2/#expeditions/60
[126] Prebiotics are substances promoting beneficial gut microflora hence improving digestion and nutrient absorption processes.
[127] Duan*et al.* Effects of dietary poly-β-hydroxybutyrate (PHB) on microbiota composition and the mTOR signaling pathway in the intestines of *litopenaeus vannamei*. *J Microbiol.* **55,** 946–954 (2017).
[128] Franke et al. Poly-β-hydroxybutyrate administration during early life: effects on performance, immunity and microbial community of European sea bass yolk-sac larvae. Sci Rep 7, 15022 (2017).
[129] https://www.newplasticseconomy.org/about/publications/global-commitment-spring-report
[130] http://www.kimointernational.org/MarineLitter.aspx
[131] http://www.onegreenplanet.org/environment/ways-plastic-pollution-impacts-animals-on-land/
[132] State of Europe's seas, EEA Report No 2/2015, ISSN 1977-8449
[133] D. Briassoulis et al. (2013) Review, mapping and analysis of the agricultural plastic waste generation and consolidation in Europe, Waste Management & Research31 (12), 1262 – 1278
[134] Beyond the Horizon: foresight in support of future EU Research and Innovation Policy- Doi: 10.2777/906131
[135] About 80% of European consumers want to buy products with a minimal impact on the environment (Eurobarometer survey, EC, 2013).

based packaging as well as raising awareness about future economic, societal and environmental challenges.

Contribution to improving the innovation capacity and the integration of new knowledge

While the generation of new knowledge is positive, it is in its industrial use that innovation capacity is increased, and impact is realised. As such, to allow generating truly market replicable innovation, the BioSupPack consortium unites the industry value chain necessary for market success. The project has integrated several transversal activities such as environmental, regulatory, safety and economic assessments, and finally business planning activities to maximise the post-project exploitation of the results. The project gives trans-sectorial industry and transdisciplinary research and technology players an opportunity to collaborate and share all required technical knowledge and market expertise. Effective knowledge transfer from the RTDs (AIMPLAS, ASU, AUA, BUT & CTB) to the participating enterprises, especially SMEs (PLT, SABIO, NFG, MICA, IRIS and ASA) often lacking own R&D capacities, laying down the basis for supporting the industry partners to carry forward the results post-project, is fundamental to increasing the likelihood of market uptake. Moreover, the engagement of large companies, for the pilot plants and as end-users, is crucial (GPI, ILAB, SAV, ULK).

As discussed in the innovation section, unique and potentially patentable bio-packaging solutions relying on **PHA for coating** formulations and **PHB compounds from BSG + broth of enzymatic recycling** will be developed to achieve optimal functionalities and allow enzymatic recycling in mild conditions. Another key novelty is the use of enzymes for PHB recycling and BSG pre-treatment in combination with plasma.

This knowledge will not only provide new insight for the scientific community who lacks systematic studies on enzymatic recycling of specific polymers, as well as also help in the shaping of new related standards but will also be used with **two industrial targets: i) support the eco-design of the products meeting target EoL scenario and ii) support organic recyclers, to ensure they sort suitably the new post-consumer waste for enzymatic recycling.** To increase the industry innovation capacity, our approach is to capitalise on a valuable opportunity for the EU bioplastic, packaging, and the end-user industries to substitute current synthetic counterparts with bio-based compounds and coatings. BioSupPack represents a **very timely development of new cost-efficient, recyclable and high-performance bio-based materials,** into growing 'green' and 'bio-based' industries.

This will result in **great business opportunities reinforced by current** bioplastics **market trends,** the growth in our target end use sectors, **increased** visibility, and policy awareness and support for such solutions**.** Indeed, we tap into the massive **EU plastic business** (€350 billion **turnover** per annum). Packaging Plastics demand in EU is **ca. 20 Mt**, **of which 9% is for olefin rigid packaging**[136] **and coatings**. BioSupPack is estimated to reach a **market penetration of 0.5% of these plastic segments** (totalling a potential of 9,000 kt per year) **by 2030 in EU**, generating close to **€45 million turnover and ~180 new jobs** (based on 1 job/250 k€ of turnover) **at the biopolymer production upstream** level. **Such figures would be reinforced at the downstream (end users) level** who benefit from new unique products. Besides direct partners' exploitation, these figures, equivalent for current limited European PHA production to increase by a factor of seven by 2024 according to European Bioplastics, correspond to an overall deployment through further licensing, business alliances with additional producers and converters. Estimates about how partners respectively expect to grow their business and create jobs with our solutions are given in section 2.2.

Impact on consumers and consumer choice: With the growing consumer awareness towards greener and wiser choices from the environmental point of view, more attention is paid to products with a lower environmental impact. While this concerns the product as a whole, it is also increasingly the case for packaging material, which is especially critical in the case of food, drink and personal care products. These products are sold in massive quantities for short-term use and the packaging must be light, resistant and impermeable to gases and liquids, maintain the organoleptic properties of its contents while keeping compatibility with existing recycling options. In terms of impact on consumers and their preferred choices, BioSupPack addresses the following aspects: i)favour the selection of products that are packed in selected biobased materials, which per se (or on the monomeric state) do not pose a safety risk to human health due to migration effects; ii) Moreover, the BioSupPack packaging solutions accommodate the growing demands of green- and health- conscious consumers concerned about the environment[135]; iii) consumers will benefit from high-quality products due to the improved preservation properties of the new packaging formulations; iv) favour the use of natural materials to counteract consumers' concerns about synthetic additives and substances migrating to food; v) consumer expectations regarding recycling/upcycling, reduction of packaging waste and environmental side effects of mishandled packaging will be fulfilled[137].

Barriers/obstacles/framework conditions.

BioSupPack has carried out an analysis using the PESTLE approach. Table 11 presents the political (P), economic (E), social (S), technological (T), legal (L) and environmental (E) factors that can have an impact on the market. The success of the project and the exploitation of the results depend on many external factors. The primary risks/barriers

[136] https://www.packagingstrategies.com/articles/90098-whats-next-for-the-rigid-plastic-packaging-market

[137] For about 95% of European consumers, it is important that the design of products/packaging facilitates plastic recycling (Special Eurobarometer 501 Survey, 2019)

related to project implementation are covered in section 3.2. The longer-term barriers and obstacles towards achieving impact relate to scaling the project results from TRL6-7, into widespread market replication, as following:

Table 11: Barriers and mitigation measures in BioSupPack

BARRIER DESCRIPTION	MITIGATION MEASURES
Economic Barriers	
Worldwide pandemic situations such as Covid-19 has led to material disruptions and changes in the operations (installations, manufacturing and supply chain).	Based on a recent study published by McKinsey138 on how the COVID-19 crisis will likely impact the packaging industry, solutions like BioSupPack relevance is expected to increase: i) More focus on hygiene aspects, less on re-usable packaging: this implies that more items will only be used once, thus increasing the need for recycling and upcycling; ii) Disrupted supply chains, need to avoid the reliance on importing fundamental goods: we use bio-based polymers for which the feedstock is sourced locally and will strengthen the EU production capacity. In contrast, the oil prices decrease makes the competition harder for bio-based materials and is a disadvantage for us, but it is not expected to continue till the market entry of our solutions. Food, hygienic and drink consumption will increase in home that use small packages in comparison with HORECA sector. Providing then waste feedstock to industries keeps up ongoing.
Market: High cost of early demonstration projects and lack of testing facilities is a challenge which requires greater government support.	BioSupPack will have their own testing places to validate the new materials and technologies developed. 5 different places (for the 3 pilot plants) will be available for validation of demonstrator's circular chain at pilot plant conditions: Czech Republic, Italy, Portugal, Belgium & Spain. Additionally, as Table 4, section 1 indicates, most of the partners take part in research & innovation projects, which valuable expertise accumulated. Wherever possible industrially available technologies will be used in the packaging conversion steps to avoid carrying the burden of large investments for PHA adopters.
Continued cost reduction: Bio-based materials need investments in research and innovation to accelerate cost reduction, particularly in technologies very close to commercialisation.	BioSupPack will contribute to reduction of equipment costs (through integration of technologies), logistic costs (i.e. the online platform will reduce cost of logistics as well as will let better exploit the waste feedstock resources), and energy consumption by, mainly, use mild conditions in the different fermentative and enzymatic steps, and improve yield production; compared to existing bio-based production of high-level materials. All those factors would both decrease the CAPEX and OPEX and increase production efficiency, further reducing the final cost of the new PHB/PHA products.
Private financing of follow-up steps: Increasing amounts of private investment will enter the Bio-based packaging sector as the technology matures. Ultimately, the decision whether or not to support the full commercialisation rests with private investors.	Key component of WP8 will be to prepare the case for post project financing from commercial or public funding sources to ensure we can complete the journey over the Innovation Gap and then, swiftly on through the Valley of Death. This will be supported by Life Cycle Costing (LCC) activities undertaken in WP6. ENCO is experienced in securing commercial investment from long term sources of finance for innovative technologies. This experience has contributed to the structure of the proposed plan of work. It has identified the required demonstration scales required for the various pathways, the setting of cost targets that must ultimately be based against commodity traded fossil fuel market prices and, of the continual requirement to monitor and mitigate risks as they arise (both technical and economic), undertaken in WP9. All processing partners has a strong investment capacity but, in view of its high diversification in terms of processing techniques, ILAB and GPI would opt first for new PHN bottles and PHA coating to be processable with existing equipment before reaching a volume warrantying dedicating an own production line.
Sociological Barriers	
Public acceptance & their support is vital to ensure that viable deployment sites can be secured. As is the case for regulators, public acceptance will grow over time as the number of successful deployments increases.	Public concern on use of durable plastics and their leakage into the environment have rose and led to the currently more stringent packaging regulations that generally can support biodegradable plastics, paper & board packaging and recycling. On previous EU projects in which partners have participated, a progressive involvement of citizenship has been checked (SAV projects in Table 4). Furthermore, the tendency of people is changing, and every time they are more aware of the climate change issues. Preparation of communication and education concepts. Involvement of science centre in EU in education.
Technological Barriers	
Technology development: new technologies and materials are needed to reduce costs.	Improvement of the yield of PHB production from BSG, barrier properties and EoL needed. Technical risks and mitigation strategies associated with the project delivery are covered in section 3.2. Any scale-up and industrialization of a technological process involves a certain level of risk. However, in addition there are risks that competing actions could result in protected IP that could limit the proposed innovation strategies. Freedom to Operate is fundamental for future exploitation. Initial IP searches undertaken in the preparation of this proposal (see section 1.4) have shown that there is no existing IP conflict and that the consortium in fact has a rich IP portfolio to build upon in the project. However, throughout the project, as part of WP8, the consortium will employ due diligence to ensure that any future identified IP risks do not become barriers.
Political, Legal and Regulatory Barriers	
Standards and certification: introducing bespoke solutions and new standards	A number of regulations are underway and expected to affect the market, such as: • Single use plastics directive (SUPD): Ban of certain single use plastic products by 2021 –shifting additional volume to fibre alternatives but also influencing the choice of suitable polymer barrier coatings. • Packaging and packaging waste directive (PPWD): could boost fibre packaging which are more recycled already to reach the overall targets. •Landfill bans of recyclable materials by 2025 (proposal) –with potential positive impact on recycling.
Policy development: European Commission last initiatives,	

[138] McKinsey & Company 'The packaging industry and the coronavirus pandemic' (April 2020)

Directives, etc.	BioSupPack will contribute to put into practice EU & national initiatives, Directives aligned with bioeconomy & circular economy. Project partners and stakeholders will foster public bodies and administration involvement, thanks to their current relationships and networks. E.g. the Circular Economy Package (CEP), Directive (EU) 2018/852 amends Directive 94/62/EC and EN 13427:2005, on Packaging & Packaging Waste; REACH regulation (EC No 1907/2006), Food Contact Materials – Plastics. Regulation (EU) No 10/2011 (last amendment by Regulation (EU) No 2017/752. In 2019, ECHA (European Chemicals Agency) published a new version of the restriction report. The report mentions a list of substances banned for use as a substitute for microplastics in cosmetics and fertilizers and opens up an opportunity for PHB, since PHB is named as an example of a reference for biodegradability.
Environmental Barriers	
Carbon footprint: The carbon emissions reduction of the whole bio-based product life cycle is complex	As seen in terms of GHG, the CO_2 eq emission of bioplastics is much less than that of fossil plastics packaging. For the specific case of the coated packaging, the carbon footprint is generally more positive for fibre-based packaging than for other materials (driven by renewable sourcing and recyclability) so they can help reducing the GHG emission of the food packaging sector. In the case of monolayer bottles, we expect a reduction of greenhouses gases reduction between 3-5 CO_2equi/Kg of package depends on selected EoL. The intelligent logistics platform together with efficient driving systems through the use of sensors put in practice in WP1 by SAV will allow a reduction of greenhouse effect emissions close to 20%, based on previous knowledge on PlastiCircle project, and IoT optimization technologies.
Life cycle impacts: Mooring and anchoring seabed disturbance impacts on marine life and species.	Plastics (and its conversion to microplastics while in the sea) will be avoided in a significant way by the self-production of bioplastics, being easier to degrade and being these ones the "possible" plastics to end in the sea. For example, when by accident plastic bottle, try or cup will allow the sea water, BioSupPack's packaging proposed will be marine degradable and therefore have a limited impact on marine species.

Additional investments

In terms of further contributions to maximizing the impact of this BBI action, significant **in-kind contributions** are planned to reflect the industrial context of the action and will definitely contribute to increasing the impact and reaching the project objectives. T**he consortium covers 38.2% (2.43M€) of the overall project budget**. This co-funding, comprising the 65% budget of the large companies and SMEs as well as a 10% for RTOs partners, is justified by the genuine conviction of the partners that they will on the short-medium term obtain a return on their investment when the BIOSUPPACK solutions reach the market. Our success securing in the consortium world leaders in the bioplastics, packaging, food and personal & homecare fields who will fully cover their own costs by themselves, and members of the AB who will make also significant in-kind contribution (in providing samples, assessment, advice, providing actual society/ consumers data, etc.) shows that we have a winning concept. In addition, some partners also provide **additional investments towards the BBI-JU's objectives** (Table 12).

Table 12: Summary and cost of the additional investments (note: no financial transfers between partners are made)

Entity	Activity	Contribution to the BBI JU objectives	Estimated private investment (M€)	Estimated value public investment (M€)	Other EU funding received
1.AIMPLAS	Improve control of recycling steps, adaptation of facilities for bioplastics. Synthesis of biopolymers (2021-2023)	KPI1 (polymer processing), KPI2 (polymer processing), KPI6 (RTD for food packaging processing)	0.55M€	Investments are supported by regional authorities via own resources and European regional funds (50%).	
3.CTB	Extension of lab facilities and pilot lines to help the reconversion towards a circular bio-based economy (2021-2022)	KPI1 (polymer processing), KPI2 (polymer processing),	1.5 M €	1.0M€ Belgian Province W-Flanders invests in the building & infrastructure	Investments supported via EU regional funds (50%)
4.ASU	Innovation Campus – Model Factory and Laboratory for sustainable food production and food packaging	KPI1 (polymer processing), KPI2 (polymer processing), KPI6 (RTD for food packaging processing)	2,0 M€	No other investment or funding expected to be received	
8.SABIO	Upgrade of the production plant and recycling plant.	KPI1 (polymer processing), KPI2 (polymer processing), KPI6 (RTD for food packaging processing)	0.05 M€	No other investment or funding expected to be received	
9.NFG	Building a 1kT capacity industrial plant to increase the capacities of PHB production (2020-2021)	This investment will contribute to the successful meeting of the project objectives. Specifically, KPI1 and KPI8 related to BSG hydrolysis and PHB production will be fulfilled together with KPI12 (business and growth).	0,33	No other investment or funding expected to be received	
10.ULK	ULK as a high distributor of Fast-Moving Consumer Goods is determined to play a key role to push all parts of the supply chain to close the circle of plastic, push forward the use of biopolymers and significantly reduce the rate of plastic disposal in the environment	R&D activities associated with plastic recycling, introduction of bioplastics and sustainable packaging that contribute to BBI-JU objectives (KPI2 and 6)	0.4M€	No other investment or funding expected to be received	

14.IRIS	Participation in projects where IRIS provides a significant cofunding (BBI: PolyBioSkin, Refucoat, Pulpacktion, Agrimax, CE: Ecobulk, SPIRE: Multicycle)	Research and innovation related with bioplastics, plastic recycling and packaging that contribute to BBI-JU objectives (KPI2 and 8) and are in fact synergistic with BIOSUPPACK	0.87M€ (2019-2022)	2.3 M€ (2019-2022)	NA.

In short, the foundation for achieving the expected impact of the project has been laid: an innovative research agenda, a resourceful action plan for the contextualisation and exploitation of the expected results, and an excellent conglomerate of actors in the consortium who are in pursuit of a shared common goal.

2.2 Measures to maximise impacts

Our vision is to lead the transition towards a sustainable post-petroleum society in the plastic sector while decoupling economic growth from resource depletion and environmental impact by serving industry with the provision of cost-effective high performance sustainable multifunctional bioplastic packaging solutions with tailored materials and mechanical/enzymatic recyclability at EoL. In order to assist EU in the transition towards a bio-economy new sustainable cross-sectorial value chains will be established providing competitive bio-based alternatives in rigid packaging, such as food & beverage trays, cups, bowls, bottles; cosmetic & homecare packaging applications. In line with a number of EU and national regulations, there is a clear need for new materials with high bio-based content and for meeting the demand for plastics with innovative EoL options. In order to realise this vision, post-project exploitation pathways have been carefully planned already during BioSupPack proposal preparation. We have devised our **DRAFT detailed strategy for the dissemination and the exploitation of the project results** (see Annex, section 4-5), vital for the successful exploitation of the project with a view to increasing business outcomes as well as social & environmental impacts.

The combined resources of industry and SMEs, academia and RTOs groups are complementary, and the multi-sectoral nature of the dissemination partners will ensure widespread dissemination among the rigid primary & secondary plastic packaging, packaging waste to secondary raw material production in several sectors and among wider industry stakeholders. BioSupPack Communication, Dissemination and Exploitation actions will:

- Focus on building a **stakeholder community** that can be sustained and increased **during and after the project's lifetime.** These actions will build a sustainable user community that will adopt and exploit the BioSupPack products and information. The **BioSupPack stakeholders** (see Annex 4-5): (1) will be engaged during the project; (2) will likely benefit from the innovations in the BioSupPack developments (new PHA-based materials & products, new logistic & management system; new integrated technologies and EoL); and (3) will be the starting point for assessing the potential industrial business implementation of BioSupPack, through consortium agreements/licences. The outcomes of BioSupPack, as well as other information about the project and its products, will be tailored for external dissemination to the wider community.
- The consortium will develop **various activities and channels for communication** to increase the visibility of the project and its outputs. Activities and their impacts will be continuously monitored during the project and these will be reported in all contractual progress reports. Dissemination activities to optimise communication of the value and impact of BioSupPack results will be undertaken by all partners with significant exploitation supports provided by WP8 (future needed certification, IPR, management, market penetration, business plan and commercialization strategy). Actions and their impacts will be continuously monitored during the project and the resulting actions will be reported in all contractual reports in months 12, 30 and 42. This is further supported by the overseeing role of the AB. This body will be directly informed of the project's main achievements and will contribute, as an external advisory and quality assurance body, to keep the project aligned with the BBI on bioplastics needs, requirements and specifications, as well as to maximise the exploitation and impact of the results.

2.2.1. Dissemination plan

BioSupPack will implement a Plan for the Exploitation and Dissemination of the Project´s Results (PEDR). It comprises a **set of actions to spread the non-confidential results** generated during the project to ensure that as many relevant specialist audiences as possible have access to them. The aim underpinning the dissemination strategy is to heighten awareness about the tangible and intangible outputs of BioSupPack, to contribute to knowledge building and advancement of innovation in EU and as well as to avoid duplication of research and demonstration work. To safeguard the commercial opportunity, the dissemination activities will be subject to a confidentiality policy. By default, and building on the consortium experience in collaborative projects, any result of the project is to be treated as confidential and can be disseminated only after agreement of the project partners following the relevant clauses of the consortium agreement (updated and signed during the grant agreement preparation phase in keeping with initial agreements on ownership reflected herein and respecting the general terms of the grant agreement). This approach allows us to retain complete control and **make sure that no "exploitable" information is disclosed before being protected**. The protection of the IPR is of invaluable importance for the participating partners. Therefore, the non-confidential results and information on the project to be publicly disseminated will be carefully identified and

specialized high-impact technical peer-reviewed journals will be chosen. The RTD participants in the consortium will publish results in peer-reviewed open access journals, which, will be an indicator of overall project success (in terms of progress vs. the scientific SoA) and an important element of dissemination towards industry, standardisation bodies, and policy makers. The industrial partners and market actors will focus on showcasing and disseminating results at conferences and trade fairs and other industrial channels such as technical/sectorial magazines, industry network meetings, and the like (to demonstrate industrial novelty and commercial outcomes potential).

The Exploitaiton & IPR Manager (Dr. Marco de la Feld from ENCO) will work closely with the project innovators to manage the identification of non-confidential disclosable information that can be published/disseminated. Our project dissemination activities will intersect dissemination strategies of the BBI-JU, via associated BIC members AIMPLAS, CTB and EUBP, as well as other EU PPP as SPIRE through IRIS and ENCO, which will contribute to the distribution of project outputs across the relevant EU and international communities and amplify the project dissemination effort. All the dissemination activities will be developed respecting the short-medium term exploitation of the project results by other parties as a result of the project´s innovation actions as well as other private initiatives which can accelerate the route towards the take-up of new technologies within the bio-based plastic sector, also beyond the BioSupPack selected applications.

Beyond use of traditional dissemination channels (presentations & posters at conferences, publications, industry magazines, newsletters and on-line platforms), we will also tap into more vanguard channels, such as social media platforms (ResearchGate, Academia.edu, SlideShare, Twitter, YouTube, Vision2020 open innovation platform), etc. Also, the project website will serve as a lasting virtual hub for disseminating downloadable public deliverables and outputs of BioSupPack, as it will house also open access publications to the peer-reviewed documents of the project.

In particular, the network of EUBP across the EU bioplastics value chain should be highly instrumental in identifying and approaching prospective business partners and potential licensees. With around 70 member companies from across the bioplastics value chain that are active in the European market, EUBP works closely together with all major stakeholders in this industry. The association also has access to several unparalleled channels of communication within the bioplastics industry, ranging from the largest European bioplastics conference, to the bioplastics-focussed newsletter with the widest circulation, to one of the most widely-followed (4000+ followers) Twitter accounts focussing on bioplastics topics. To amplify our impacts, we also capitalise on our AB communication channels to reach the companies & citizens all over the cycle value chain (from material production to packaging end-users). As shown in Table 13, the project will disseminate the results through the participation in international conferences, project presentations and publications in journals/technical magazines. The ultimate goal is to demonstrate the added value and the innovation of BioSupPack technologies/materials. All partners in technical WPs are responsible for both academic and industrial publications.

Table 13: BioSupPack Dissemination Plan

Target Audience	Tools & Channels	How contributing in achieving Impacts
Consumers, Retailers, General Public	Project Leaflet, Video Clips, Website, Blogs and Magazines e.g. Food Production Daily, Bio-Based World News, PlastEurope, ESOF (EuroScience Open Forum)	Market pull. Public acceptance of the proposed innovative bio-based plastic applications that impact daily life stimulating the feeling they are getting value for money from public funded research.
Monomers producers, Copolymers producers, Coating manufacturers, Compounders, Agro feedstock suppliers	Trade fairs/exhibitions: EFIB, Compounding World Expo, 2nd PHA Platform World Congress, K Trade Fair. Open Access Journals (OAJ): Polymers, Unitex journal, Bioplastics newsletter and magazine, Journal of coatings technology and research, Surface and Coatings Technology, European Coatings Journal, Journal of Polymers and the Environment., Biomacromolecules, Frontiers in Materials, Journal of Renewable Materials, Coatings, Progress in Organic Coatings, Packaging Technology and Science, Packaging Europe, Packaging Journal, Green Chemistry, Chem Sus Chem. Plastics & Recycling Technology, Advances in Materials Science and Engineering, Polymers, Polymer Degradation and Stability, Journal of Applied Polymer Science, Polymer Testing, Materials, Packaging Technology and Science, Food Packaging and Shelf Life, International Journal of Polymer Science.	Widespread of the use of the project innovation. Multiply success of the project by inspiring new initiatives based on the project results.
Packaging converters (plastics, polymers and coatings). Plasma technology, Biotech-nology industry, Cosmetics & personal care, home care companies, other packaging users, other potential users (pharma or health food	Trade exhibitions and conferences: Interpack (GE). International Conference and Exhibition on Food Processing & Technology (UK); Alimentaria (ES); Cosmetics Business (DE), Anuga FoodTec&Anuga (DE); Bioplastics Business Breakfast (GE), Fakuma International trade fair for plastics processing; European coatings show; FACHPACK (Nürnberg); Interpack (Düsseldorf) Anuga FoodTec (Köln); BiPoCo (Hungary); Innovations in Food Packaging, Shelf Life & Food Safety Conference; Pollination	Foster new supply chains and new cross-sectorial value chains (which will be reinforced by the presence of our expert Association EUBP) Raise awareness about BioSupPack materials and suitable end of life options tailored on a vase by case basis. Multi-actor approach to support the transition

sectors).	Day. Magazines: Plastics News Europe, Nature Sustainability, Bioresource Technology, Green Chemistry, Plasma Processes and Polymers, Applied Microbiology and Biotechnology. Project's training events.	towards a post-petroleum economy
Industry Multipliers- Sectorial Associations (Plastics Europe, CEFIC, Plastics Europe, PAFA - Packing and Film Association, and other associations. Citizens & Education fields	Conferences, seminars and workshops (participation in organising, scientific committees, with poster, papers, presentations, or chairing): Annual World Food and Innovation Forum; Bio-based World, Biopolpack, Biopol, Sustainable Bioplastics, Bioplastics, Biorefining, Bio-based Materials, etc. Features in Newsletters and e-bulletins (plastics today, food production daily, plastic news, agronotizie, CosmeticsMag, Social responsibility and quality for sustainability, World Usability Day, etc.)	Dissemination of new knowledge and technologies for transitioning to a bio-based economy. Communication to retailers and consumers about the bio-based materials used in BioSupPack Increase target industry sustainability for consumer and environmental protection. Building new value chains and cross sectorial interconnections among clusters.
Standardisation bodies (CEN, ISO, DIN CERTCO, Vinçotte), and policy stakeholders, national food safety authorities, EFSA, EC, Expert Group for Bio-based Products (E02886), National governmental bodies	Policy dialogue: joining roundtables fostering multilevel and horizontal discussion on BioSupPack results. Collaboration with public administration will drive better informed decision-making processes across Europe also with a view in improving waste management legislation and standards.	To increase the influence of the project and its results in policymaking towards the creation of a positive framework for the take up of BioSupPack (and in general bio-based plastics with tailored EoL).
Public/Private waste management & logistics, Recyclers (enzymatic and mechanical recycling)	Conferences: ISWA World Congress, Waste Management, Ecomondo, Waste & Recycle Conference, Plastics Recycling Conference; International Conference on Sustainable Solid Waste Management. OAJ Articles: Waste Management & Research, Waste Management World Magazines: Waste Management World (ISWA Magazine)	To support the transition towards a post-petroleum economy improving circularity Market & public organisation push
Research and academic stakeholders	Magazines: Pack World, Bio-based News, Plastics films, plasticseurope.it plasticsnewseurope.com, Bioplastics Magazine, Sustainable Plastics, Labiotech. Plasticeurope, Plastiker, Bio-based World News, Farmers Journal (IE)	Effective knowledge sharing among the research and scientific communities in order to accelerate the paradigm shift from oil-based towards bioeconomy.
Early stage researchers, e.g. on LCA.	OA Journal Articles, Advances in Materials Science and Engineering; International Polymer Science & Technology, Food packaging and shelf life (Elsevier); Packaging technology and science (Wiley); Polymers (MDPI); Kunststoffe international; Lebensmitteltechnik; Plastverarbeiter. Specific events as Open day at Universities, Researchers' Nights, Knowledge Open Festivals. Project training events.	Attract more young people to scientific and research careers, enriching their curricula with knowledge from the project. Early stage researchers inspired by the results, will be motivated to remain in EU- research and possibly work in areas that are beneficial for society, BBI, sustainability and innovation.
Other H2020 BBI projects and international/ national projects	BIOnTOP, REFUCOAT, SuperBIO, Open-Bio, BioSTEP, InnoNet, BIOCOMPLACK, CommeBeBiz, AGRIMAX, PolyBioSkin, Hyperbiocoat, MultiCycle	Cooperation, knowledge interchange and reuse between previous and future projects.

*****Partners will attend to a minimum of 15 conferences/trade shows as speaker or poster in EU & international level & will engage to launch a minimum of 30 publications in high impact journals.***

2.2.2.Business, Innovation & Commercialization Potential Plan

- **Exploitation and Business Plans for BioSupPack Key Exploitable Results (KERs)**

A market-oriented approach will drive BioSupPack innovation activities capitalising on the participating rigid PHAs packaging end users' and the AB members' expertise, and Tasks 3.4/ 6.3; ensuring that sufficient knowledge is collated to steer the innovation towards delivering **consumer-oriented solutions** with good market potential. BioSupPack integrates all exploitation pathways for the results:

Commercial exploitation: a key exploitation route is the industrial use of the project knowledge for the creation of new innovative bio-based and biodegradable packaging materials, within and beyond the validated applications, new processes and advisory services to deliver high competitiveness and growth impacts.

Education and training: the use of the new knowledge generated for upskilling and for enriching curricula in EU education will have a high impact for raising the EU knowledge base to prepare rigid biopackaging, and end user industry sectors for the transition to the bio-based economy; several training events & workshops will be organised.

Further research: planning future spin-off research and development projects especially in areas that will contribute to the transferability of the project results to new applications.

Contribution to standards: while making available new bio-based materials with innovative EoL, the project results can on the medium- term contribute towards new regulations in the field of disposable plastic waste.

Detailed exploitation & business plans will be developed in Task 8.1 & 8.2., generating a Business and exploitation planning, and a **Plan for the Exploitation and Dissemination of results (PEDR)**, in order to provide the details on how the project's results will be used in commercial exploitation activities, for training purposes, further research as well as standardisation activities. Further measures will be planned in the framework of this task to increase the continuity of the established ecosystem and value chains, as well as market uptake of BioSupPack KERs for EU industrial sustainability, competitiveness, growth, and jobs.

The exploitation routes and corresponding business models will be investigated in WP8. The route for future commercialization of the results can be logically identified from the BioSupPack WP structure and work plan, from the logical flow of the project activities and their clear role.

As previously mentioned, ENCO will act as coordinator of the IPR & Exploitation, Business issues (as Exploitation & IPR manager) to ensure that all IP-owning partners as well as potential and confirmed exploitation partners will work towards a successful technology and knowledge transfer to ensure efficient market uptake. The role of the Commercial & Exploitation manager will be to assess the commercial potential of the results and liaise with industrial companies to maximize their commercial use. ENCO will follow the progress made in BioSupPack and will develop full exploitation plans as the unique values of the project results are confirmed. Dr. Marco de la Feld from ENCO, helped with the chairing by EUBP (WP7 leader), and with the contribution of all the other partners, will constitute the Innovation Committee (IC) members. IC will work to establish targeted exploitation and dissemination strategies, ensuring that they are fully aligned, when the project outcomes reach enough maturity. The entire project Consortium will discuss this strategy and the corresponding document will be delivered for the mid-term assessment. An updated and more detailed exploitation plan will be submitted thereafter (Final PEDR-Exploitation Plan & Communication & Dissemination Plans, by combining inputs from WP7 & WP8).

Furthermore, a successful maximisation of the project impact will contribute, in the mid-term, as an outcome from the project, to speed up the exploitation, in the different European areas. In a second stage globally including, for instances, North America, Japan and South East Asia.

All technologies, materials and procedures developed by the partners will be considered for exploitation. A summary of anticipated Key exploitable results (KER) is reported in table 14. Exploitation issues will be brought to the attention of all partners at an early stage in the project ensuring that issues regarding market and commercialization, identified during exploitation analysis, can be fed into the demonstration actions of the project. Additional KERs are expected to be generated under the project development:

*Table 14: BioSupPack Key Exploitable Results (Knowledge, products or technologies) *Years after project end*

Please note that the Timetable after the end of the project, will take into account the gap between the demo sites development and the needed optimization of the installations and equipment adaptations, as well as the suitability and availability of feedstock; having as a contingency plan the option to use mixtures with commercial PHAs & BSG source, until the full industrial process is ready.

WP	KER	Owner	Envisaged IPR Tool	Type of exploitation	Target end-users	Timetable after the end of the project *
1	PHB Bio-based formulated materials & its production process	SABIO	Confidential know-how	Own production	Polymer converter	First year of the project
	Material scale up production	SABIO	Patent	Licensing platform	Other compounders	1or 2 years
	Recycling Process	SABIO	Confidential know-how	Sales of products	Recyclers	Direct post-project
	Production of biobased material PHB	NFG	Confidential know-how	Own production	Nafigate Cosmetics	From the second year of the project and post the project
	Production of biobased material PHB	NFG	Patent or trade secret	Own production, sales of products	Medical industry	5 years
	Production of biobased material PHB	NFG	Confidential know-how	Own production, sales of products	Nafigate Cosmetics and other cosmetics packaging producers	2-3 years
2	PHA coated/ grafted fibre based packaging	GPI	Patent if possible or trade secret.	Direct sales	Food and service food sector	2 or 3 years or when PHA is commercially available
	Packaging development and production process	ILAB	Confidential know-how	Services	Brand-owners	Direct post-project
	Nano-scale grafted fiber-based packaging	ASU	Confidential know-how	Consulting of packaging manufacture	Producer fiber-based packaging materials	Second Year of the Project

3	Support PHA enzymatic recycling	PLT	Patents	Sales of equipment	Polymer producers	2/3 years
	BSG pretreatment	PLT	Confidential know-how	Sales of equipment	Plasma biomass treatment	1-3 years post project
	Monitoring prototype for sorting application	IRIS	Patent/trade secret	Sales of equipment	Waste managers	1year post-project
	Modelling and data analysis	IRIS	Confidential know-how	Consulting service, commercialization	Waste managers, quality control	Directly after the project
	Validation of packaging prototypes	NFG	Confidential know-how	Own usage and consultancy	NFGC and other cosmetics producers	Directly post the project
	Study of consumer perception and acceptance	NFG	Confidential know-how	Own usage	Nafigate Cosmetics	Directly post the project
4	Logistic model	SAV	Confidential know-how	License in platform	BSG producers and feedstock end-users.	1or 2 years. Tailor made design.
5	PHB-depolymerizing enzymes	ASA	Confidential know-how	Production ans sales of enzymes	Recycler	Directly after the project
	PHB from 3HA after enzymatic hydrolysis	NFG	Confidential know-how	Own production	The same as from PHB in WP1	Directly post the project
	PHB-degrading Microorganisms	AIMPLAS	Confidential know-how	Sales of products	Recyclers	From the end of the project
	Improved PHB-degrading enzymes	AIMPLAS	Confidential know-how	Sales of products	Recyclers	1-3years post-project
6	Knowledge of the LCA and LCC	ASU	Confidential know-how	Consulting on sustainability, LCA and LCC of PHA packaging	Producer of food and packaging materials	After the project end
	Use of PHB-based packages in food (ketchup, mustard) and homecare (wash liquid) end-products	UNILEVER	Confidential know-how	Sales of products	Food & personal care products to EU28 consumers	Approx. 2/3 years (PHB-based packages are commercially available)
7	Optimized water-based PHA coating formulation	CTB	Patent	Licensing to formulator or GPI	Formulators, paper coating companies	1-2 years

- **Plan for Exploitation & Dissemination of the Results (PEDR).**

A detailed business plan will be developed within WP8 and integrate into the **Plan for Exploitation & Dissemination of the Results (PEDR).** The exploitation of project results addresses a chain of value ranging from technological innovations (upstream) for the benefit of several technology-based user segments, to innovations on products and services (downstream) that foster the biopackaging business sustainability through better risk management, decision-making and overall business assessment. The Business Plan section will explain the commercialisation of different technological solutions resulting from the project and other exploitation measures. BioSupPack workshops in the framework of relevant EU scientific and business conference should facilitate new research/services/business opportunities based on BioSupPack demonstration data and products for commercial applications. Actions organized under “community development”, “capacity building” and “clustering” are expected to have a positive impact on this.

Besides cost-efficiency indicators, Key Performance Indicators (KPI) will be used in the innovation process moving towards higher TRL level that supports fast and easy uptake of the BioSupPack technologies and their penetration in relevant markets. The structure of the project Work packages has been designed to facilitate this process, including:

(1) **Refinement of the requirements and final applications to ensure the planned development is fit-for-purpose**. KPI: Positive feedback (satisfaction level) from the relevant user groups on the fit-to-purpose of the proposed BioSupPack materials and solutions for new packaging in food&beverage sectors, cosmetic & homecare.

(2) **Production and Demonstration of the new packaging materials and products with the BioSupPack technologies**. KPI: Positive feedback (satisfaction level) from the relevant user groups on the capability of the novel developments to replace existing, more unsustainable technologies (market replication)

(3) **Uptake of BioSupPack new management system by the Industry**, including the project relevant partners. The detailed assessment of BioSupPack overall solution in a specific demos site will be relevance. The feedback of industrial end-users (public and private biotechnology & biobased companies), public authorities and other sectors (such as waste managers, plasma & enzymatic technologies, pharma, healthy food, etc.) will be critical. KPI: Number of positive feedback (satisfaction level) from targeted industrial and societal players (including project partners).

Business plans will be developed to support the commercialization of each KER and partner. The Business Model and Plan developed in WP8 will integrate IPR, innovation protection, commercialization and exploitation strategies, and the accompanying techno-economic analysis, safety, regulatory and LCA studies developed in WP6. This will

pave the way for the successful exploitation of the project results. The expected impact based on a conservative scenario with the capacity to sell 5-10 Million bottles/year in the period 2025-2029 is described in qualitative and quantitative terms in table 15- below.

Table 15: BioSupPack Estimated economic impact & business model proposed by industrial partners per year after 2024

Industrial Partner	Product, application or system to be exploit	Expected Turnover (M€/y)	Direct Employ-ment[139]	Required investment	Business model
6.PLT-a	Plasma bubble biomass reactors	2	3	< 250K€ in new sales of plasma technology for biomass processing	Selling scaled plasma bubble reactor technology
6.PLT-b	Plasma packaging surface modification	1.5	2	< 100K€ in new sales of plasma technology for polymer processing	Selling scaled plasma surface modification technology
8.SABIO	New grades of bio-based sustainable materials	0.3-0.5	4	± 0.1 M€	Producing and selling new more sustainable materials.
13.GPI	PHA coated trays, bowls or cups	6	15	0.5M€ to move forward from TRL7 to TRL9 and minor fine tuning of reallocated existing manufacturing facilities of PE-coated packaging	Selling new more sustainable packaging
14.IRIS	Hyperspectral imaging solution for sorting	1-1.5 M	5	< 150K€ Personnel costs for optimization & adaptation of the chemometrics models, sales force engagement, minor adaptation to TRL 9	Direct commercialization to users or OEMs
16.ASA	PHB-depolymerizing enzymes	0,5 – 1,0	2	No investment necessary.	Production and sales of enzymes
17.SAV	IoT logistic platform for the mobilization of waste to bioprocesses.	0.2-0.35	2	<€0.2M	Revenues from the IoT logistic platform
10.ULK	Introduce PHB-based packages in food (ketchup, mustard) and homecare (wash liquid) end-products	0.7 considering only the plant in Greece	1-2	Qualified staff to integrate the use of PHB-based packages and validate consumer perception and acceptance of the new packages (<€0.5M)	Commercialize the PHB-based packages for sauces and wash liquids using BIOSUPPACK innovations developed to end users via exploitation of consumer perception evaluation study.
9.NGF	Several equipment adapted to obtain the new PHB (such as centrifuge, fermenters, etc.)	1	2	>180K€ for the new tool that enables to increase the current production capacities, >135K€ into the extension of current production lines.	Commercializing new feedstock, Licensing production to potential partners for large production volumes.

For the sake of clarity, a Business Model Canvas has been prepared for the BioSupPack materials, featuring core KERs along the value chain (Business plan annex). Similar tools will be completed for each KER in WP8. This will be instrumental to develop and pivot the related business models during the project. The final business plans will provide the details on how the project's results will be used in commercial exploitation activities, for training purposes, for further research activities as well as for standardisation activities. It will also specify in detail for each result both joint and individual exploitation strategies in terms of technology roadmaps, finance requirements, commercial opportunities and related business roadmap, taking into account IPR.

Figure 13 represents the expected full scale coordinated deployment of the different materials and finished products along the supply and value chain to reach our target markets within 3 to 5 years post-project (in different stages, being launched the first packaging around 20226-2027), as well as the adoption to new applications. In terms of the measures that will be in place in order to effectively plan for the exploitation of the project results, the robust industry involvement in the consortium backed by strong knowledge management, knowledge transfer and market exploitation plans, and their key role in the project activities will boost an industry-driven approach to the project respecting market, consumer and regulatory requirements. This will ensure that BioSupPack is successful in transitioning knowledge towards the market overcoming potential barriers that may hinder the swift uptake of the

[139] Job calculation between 250-450 k€ turnover/job. Average turnover increase in the chemical/engineering industry to create a new employee.

results of BioSupPack innovations in the post-project phase. Effective knowledge transfer, under Task 7.3., from technical partners to SMEs (internally), third-party enterprises and end-users (externally) will also be fundamental for paving the way towards commercialization, guaranteeing that sufficient knowledge and competencies to exploit results are acquired.

To this end, the involvement of RTDs of international renown in the fields of biomass fermentation processes, biopolymers production, polymer formulations, plastic processing & synthesis, coating technologies and interaction with the food and packaging; and who have a track record in working closely with and in transferring research results to industry, as well as providing training and educational programmes, is a valuable asset, which will ensure that we bridge the innovation gap that frequently prevents industry adoption of R&D outputs. In particular, AIMPLAS performs very frequent broad scope trainings for its over 650 associated companies in the plastic sector. The non-industrial partners will exploit the results of BioSupPack to enhance their educational offer with the novel concepts and system design principles that will be identified and developed during the project. Some of the topics tackled in the project shall become part of Master degree courses, as well as research topics for Master & PhD theses, offering students a taste of research problems that have practical impact and interest, and the opportunity to perform research in a high-quality international environment.

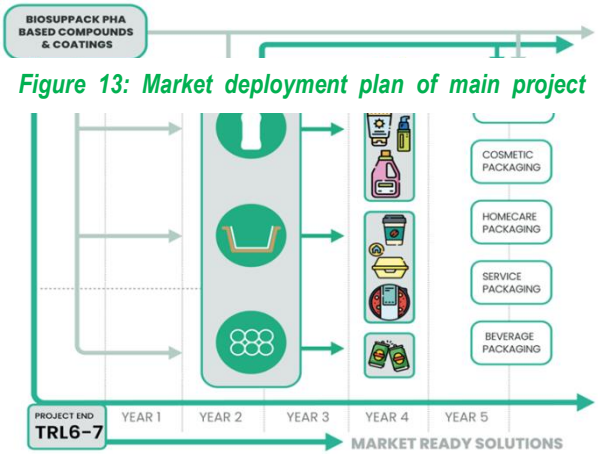

Figure 13: Market deployment plan of main project

The inclusion of ENCO with extensive experience in translating R&D output into actual commercialisation and market penetration, is another important added value for the consortium and will contribute to ensuring the post-project stage. Importantly, we will work towards very high bio-based content of our formulations (close to 100% including different bioplastics and board fractions) and we will follow a cautious and transparent communication policy both in terms of sourcing and EoL, backed up by consequential LCA results, which will contribute to raising industry confidence delivering a positive breakthrough in the context of environmental-friendly food packaging, as well as single use service packaging.

Sound management of the knowledge generated during the project, active planning in relation to post-project exploitation activities, further research work for escalating along the TRLs, as well as scouting for possible viable financing routes to commercialisation that will allow us to quickly scale up, bridge the innovation gap, and rapidly overcome the critical stage between pre-competitive results and the market according to market deployment timeline.

How the participants will manage the research data generated and/or collected during the project.

Although the exploitation interests will be favoured by default, BioSupPack will participate in the Pilot Open Access Research Data scheme for a selected amount of research data generated by the project. A data management plan (DMP) has been included in WP9. Non-confidential results will be widely disseminated in open access media and repository (after anonymising critical information e.g. on composition). All publications and selected results will be considered for this so as to provide the scientific community with a better understanding of e.g. the capability of PHAs in rigid packaging and their improved properties, thanks to novel technologies applied; allowing excluding the commercially sensitive details in case of selected successful formulations. All publicly available data will be protected by copyright statements.

The Knowledge Management Strategy (Figure 14) which will be followed throughout the duration of BioSupPack is heavily interconnected with the exploitation and dissemination activities to pave the way for post-project exploitation. An integrated approach will be followed for the identification, capture, retrieval, distribution, sharing, use and reuse of BioSupPack information and knowledge assets. It will be essential to drive the value and benefits of these information and knowledge assets and contribute to maximising the innovation impact of the action. Knowledge management sessions will take place regularly during the project meetings. These sessions will be chaired by the Project Coordinator (Ms. Rosa González, AIMPLAS) and the IPR Manager (Dr. Marco de la Feld, ENCO), attended by all project beneficiaries, and will facilitate the disclosure of the IP and the novelty of the project results. The aim is to review: i) expected results per WP, ii) evaluation of IP (technology watch/prior art review), iii) evaluation of protection options (IPR tools evaluation), iv) third party components included in the project activities including tools and datasets, v) confidentiality of project results, vi)

Figure 14: Knowledge Management

expected publications per partner and vii) exploitation intentions per partner.

The Partners have signed a Consortium Agreement (CA) prior to the signature of the Grant Agreement with the EC in which all the rules for ensuring confidentiality (procedure for granting publication of the results), ownership regime and IPR issues (background included and excluded, ownership, licensing terms, etc) are detailed. Some previous knowledge and background are detailed in Annex-Business Plan. The CA is based on the DESCA model. The CA includes the initial KERs of the partners and patent interest list, which is the starting point for IPR protection and Exploitation Plans.

The following three areas of activity have been identified to ensure proper IP management and maximize the impact and market relevance of the knowledge generated in the project: Proper assessment of the pre-existing knowledge of the various project partners, their potential contribution to the project results IP, and potential overlap of IP. The document will be upgraded during the project to include new results unforeseen at the beginning and included as an integral part of the final PEDR.

The overall IPR strategy focuses on the knowledge generated by the project, the IP rights, the patentability and optimal IPR protection options. Any disputes over IPRs should be settled according to the CA. Legal requirements for commercialization will also be explored, in particular **through "freedom to operate" analysis** as well as individual patent filing strategies and drafting of patent applications for the most promising project results.

Background and Foreground: IPR and access rights, Results definition, protection and ownership.

Each partner in the consortium has made a significant effort in order to define as detailed as possible their exploitation interests and possibilities (preliminary KER -Table 14). For the results obtained, ownership, protection and use will be considered in detail following the recommendations detailed in EU IPR helpdesk document[140] (July 2018). So, the basic IPR framework agreed by the partners considers:

Background: Background, that is, information and knowledge (including inventions, databases, etc.) held by the partners prior to their accession to the EC grant agreement, as well as any intellectual property rights which are needed for carrying out the project or for using results held by the consortium, or indeed by third parties, is defined in the proposal preparation phase (pre-Grant Agreement). Access rights necessary for the preparation of the proposal and the execution of the BioSupPack project will be granted on a royalty free basis. Most of the partners in the consortium will contribute to the project with their respective pre-existing know-how. At this stage no positive or negative list regarding the background access has been developed, and no partner has explicitly required it. Background needed to use the results of the project will be provided either royalty free or under fair and reasonable conditions. The need of such lists has been again considered when preparing the Consortium Agreement.

Foreground: Selected results from KERs, along with the expected scenarios of results ownership. This table will apply for all project results, and it will be periodically revised and updated during the project in line with developments. The project results will be continually assessed in terms of novelty, patentability and protection through WP8 tasks, as part of an effective knowledge management strategy. The choice of the most suitable form of IP protection, as well as the duration and geographical coverage depends on the results at stake, but also the business plans for their exploitation and legitimate interests of consortium partners. For the future exploitation of the results generated in this project, an agreement among the partners will be reached to transfer the ownership and grant licenses, based on the assessment provided by the Innovation Committee.

The protection of every specific development will in any case be previously evaluated against the interest of the whole consortium, to assure other results exploitation will not be impaired due to a lack of foresight. For each result an evaluation will be made of whether the results may be licensed or shared with third parties, with the objective of giving all partners the widest possible autonomy after the project without compromising the others.

A first draft of the Exploitation Plan will be available by M12 and will be continually revised and edited throughout the project, and a final version will be signed by all partners at M36 (annexed to D8.6.).

Open access to peer-reviewed scientific publications

BioSupPack partners will provide peers and end-users with free of charge online access to the project peer-reviewed scientific research articles. Partners can follow two routes towards open access: (1) by self-archiving their journal articles in an open access repository, also known as 'green' open access, or (2) by publishing in an open access journal, known as 'gold' open access. Whereas fees to cover publishing in gold open access journals have been budgeted, BioSupPack will privilege green open access, whereby the partners will publish in peer-reviewed scientific publications and then self-archive a version in repositories (using only those indexed in specialised search engines like Scopus). The project website will complementarily house the 'BioSupPack Green Channel' for open access to the peer-reviewed publications of the project. In relation to expected publications from the project, these include but are not limited to: results from food bio-packaging; demonstration of recyclability, biodegradability, LCA; novel

[140] www.iprhelpdesk.eu/sites/default/files/newsdocuments/Fact-Sheet-Background-in-H2020.pdf

results and applications of PHAs and copolymer blends, as well as the novel technologies integrated.

- Communication activities

Communication activities are of the outmost importance for effectively promoting our action and outputs towards the stakeholders that we want to reach to fully exploit this BBI action. To successfully connect and cooperate and acquire positive support from relevant stakeholders and the public at large, a strategic plan will be implemented in Task 7.1., on top of the preliminary outline presented in Figure 15. The overall communication campaign, which covers the whole project duration, will be led by the Innovation, Communication & Dissemination Manager, Mr. Christian Schulz (EUBP), who holds an established extensive network and communication capacity to ensure that benefits and impacts of the project are genuinely conveyed to key stakeholders and the general public (in particular through brand owners retailers and fast food chains (for the case of service packaging). Our aim is to i) generate understanding of what BioSupPack seeks to achieve, ii) stimulate interest and support, and iii) motivate the collaboration of actors along the value chain as well as of the general public in order to boost the transition to the bio-based economy in EU. The BioSupPack communication campaign leverages one-way, bilateral, and multilateral strategic channels. Following a preliminary stakeholder analysis (which will be developed in Task 7.2), the target audiences of BioSupPack communication activities comprise; monomers & biopolymer suppliers, agro-food processors, plastic packaging manufacturers and converters, personal care and food manufacturing industry and retailers, other packaging end users, consumers & consumer organizations, agricultural associations, public authorities, standardisation & safety agencies/organisations, scientific communities, and recycling & industrial composting companies. Particular efforts will be made to engage with and raise awareness of the general public. It is important to address consumers' concerns by informing the public about our strong dependence on fossil-based conventional plastics to which renewably sourced, bio-based, high-performance, and cost competitive bioplastics can be the more sustainable alternative. The awareness raising actions carried out will aim to generate acceptance and open the path to building trust towards innovative biopackaging solutions, through optimal communication and transparency using appropriate language enabling EU consumers to make informed purchasing and consumption decisions. A set of key messages will be developed specifically to popularise complex subjects and to bring the issue closer to people's lives by making it more understandable. The campaign will be rolled out through various channels and tools (figured in Fig.-14), with the following approach: 1. 'Push' communication (graphic identity, newsletters, press releases, promotional materials, gadgets, etc.) and 2. 'Pull' communication and interaction (website, social media, liaison with other initiatives and projects, joint communication, etc). The fact that our demonstrators are broadly focused on consumers applications like standard and service packaging (which in turn was a prominent reason for their selection) will allow maximising the awareness of the impact of the bioeconomy, and indeed BBI programmes, on their everyday life.

C. Tool	Month X-initial	Target-KPI
Graphic Identity (templates, logo)	M1	N/A
Website	M3	>20,000 hits
Gadgets	M18	>500 Gadgets distributed
Promotional materials (leaflets, roll-ups, posters)	M5	>3000 leaflets; > 15 posters
Audiovisual materials (video-Youtube, demo videos)	M6	>1500 visualizations; 150 sharings
Social media	M2	>1000 followers
Media Presence (TV, radio, Press release)	M1	>30 press releases; >10 interviews
Newsletter (yearly by email)	M12	>400 suscriptors
Joint communication (BBI, EIP-AGRI, etc)	M6	>8 joint actions
Synergies (with other projects & initiatives)	M2	>8 joint actions

Figure 15: Project Communication Tolbox

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

The project objectives have been the starting point for preparing a robust plan of the work to be implemented within this Innovation Action (Demo) project (Section 1.1), building upon past successful research. All actions have been analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the experts in keeping with their relevant fields of knowledge, infrastructure, resources and in the context of brewery industry, bioplastics supply, packaging industry value chains and packaging waste management. A **clear, comprehensive and realistic work plan** has been split into 9 WPs logically distributed along the value chain, with clear tasks, many interdependent, with very close interaction between the WP and task leaders (see Pert Diagram and Gantt Chart). Logistical matters like the time needed for ordering raw materials, shipping new bio-based materials/prototypes, installations, and possible scenarios of delays have been considered. To ensure the focus **towards the target applications** all the technical WPs are led by industrial partners.

WP1 deals with the production of new **PHA based compounds and coatings** for the specific packaging applications (food, beverage, homecare and cosmetics). Transfer of previous knowledge of the industrial partners to BioSupPack feedstock, derived PHB and PHA coatings will take place, considering a lab scale initial period for it followed by the biobased materials production at pilot scale. **Plasma technology** for feedstock´s pre-treatment and increasing fermentation yields will be implemented at pilot scale. In **WP2**, led by plastic converters partners, **high-performing packaging prototypes** will be produced at large scale using conventional technologies in the plastics industry, after an assessment of the processing ability at pilot plant of developed compounds and coatings in WP1 and with the